GE Medical Systems

Technical Publications

Direction 2223753

Revision 2

Signa® MR/*i* & CV/*i* System Upgrades

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785–5052/8*323–5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section “S” of the Policy & Procedure Bulletins.

3/12/92

GE Medical Systems

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GE Medical Systems — Europe: Telex 261794
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WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0	Mar 4, 1999	Initial release.
1	Jun 11, 1999	Section 1 – Updated catalog information and renewal parts references. Updated flowchart in Section 3 to include new references to CD–ROM procedure locations. Tab 2 – updated upgrade procedures to include latest design changes and other adjustments that would aid in installation.
2	Sep 10, 2010	Overview Tab, Section 1 – Revised Ferrous Material Handling warning, Tab 2– Revised Ferrous Material Handling warnings. Added warnings for movement of rear pedestal. Revised Warnig for removing Dock Assembly when magnet is ramped. (CAPA 1859293)

LIST OF EFFECTIVE PAGES

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Title Page	2
Damage in Trans.	–
Direction 2128126	0*
Direction 2147446	1*
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1–6 to 1–25	1
2–1	0
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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – SYSTEM UPGRADE INFORMATION

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1–1 INTRODUCTION

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

The Upgrade configurations documented throughout this publication are:

CURRENT SYSTEM CONFIGURATION		UPGRADED SYSTEM CONFIGURATION
Signa Horizon LX with LCC Magnet	➧	Signa MR/*i* with LCC Magnet & WideOpen Enclosure
Signa CVMR with LCC Magnet	➧	Signa CV/*i* with LCC Magnet & WideOpen Enclosure

NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

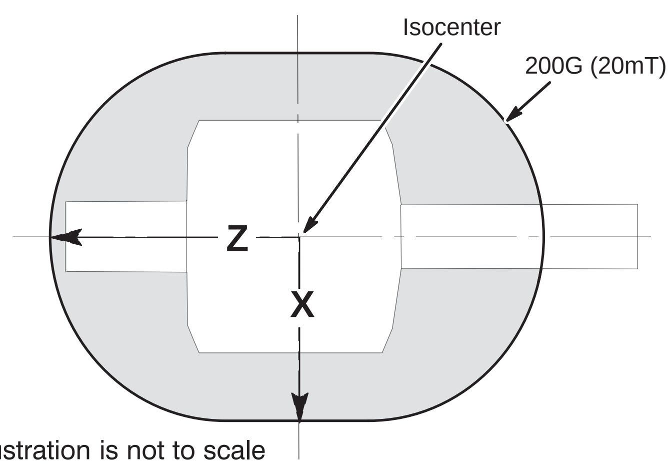


Illustration is not to scale

Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in)[1]	205 cm (81 in)[1]
Unshielded Magnet	Measure [2]	

Note 1:
These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed.

Note 2:
For unshielded magnets, the magnetic field must be measured using a gauss meter.

1–2 SITE READY CHECK

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2223751, Signa MR/i & CV/i Upgrade Pre–installation,* Pre–installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon 8.x system are listed in *Direction 2223751, Signa MR/i & CV/i Upgrade Pre–installation*, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit

1–3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator’s Room
- Orange – Magnet Room
- Yellow – Post–installation (Phantoms, Coils).

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1090NC is for the Signa Horizon EchoSpeed Upgrade). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summery of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1–4 PRODUCT LOCATOR

Product Locator System tracks creation, shipment, trans–shipment, and installation location of serialized models.

The installation card is not a “warranty card”, therefore it is to be processed as soon as the installation starts.

The PLC system also includes Remanufacture, Scrap, and Transfer cards. They should be obtained and completed for all serialized cabinets removed from the system according to Policy and Procedure.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. Process just the installation card and discard any extra shipment cards and labels.

System upgrade serial–number rating plates are furnished. They provide the means to track systems with various hardware versions of the System Upgrade. Kit added rating plates are furnished with various upgrade kits. They signify that model numbers on the major component have changed function and compatibility and have become a “look–alike” for current production in performance and compatibility. These rating plates do not have shipment cards since serial numbers are not assigned.

1–5 DISCARDED MATERIAL RETURN POLICY

Items that are “discarded” or put on a “return pile” when installing the Signa WideOpen upgrade or new options must be returned in accordance with GE recycling policy. Refer to Section 5 for details.

1–6 UPGRADE SCHEDULE CONSIDERATIONS

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options (Multi–coil, Spectroscopy, etc.) will directly add to the man–hour and system down–time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1–7 UPGRADE SEQUENCE

Follow the Horizon system Upgrade Flow chart in SECTION 3 for best installation efficiency.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

Alternate tasks are noted above the task block to set them aside from parallel tasks.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

  Aside: Removing and installing cables is more efficient and accurate when the rooms are un–cluttered. It is strongly recommended that this be done after equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1–8 CATALOG INFORMATION

Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements.

- SECTION 1–9, UPGRADE TO SIGNA MR/*i* FROM SIGNA HORIZON LX WITH LCC MAGNET
  - Fixed, Transportable, or Relocatable sites
  - True Mobile sites
- SECTION 1–10, SIGNA MR/*i* SYSTEM OPTIONS
- SECTION 1–11, UPGRADES TO SIGNA CV/*i* FROM 1.5T SIGNA CVMR WITH LCC MAGNET
  - Fixed, Transportable, or Relocatable sites

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Pre–Installation document uses the technical document names.

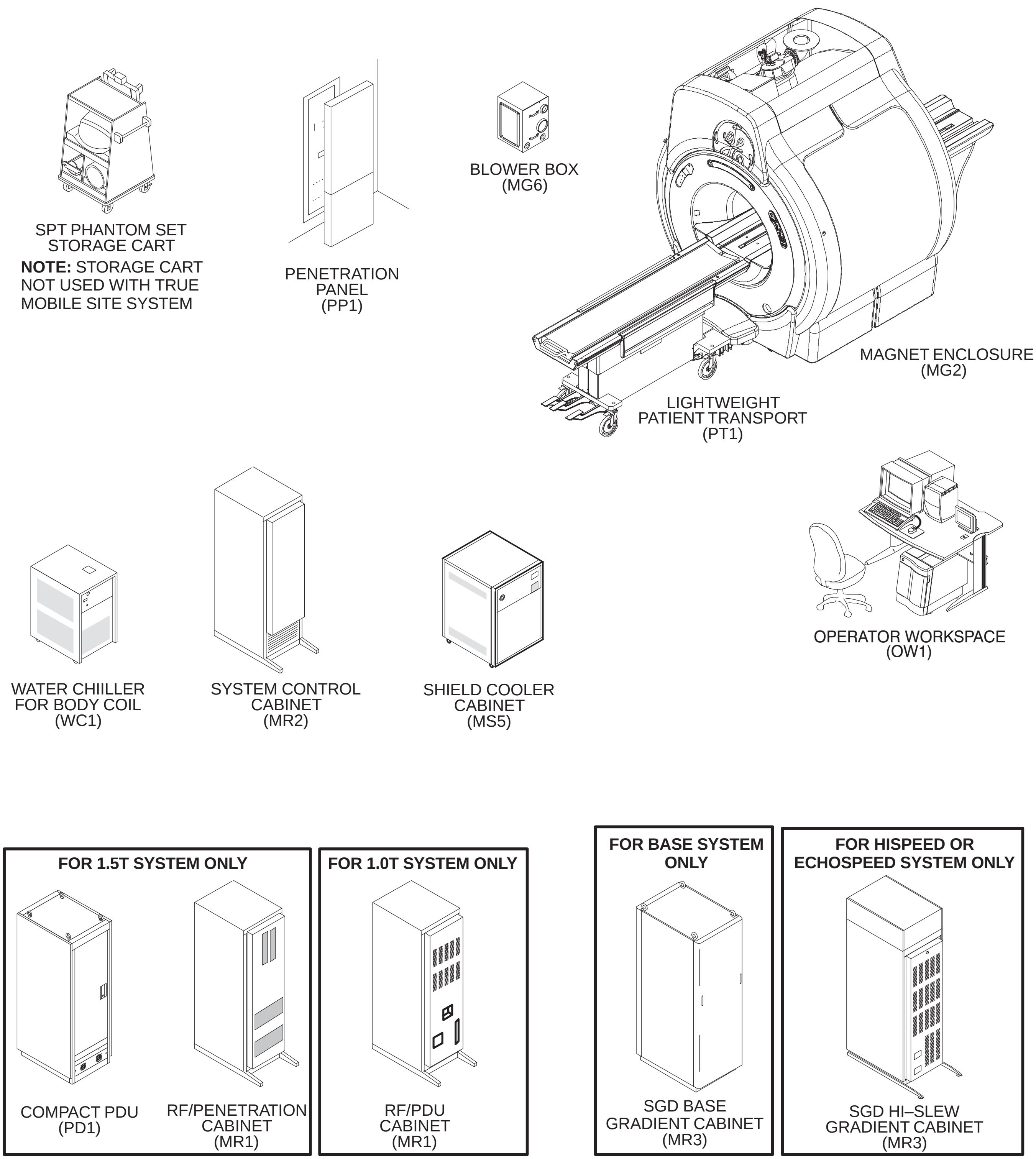
SALES LITERATURE NAME	=	TECHNICAL DOCUMENT NAME
CX 100 Magnet with K4 Technology (CXK4)	=	1.0T LCC (Active Shield) Magnet
CX 150 Magnet with K4 Technology (CXK4)	=	1.5T LCC (Active Shield) Magnet
SX Gradient Subsystem	=	SGD (Scaleable Gradient Driver)

Illustration 1–1 shows the system's major equipment after a Signa MR/*i* Upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final product configuration. The following sub–sections provide an overview of the various Signa Horizon LX with SGD upgrade hardware.

Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

1–8 INTRODUCTION (Continued)



SIGNA MR/*i* SYSTEM MAJOR EQUIPMENT

ILLUSTRATION 1–1

1–9 UPGRADE TO SIGNA MR/*i* FROM SIGNA HORIZON LX WITH LCC MAGNET

The following tables lists the Signa MR/*i* upgrade catalogs for Signa Horizon LX system with an LCC Magnet. Illustration 1–2 shows major equipment impacted (added, replaced, and modified) by the listed catalogs.

TABLE 1–1
SIGNA MR/*i* UPGRADE SITE COLLECTOR KIT (SELECT ONE)

CATALOG	DESCRIPTION
M1090YL	1.5T/1.0T LCC Magnet WideOpen Enclosure Upgrade for fixed site, Transportable, or Relocatable ●Bridge for WideOpen Enclosure with BRM Gradient Coil ●New Rear Pedstal and new Rear Pedestal covers (electronic parts are transferred from Rear Pedestal) ●Remote PAC Interface Assembly (mounts to new enclosure) ●Fixed site enclosure trim and removable cover pieces. ●New fixed site Bore Light Assembly ●Dock Cover Assembly ●PDU modification to power Water Chiller for Gradient Coil Cooling
M1090YN	1.5T/1.0T LCC Magnet WideOpen Enclosure Upgrade for True Mobile ●Bridge for WideOpen Enclosure with BRM Gradient Coil ●New Rear Pedstal and new Rear Pedestal covers (electronic parts are transferred from Rear Pedestal) ●New Dock–to–magnet mounting bracket ●Remote PAC Interface Assembly (mounts to new enclosure) ●True Moibile system removable cover pieces. ●New Mobile Bore Light Assembly ●Dock Cover Assembly

TABLE 1–1
SIGNA MR/*i* UPGRADE MAGNET ENCLOSURE COLLECTOR (SELECT ONE)

CATALOG	DESCRIPTION
M1090YW	Fixed, Transportable, or Relocatable site LCC Magnet WideOpen Enclosure Upgrade
M1090YY	True Mobile LCC Magnet WideOpen Enclosure Upgrade

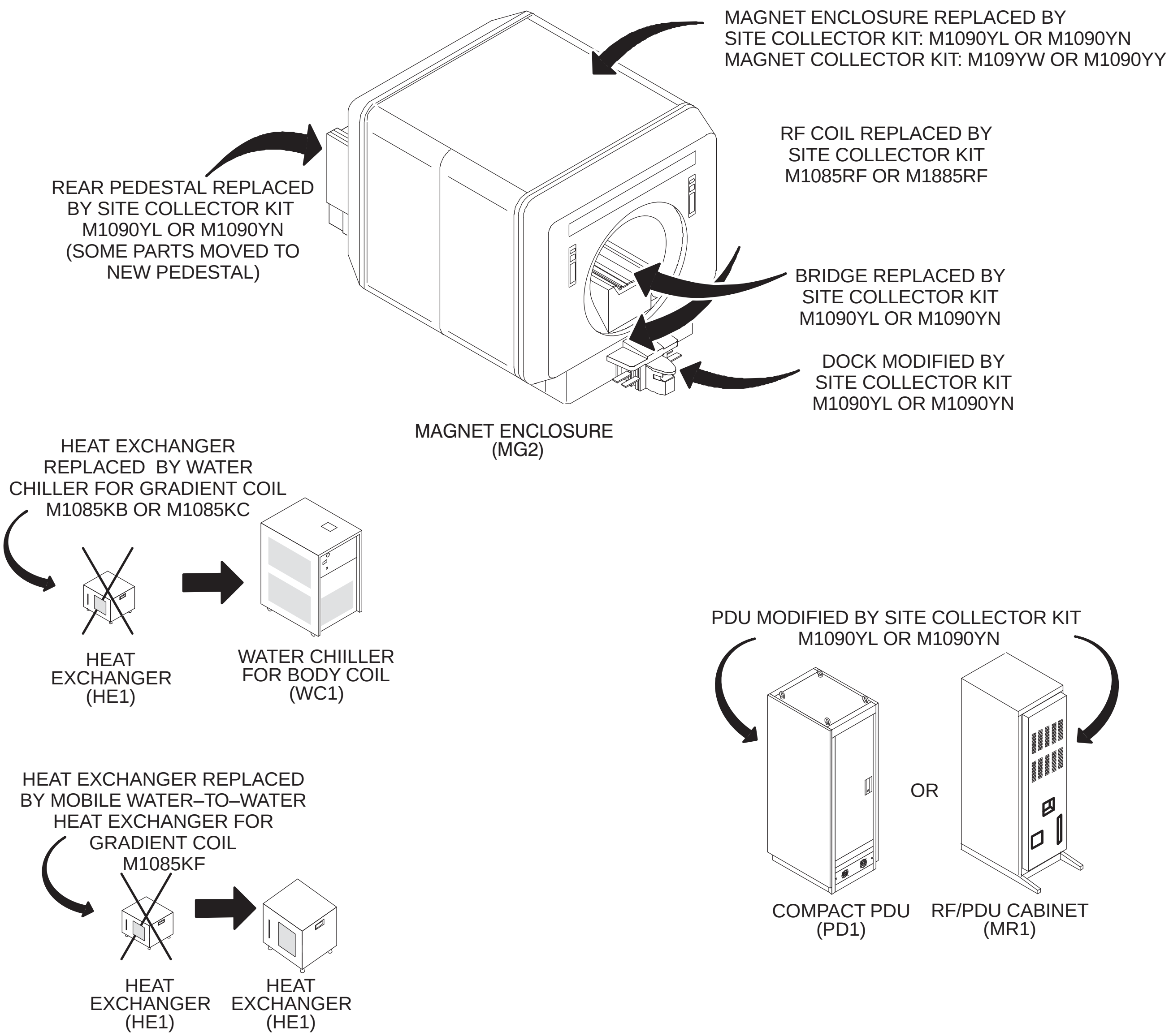
TABLE 1–2
WATER CHILLER FOR GRADIENT COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085KC	●60 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KB	●50 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KF	●50/60 Hz Water–to–Water Heat Exchanger for Mobile systems Gradient Coil Cooling *See Note 2*
Note 1 Fixed Site must order Water Chiller for their power frequency (60 or 50 Hz). **Note** 2 Mobile (True, Transportable, Relocatable) systems require this heat exchanger. The Mobile van will provide chilled water for the heat exchanger operation.	

TABLE 1–3
WIDEOPEN ENCLOSURE COMPATIABLE RF COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085RF	1.5T RF Coil compatiable with WideOpen Enclosure
M1885RF	1.0T RF Coil compatiable with WideOpen Enclosure

1–9 UPGRADE TO SIGNA MR/*i* FROM SIGNA HORIZON LX WITH LCC MAGNET (Continued)



SIGNA HORIZON LX SYSTEM WITH LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–2

1–10 SIGNA MR/*i* SYSTEM OPTIONS

There are no system options specific to the Signa MR/*i* Upgrade. Refer to Direction 2223170 Signa MR/*i* Pre–Installation for typical system options information.

1–11 UPGRADE TO SIGNA CV/*i* FROM SIGNA CVMR WITH LCC MAGNET

The following tables lists the Signa CV/*i* upgrade catalogs for Signa CVMR system with an LCC Magnet in a fixed site or Transporatable/Relocatable Mobile configuration. Illustration 1–2 shows major equipment impacted (added, replaced, and modified) by the listed catalogs.

Note
Signa CV/*i* upgrade is not available for True Mobiles or 1.0T system configurations.

TABLE 1–4
SIGNA CV/*i* UPGRADE SITE COLLECTOR KIT

CATALOG	DESCRIPTION
M1091YM	1.5T/1.0T LCC Magnet WideOpen CV/*i* Enclosure Upgrade for fixed site, Transportable, or Relocatable ●Bridge for WideOpen Enclosure with CRM Gradient Coil ●New Rear Pedstal and new Rear Pedestal covers (electronic parts are transferred from Rear Pedestal) ●Remote PAC Interface Assembly (mounts to new enclosure) ●Fixed site enclosure trim and removable cover pieces. ●New fixed site Bore Light Assembly ●Dock Cover Assembly ●PDU modification to power Water Chiller for Gradient Coil Cooling ●Signa CV/*i* RF Coil compatiable with WideOpen Enclsoure

TABLE 1–1
SIGNA MR/*i* UPGRADE MAGNET COLLECTOR

CATALOG	DESCRIPTION
M1090YP	Fixed, Transportable, or Relocatable site LCC Magnet WideOpen Enclosure Upgrade

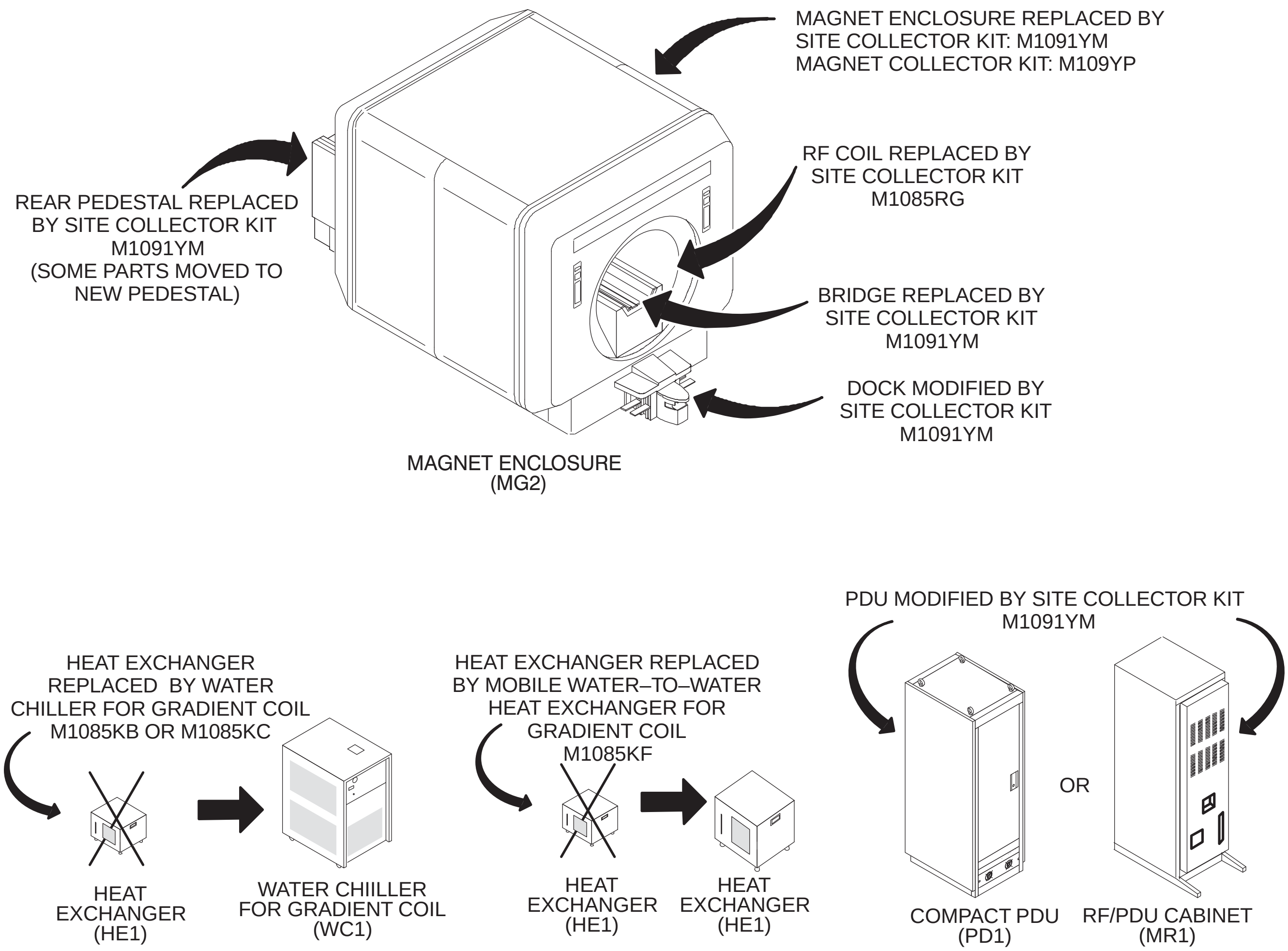
TABLE 1–2
WATER CHILLER FOR GRADIENT COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085KC	●60 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KB	●50 Hz Water–to–Air Chiller for Gradient Coil Cooling *See Note 1*
M1085KF	●50/60 Hz Water–to–Water Heat Exchanger for Mobile systems Gradient Coil Cooling *See Note 2*
Note 1	Fixed Site must order Water Chiller for their power frequency (60 or 50 Hz). Transportable and Relocatable systems do not require this chiller, Gradient Coil chilled water will be provided by Van chilled water system.
Note 2	Transportable and Relocatable systems require this heat exchanger. The Transportable/ Relocatable van will provide chilled water for the heat exchanger operation.

TABLE 1–3
WIDEOPEN ENCLOSURE COMPATIABLE SIGNA CV/*i* RF COIL (SELECT ONE)

CATALOG	DESCRIPTION
M1085RG	●Signa CV/*i* RF Coil compatiable with WideOpen Enclosure

1–11 UPGRADE TO SIGNA CV*i* FROM SIGNA CVMR WITH LCC MAGNET (Continued)



SIGNA CVMR SYSTEM WITH LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–3

1–12 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable Signa Horizon LX system with LCC Magnet or Signa Horizon CVMR sytem with LCC Magnet upgrade catalogs are listed in the previous upgrade sections.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The van/system upgrades can typically be preformed at the customer site. It is customer responsibility contact van manufacturer to confirm viability to perform upgrade at customer site.

- Customer must contact Van Manufacturer to verify upgrade can be performed at customer site. Refer to Section 1–14, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- System check and complete calibrations are performed by GE Field Service.

1–13 TRUE MOBILE UPGRADE

True Mobile Signa Horizon LX system with LCC Magnet upgrade catalogs are listed in the Section 1–9 UPGRADE TO SIGNA MR/*i* FROM SIGNA HORIZON LX WITH LCC MAGNET.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The True Mobile van/system upgrade are preformed at the Van Manufacturer facility.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility. Refer to Section 1–14, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility.
- System check and complete calibrations are performed by MR Manufacturing.

1–14 VAN MANUFACTURERS CONTACT INFORMATION

• Calumet Coach Company		
2150 East Dolton Road, PO Box 1369	Telephone:	708–868–5070
Calumet City, IL 60409–1411	FAX;	708–868–5101
• Ellis & Watts		
4400 Glen Willow Lake Lane	Telephone:	513–752–9000
Batavia, OH 45103	FAX:	513–752–4983
• PDC Facilities		
700 Walnut Ridge Drive, PO Box 900	Telephone:	414–367–7700
Hartland, WI 53029–0900	FAX:	414–367–7744
• AK Associates Inc.		
16745 South Lathrop Avenue		
Harvey, IL 60426	Telephone:	708–596–5066
USA	FAX:	708–596–2480

1–15 WideOpen UPGRADE RENEWAL PARTS

Since considerable hardware remains from initial installation, (electronics cabinets, operator workspace, etc.) retain *Direction 2138247 Signa Renewal Parts Release 5.x & 8.x.*

The following parts lists are specific to the WideOpen upgrade.

CATALOG M1090YL REV 5 1.5/1.0T FXD LCC WideOpen U/PGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2202900–10	N	ASSEMBLY	1	WideOpen UPGRADE FIXED
2	2230960	2	UPGRADE KIT	1	COMPACT PDU GRAD WATER CHILLER ADDER
3	2239133	2	CONTAINER	1	5 GALLON ROUND PAIL W/BUILT–IN HANDL
4	2239134	2	WARN LABEL	1	ETHYLENE GLYCOL WARNING LABEL
30	2228972	N	COLLECTOR	1	WideOpen U/G TECH PUBS COLLECTOR
31	2209600	N	KIT OF PARTS	1	8.X APPLICATIONS TECH PUBS COLLECTOR

CATALOG M1090YN REV 6 1.5/1.0T MOB LCC WideOpen UPGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2202900–11	N	ASSEMBLY	1	WideOpen UPGRADE MOBILE
2	2138791	1	ANTIFREEZE	12	BODY COIL COOLANT,1 GALLON CONTAINER
3	2239133	2	CONTAINER	1	5 GALLON ROUND PAIL W/BUILT–IN HANDL
4	2239134	2	WARN LABEL	1	ETHYLENE GLYCOL WARNING LABEL
30	2228972	N	COLLECTOR	1	WideOpen U/G TECH PUBS COLLECTOR
31	2209600	N	KIT OF PARTS	1	8.X APPLICATIONS TECH PUBS COLLECTOR

CATALOG M1090YP REV 0 CARDIAC LCC FIXED WideOpen UPGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2235245	N	CARDIAC ENCL	1	CARDIAC WO UPGRADE FOR CXK4
2	2230707	N	LADDER/PLATF	1	LADDER/PLATFORM,WideOpen

CATALOG M1090YW REV 1 LCC FIXED WideOpen UPGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2224743	N	ENCLOSURE BR	1	ENCLOSURE,BRMF UPGRADE,LCC, FIXED
2	2230707	N	LADDER/PLATF	1	LADDER/PLATFORM,WideOpen

CATALOG M1090YY REV 1 LCC MOBILE WideOpen UPGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2224743–2	N	ENCLOSURE BR	1	ENCLOSURE UGDRADE,MOBILE CXK4
2	2230707–2	N	LADDER/PLATF	1	LADDER/PLATFORM,WideOpen,MOBILE

CATALOG M1091YM REV 0 CARDIAC LCC FIXED WideOpen UPGRADE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2202900–12	N	ASSEMBLY	1	WideOpen CRM UPGRADE FIXED
30	2228972	N	COLLECTOR	1	WideOpen U/G TECH PUBS COLLECTOR
31	2209700	N	KIT OF PARTS	1	CV/I APPS TECH PUBS COLLECTOR
61	2230960	N	UPGRADE KIT	1	COMPACT PDU GRAD WATER CHILLER ADDER
62	2239133	N	CONTAINER	1	5 GALLON ROUND PAIL W/BUILT–IN HANDL
63	2239134	N	WARN LABEL	1	ETHYLENE GLYCOL WARNING LABEL
64	2240964	2	ADAPTER CONN	6	1/0G TO 2G INLINE SPLICE

WideOpen U/G TECH PUBS 2228972 REV 1

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2138247	N	DIRECTION	1	SIGNA RENEWAL PARTS RELEASE 5.X/8.X
4	2160623–3	N	DIRECTION	1	SIGNA RELEASE 8.X SERVICE METHODS
6	2153389	N	DIRECTION	1	SIGNA RELEASE 8.X BLOCK DIAGRAMS
9	2223753	N	DIRECTION	1	SIGNA MR/I SYSTEM UPGRADE INSTALL

WideOpen APPLICATION PUBS 2209600 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2148091–148	N	OPERATOR MAN	1	SIGNA LX SCAN ASSISTANT MULTI–LANGS.
3	2198119–100	N	OPERATOR MAN	1	SIGNA HORIZON LX2 PSD
5	2219459–100	N	OPERATOR MAN	1	SIGNA 1.5T LX 8.2 PROTOCOLS
6	2216719–100	N	OPERATOR MAN	1	PROBE RECON & QUANTITATION CALCULAT
7	2211602–148	N	OPERATOR MAN	1	SIGNA HORIZON LX DW–EPI OP MAN.
8	2217927–100	N	OPERATOR MAN	1	1.0T LX 8.2 PROTOCOLS
9	2229280–100	N	OPERATOR MAN	1	MRI TOTAL BODY ATLAS VOL. III
10	2207505–248	N	OPERATOR MAN	1	LX 8.2 CD–ROM OMS & CBT_ALL LANGUAGE
11	2222073–100	N	OPERATOR MAN	1	WideOpen NEW FEATURES OP MAN.
12	2222071–148	N	OPERATOR MAN	1	ENCORE EXPRESS NEW FEATURES OP MAN
13	2182101–148	N	OPERATOR MAN	1	LX 8.2.5 ERRATA/REL. NOTES
14	2148092–100	N	OPERATOR MAN	1	SIGNA LX 8.1 CD W/SCAN & CBT QTY1

CATALOG M1085KB REV 0 WideOpen CHILLER – 50 HZ

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2222564–3	1	CHILLER	1	50 HZ GRADIENT CHILLER
3	2223940–35	2	PKG LBL COLL	AR	PACKAGING LABELS FOR M1085KB

CATALOG M1085KC REV 0 WideOpen CHILLER – 60 HZ

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2222564–2	1	CHILLER	1	60 HZ GRADIENT CHILLER
3	2223940–36	2	PKG LBL COLL	AR	PACKAGING LABELS FOR M1085KC

CATALOG M1085RF REV 0 1.5T BRM–F RF TUBE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2220090	1	PART	1	RF COIL ASM,600 X 716_1.5 T (BRM–F)
2	2223940–25	2	PKG LBL COLL	AR	PACKAGING LABELS FOR M1085RF

1.5T RF COIL ASM (BRM–F) 2220090 REV 2

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
3	2218111	1	PART	1	1.5T SHORT RF COIL ASS
4	2221294	2	PART	4	PAD
5	2221633	2	PART	4	PAD,MOUNTING,UPPER
6	2221634	2	PART	4	SET SCREW
7	46–220312P1	2	ADHESIVE	AR	LOCTITE SUPERBONDER 416,
11	2221632	2	PART	2	CONSTRAINT,AXIAL,RF TUBE
12	2143771–2	N	TEFLON TAPE	12	2” TEFLON TAPE
35	46–282805P2	2	SHOCK MOUNT	2	BRIDGE SUPPORT URETHANE PAD, 2X27.3”
40	2139560	N	BULKHEAD ASM	1	BULKHEAD ASSEMBLY

CATALOG M1885RF REV 0 1.0T BRM–F RF TUBE

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2220090–2	1	PART	1	RF COIL ASM,600 X 716_1.0 T (BRM–F)
2	2223940–26	2	PKG LBL COLL	AR	PACKAGING LABELS FOR M1885RF

1.0T RF COIL ASM (BRM–F) 222009–2 REV 2

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
3	2218111–2	1	PART	1	1.0T SHORT RF COIL ASS
4	2221294	2	PART	4	PAD
5	2221633	2	PART	4	PAD,MOUNTING,UPPER
6	2221634	2	PART	4	SET SCREW
7	46–220312P1	2	ADHESIVE	AR	LOCTITE SUPERBONDER 416,
11	2221632	2	PART	2	CONSTRAINT,AXIAL,RF TUBE
12	2143771–2	N	TEFLON TAPE	12	2” TEFLON TAPE
35	46–282805P2	2	SHOCK MOUNT	2	BRIDGE SUPPORT URETHANE PAD, 2X27.3”
40	2139560	N	BULKHEAD ASM	1	BULKHEAD ASSEMBLY

BULKHEAD ASSEMBLY 2139560 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2112592	2	MHV ST ADAPT	2	JACK TO JACK STRAIGHT ADAPTER
3	2138711	2	FRT BULKHEAD	1	FRT BULKHEADRF COIL
4	2109873–6	2	SCREW	4	SCREW PAN HEAD 2.5 MM 10 MM

1.5/1.0T LCC WideOpen–FIXED 2202900–10 REV 3

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	46–287908G1	N	CABLE	1	RUN 421, MG2–A33–J3 TO MG2–A31–J1
3	2205100–13	N	ASSEMBLY	1	WideOpen UPGRADE BRIDGE BRM
4	2199000–5	N	ASSEMBLY	1	REAR PEDESTAL FOR UPGRADES
5	2222541–2	N	COLLECTER CO	1	WideOpen ENCL COVERS ARREM MKE
6	2222541–10	N	COLLECTER CO	1	WideOpen ENCLOSURE COVERS MKE
7	2222541–9	N	COLLECTER CO	1	WIDE OPN ENCL COVER MOBILE MKE
10	2230405	N	ASSEMBLY	1	DOCK COVER ASSEMBLY WideOpen
22	2226264	2	BRACKET ASM	1	WideOpen DOCK BRACKET ASSEMBLY
23	2223440–2	N	COLLECTOR	1	WideOpen PAC/SRI/PLATFORM ASM
26	2227142	N	REMOTE PAC I	1	WideOpen REMOTE PAC INTERFACE ASM
29	2222547	N	LIGHT ASM	1	WideOpen BORE LIGHT ASSEMBLY–FIXED
30	2228970	N	PART	1	COVER PLATE, REAR ENDBELL
31	2228970–2	N	PART	1	COVER PLATE, REAR ENDBELL

1.5/1.0T LCC WideOpen–MOB 2202900–11 REV 3

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	46–287908G1	N	CABLE	1	RUN 421, MG2–A33–J3 TO MG2–A31–J1
3	2205100–13	N	ASSEMBLY	1	WideOpen UPGRADE BRIDGE BRM
4	2199000–5	N	ASSEMBLY	1	REAR PEDESTAL FOR UPGRADES
5	2222541–5	N	COLLECTER CO	1	WideOpen ENCL COV COL MOB ARREM MKE
6	2217457	N	TOP FRONT CO	1	WideOpen TOP FRONT COVER
7	2222541–9	N	COLLECTER CO	1	WIDE OPN ENCL COVER MOBILE MKE
10	2230405	N	ASSEMBLY	1	DOCK COVER ASSEMBLY WideOpen
22	2226264	N	BRACKET ASM	1	WideOpen DOCK BRACKET ASSEMBLY
23	2223440–2	N	COLLECTOR	1	WideOpen PAC/SRI/PLATFORM ASM
26	2227142	N	REMOTE PAC I	1	WideOpen REMOTE PAC INTERFACE ASM
29	2222547–2	N	LIGHT ASM	1	WideOpen BORE LIGHT ASSEMBLY–MOBILE
32	46–320868P1	N	PIPE INSUL	2	POLYETHYLENE CLOSED CELL MATERIAL
33	2202900APR		ASSEMBLY	1	1.0T FIX WideOpen W/O PA

1.5T CARDIAC WideOpen–FXD 2202900–12 REV 3

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	46–287908G1	2	CABLE	1	RUN 421, MG2–A33–J3 TO MG2–A31–J1
3	2205099–9	N	ASSEMBLY	1	CRM BRIDGE WideOpen UPGRADE
4	2199000–5	N	ASSEMBLY	1	REAR PEDESTAL FOR UPGRADES
5	2222541–2	N	COLLECTER CO	1	WideOpen ENCL COVERS ARREM MKE
6	2222541–10	N	COLLECTER CO	1	WideOpen ENCLOSURE COVERS MKE
7	2222541–9	N	COLLECTER CO	1	WIDE OPN ENCL COVER MOBILE MKE
10	2230405	N	ASSEMBLY	1	DOCK COVER ASSEMBLY WideOpen
22	2226264	N	BRACKET ASM	1	WideOpen DOCK BRACKET ASSEMBLY
23	2223440–2	N	COLLECTOR	1	WideOpen PAC/SRI/PLATFORM ASM
26	2227142	N	REMOTE PAC I	1	WideOpen REMOTE PAC INTERFACE ASM
29	2222547	N	LIGHT ASM	1	WideOpen BORE LIGHT ASSEMBLY–FIXED
30	2228970	N	PART	1	COVER PLATE, REAR ENDBELL
31	2228970–2	N	PART	1	COVER PLATE, REAR ENDBELL

REAR PEDESTAL–UPGRADES 2199000–5 REV 3

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2199001	N	ASSEMBLY	1	REAR PEDESTAL FRAME WELDMENT
3	2221408	2	BAR	2	REAR PEDESTAL COVER SUPPORT BAR
4	2221407	2	END CAP	4	REAR PEDESTAL END CAP FOR BAR
8	2128927–2	2	BRACKET ASM	1	CABLE BRACKET ASM FOR PEDESTAL
10	2221414	2	INSULATOR	2	INSULATOR FIBERGLASS
11	2219098–2	2	BRACKET	2	REAR PEDESTAL BAR MOUNTING BRACKET
13	2221418	2	WASHER	14	WASHER PLAIN – LARGE 6.8 MM 38.1 MM
14	2221412	2	SPACER	3	REAR PEDESTAL INSULATED SPACER
42	2223190	2	PART	1	BRACKET, IQ MOUNTING
63	46–271439G5	2	LONGITDL DRV	1	LONGITUDINAL MOTOR DRIVE, MG3 A7
94	2223171	2	PART	2	INSULATOR, REAR PED
97	2109866–10	2	SCREW	7	SCREW HEXAGON HEAD 6 MM 12 MM
98	2109866–13	2	SCREW	11	SCREW HEXAGON HEAD 6 MM 25 MM
100	2109866–17	2	SCREW	4	SCREW HEXAGON HEAD 6 MM 50 MM
102	2109880–3	2	WASHER	2	WASHER LOCK – SPRING 6.1 MM 11.8 MM
103	2109879–2	2	WASHER	6	WASHER PLAIN – LARGE 6.4 MM 12 MM
107	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
108	2109866–12	2	SCREW	6	SCREW HEXAGON HEAD 6 MM 20 MM
109	46–271110P1	2	LABEL	1	PROTECTIVE EARTH (GROUND)
110	2222871	2	PART	1	GROUND WIRE
111	46–208599P19	2	WASHER	1	1/4 EXTERNAL TOOTH PHOS, BRONZE
112	46–208758P2	2	TY–RAP	1	5.5” X .140” SELF–LOCKING CABLE TIE.
113	2229552	2	BRACKET	1	RF COIL REAR BULKHEAD
120	2235607	2	LOCATOR	2	BRIDGE LOCATOR

CRM BRIDGE WideOpen U/G 2205099–9 REV 1

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
5	2186362–4	N	PART	1	BRIDGE, S SERIES/CRM
6	2226929	N	COVER	1	WideOpen BRIDGE COVER W/LABELS
7	2131304	N	COVER,GASKET	1	GASKET, COVER, BRIDGE
8	2109873–26	2	SCREW	4	SCREW PAN HEAD 6 MM 20 MM

WideOpen UPGRADE BRIDGE 2205100–13 REV 2

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
5	2219401	N	BRIDGE	1	WideOpen, 60 CM BRIDGE
6	2226929	N	COVER	1	WideOpen BRIDGE COVER W/LABELS
7	2131304	N	COVER,GASKET	1	GASKET, COVER, BRIDGE
8	2109873–26	2	SCREW	4	SCREW PAN HEAD 6 MM 20 MM
9	46–214708P38	2	PLUG, HOLE	4	HEYCO PLUG 2754 WHITE 1–3/8

BRIDGE COVER W/LABELS 2226929 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2218112	2	PART	1	REAR PLATE, BRIDGE
3	46–302200P7	N	RATING PLATE	1	SERIAL #, MODEL#
4	2142857	N	ETL LABEL	1	ETL LISTING LABEL FOR GLOBAL
5	2117390	N	LABELS	1	CE INTERNATIONAL APPROVAL LABEL
6	2132620	N	LABELS	1	CE MARKING LABEL CLASS A GROUP 1

WideOpen ENCL COVERS 2222541–2 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2217314	2	TOP LEFT COV	1	WideOpen TOP LEFT COVER
2	2217313	2	TOP RIGHT CO	1	WideOpen TOP RIGHT COVER
3	2217308	2	RIGHT FOOT C	1	WideOpen FRONT RIGHT FOOT COVER
4	2217310	2	LEFT FOOT CO	1	WideOpen FRONT LEFT FOOT COVER
5	2217311	2	BASE TRIM FR	1	WideOpen BASE TRIM FRONT COVER
6	2217312	2	BASE TRIM RE	1	WideOpen BASE TRIM REAR COVER
7	2225982	2	ASSEMBLY	1	WideOpen BASE TRIM REAR RIGHT ASM
8	2225982–2	2	ASSEMBLY	1	WideOpen BASE TRIM REAR LEFT ASM

WideOpen ENCL COVERS–MOB 2222541–5 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
3	2217308	2	RIGHT FOOT C	1	WideOpen FRONT RIGHT FOOT COVER
4	2217310	2	LEFT FOOT CO	1	WideOpen FRONT LEFT FOOT COVER

WideOpen ENCL COVERS–MOB 2222541–9 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
3	2217705	2	PEDESTAL COV	1	WideOpen LEFT PEDESTAL COVER
4	2217704	2	PEDESTAL COV	1	WideOpen RIGHT PEDESTAL COVER

WideOpen ENCL COVERS 2222541–10 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2217457	2	TOP FRNT COV	1	WideOpen TOP FRONT COVER
2	2217458	2	TOP REAR COV	1	WideOpen TOP REAR COVER

WideOpen BORELIGHT ASM–FXD 2222547 REV 2

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2222548	2	LIGHT ASM	1	WideOpen BORE LIGHT SOURCE ASM
2	2222376	2	PART	1	PLATE,MOUNTING,BORE LIGHT
3	46–313195P1	2	TERMINAL	1	2 POSITION TERMINAL STRIP, DUAL ROW,
6	2222550	2	LIGHT HOLDER	1	WideOpen FIBER OPTIC CABLE HOLDER
7	46–170209P1	2	TAPE	AR	0.034 X 1.00 X 100 FT
8	46–287268P1	2	LABEL	2	CAUTION, EXTREMELY HOT HALOGEN LAMPS
9	2109873–5	2	SCREW	4	SCREW PAN HEAD 2.5 MM 8 MM
10	2109875–2	2	NUT	2	NUT HEXAGON 4 MM
11	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
12	46–328424P1	2	SCREW	2	SCREW SET SCREW 2.5 MM 4 MM
13	46–208758P2	2	TY–RAP	3	5.5” X .140” SELF–LOCKING CABLE TIE.
14	2184145	2	CABLE	1	(300) PP1–J15 TO MG2–A15 BORE LIGHT
15	46–136246P26	2	TAPE	2	VELCRO,LOOP,1IN.W,BLK W/PSA
16	46–136246P29	2	TAPE	2	VELCRO HOOK FASTENER

WideOpen BORE LIGHT–MOB 2222547–2 REV 1

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2222548	2	LIGHT ASM	1	WideOpen BORE LIGHT SOURCE ASM
2	2222376	2	PART	1	PLATE,MOUNTING,BORE LIGHT
3	46–313195P1	2	TERMINAL	1	2 POSITION TERMINAL STRIP, DUAL ROW,
6	2222550	2	LIGHT HOLDER	1	WideOpen FIBER OPTIC CABLE HOLDER
7	46–170209P1	2	TAPE	AR	0.034 X 1.00 X 100 FT
8	46–287268P1	2	LABEL	2	CAUTION, EXTREMELY HOT HALOGEN LAMPS
9	2109873–5	2	SCREW	4	SCREW PAN HEAD 2.5 MM 8 MM
10	2109875–2	2	NUT	2	NUT HEXAGON 4 MM
11	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
12	46–328424P1	2	SCREW	2	SCREW SET SCREW 2.5 MM 4 MM
13	46–208758P2	2	TY–RAP	3	5.5” X .140” SELF–LOCKING CABLE TIE.
14	2184145–2	2	CABLE	1	(300M) PP1–J15 TO MG2–A15 BORE LIGHT
15	46–136246P26	2	TAPE	2	VELCRO,LOOP,1IN.W,BLK W/PSA
16	46–136246P29	2	TAPE	2	VELCRO HOOK FASTENER

WideOpen PAC/SRI/PLATFORM 2223440–2 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2122268–4	2	PAC–II ASM	1	PHYSIOLOGICAL ACQUISITION CONTROLLER
3	2219777	2	PAC–SIR ASM	1	PAC–SRI–WideOpen

LCC FXD ENCLOSURE UPGRADE 2224743 REV 6

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2217702	2	WRAP COVER R	1	WideOpen WRAP COVER RIGHT
2	2217703	2	WRAP COVER L	1	WideOpen WRAP COVER LEFT
5	2218146–3	2	LOWER CLAMP	2	WideOpen LOWER CLAMPING BRKT UPGRADE
6	2218146–4	2	LOWER CLAMP	2	WideOpen LOWER CLAMPING BRKT UPGRADE
10	2235935	2	BRIDGE SUPPO	2	WideOpen BRIDGE SUPPORT
11	2218898	2	TOP RIGHT FR	1	WideOpen TOP RIGHT FRAME
12	2218897	2	TOP LEFT FRA	1	WideOpen TOP LEFT FRAME
15	2232260	2	FRONT END BE	1	END BELL ASM,WO UPGRADE FRONT
16	2232234	2	REAR END BEL	1	END BELL ASM,WO UPGRADE REAR
17	2217461	2	ARC FRONT RI	1	WideOpen ARC COVER FRONT, RIGHT
18	2217462	2	ARC FRONT LE	1	WideOpen ARC FRONT LEFT COVER
19	2217463	2	ARC REAR RIG	1	WideOpen ARC REAR RIGHT COVER
20	2217464	2	ARC REAR LEF	1	WideOpen ARC COVER REAR, LEFT
21	2218899	2	BORE LIGHT E	2	WideOpen BORE LIGHT ENDCAP
26	46–282668P18	2	HOSE, NYLON	1	2 INCH DIAMETER, 132 INCH LENGTH
27	46–208765P28	2	CLAMP	1	DIA RANGE 1–13/16 – 2–3/4, 302/305
28	2222549	2	LIGHT PANEL	2	WideOpen LUMITEX LIGHT PANEL
35	46–170617P1	2	ADHESIVE	1	RTV SILICON RUBBER ADHESIVE SEALANT.
39	2221945	2	CABLE	1	MG2–A4–J5 TO MG2–A11–J4 PAC/DISPLAY
40	2221948	2	CABLE	1	MG2–A4–J3 TO MG2–A33–J4 DISPLAY/SRI
41	2221947	2	CABLE	1	MG2A4J4 TO MG2A33–5,6,7,8,9 DSPL/SRI
42	2221946	2	CABLE	2	MG2A4–J1/J2 TO MG2A5–R/L SW PNL/DSPL
46	2215716–4	1	OPR BP/D KIT	1	WideOpen MAGNET HOUSING CNTR DISPLAY
48	2222329	2	PART	1	RF COIL FEED THRU COVER
64	2174929–6	2	CONFOR FOAM	2	OPEN CELL POLYURETHANE FOAM, 45” LG
73	2224551	2	LATCH	4	WideOpen END BELL LATCH
80	2222806–2	2	PART	1	FOAM BLOCK, WideOpen REAR ENDBELL
86	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
87	2226752	2	PART	4	FOAM GASKET, BRIDGE SUPPORT SPACER
88	2239902–3	2	FOAM SEAL	1	CONFOR FOAM
93	2227395	2	PART	1	CABLE/HOSE CLAMP, TOP
94	2227396	2	PART	1	CABLE/HOSE CLAMP, BASE
101	2227708	2	SPACER	1	WideOpen RF TUBE LOCATING SPACER
103	2222353	2	PART	4	BRIDGE SUPPORT SPACER
105	2227841	2	PART	2	FOAM GASKET, HOSE/CABLE CLAMP
106	2228286	2	PART	2	FOAM SEAL, CABLE/HOSE CLAMP
107	2228287	2	PART	2	FOAM BLOCK, CABLE/HOSE CLAMP
112	46–220191P1	2	CLAMP	14	CLAMP FOR FLAT CABLE
120	2235604	2	SPACER	1	ENDBELL PLATE MOUNTING SPACER
121	2235605	2	PLATE	1	ENDBELL ANCHOR PLATE
122	2235606	2	BLOCK	1	WASHER BLOCK
226	46–294151P11	2	ADHES SPEC	AR	LOCKTITE 271, LOCKTITE NO. 27121,
227	2232030	2	HARDWARE KIT	1	KIT,WO ENCLOSURE UPGRADE HARDWARE
229	2239902–2	2	FOAM SEAL	1	CONFOR FOAM
231	2239904	2	PIPE INSULAT	6	PIPE INSULATION
232	2239904–4	2	PIPE INSULAT	6	PIPE INSULATION
233	2237254	2	TROUGH	1	TROUGH WITH SPONGE
234	2179397–3	2	BRM ISOLATIO	2	OPEN FOAM CAULKING STRIP – .500” THK
300	2214162	2	CLAMP	3	GRADIENT CABLE CLAMP
301	2239148–2	2	BLOCK	3	SPACER BLOCK AND NUT STRIP BLOCK
400	2226534	2	TUBING	1	ZIPPER TUBING FOR GAS LINES 1–1/2 DI

LCC MOB ENCLOSURE UPGRADE 2224743–2 REV 5

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
5	2218146–5	2	LOWER CLAMP	1	WideOpen LOWER CLAMPING BRACKET—MOB
6	2218146–6	2	LOWER CLAMP	1	WideOpen LOWER CLAMPING BRACKET– –MOB
10	2235935	2	BRIDGE SUPPO	2	WideOpen BRIDGE SUPPORT
11	2218898–2	2	TOP RIGHT FR	1	WideOpen TOP RIGHT FRAME – MOBILE
12	2218897–2	2	TOP LEFT FRA	1	WideOpen TOP LEFT FRAME – MOBILE
15	2232260	2	FRONT END BE	1	END BELL ASM,WO UPGRADE FRONT
16	2232234	2	REAR END BEL	1	END BELL ASM,WO UPGRADE REAR
17	2217461–2	2	ARC FRONT RI	1	WideOpen ARC COVER FRONT,RIGHT–MOBILE
18	2217462–2	2	ARC FRONT LE	1	WideOpen ARC FRONT LEFT COVER– –MOBILE
21	2218899	2	BORE LIGHT E	2	WideOpen BORE LIGHT ENDCAP
26	46–282668P18	2	HOSE, NYLON	1	2 INCH DIAMETER, 132 INCH LENGTH
27	46–208765P28	2	CLAMP	1	DIA RANGE 1–13/16 – 2–3/4, 302/305
28	2222549	2	LIGHT PANEL	2	WideOpen LUMITEX LIGHT PANEL
35	46–170617P1	2	ADHESIVE	1	RTV SILICON RUBBER ADHESIVE SEALANT.
39	2221945	2	CABLE	1	MG2–A4–J5 TO MG2–A11–J4 PAC/DISPLAY
40	2221948	2	CABLE	1	MG2–A4–J3 TO MG2–A33–J4 DISPLAY/SRI
41	2221947	2	CABLE	1	MG2A4J4 TO MG2A33–5,6,7,8,9 DSPL/SRI
42	2221946	2	CABLE	2	MG2A4–J1/J2 TO MG2A5–R/L SW PNL/DSPL
46	2215716–4	1	OPR BP/D KIT	1	WideOpen MAGNET HOUSING CNTR DISPLAY
48	2222329	2	PART	1	RF COIL FEED THRU COVER
64	2174929–6	2	CONFOR FOAM	2	OPEN CELL POLYURETHANE FOAM, 45” LG
73	2224551	2	LATCH	4	WideOpen END BELL LATCH
80	2222806–3	2	PART	1	FOAM BLOCK, WideOpen MOBILE INSTALL
86	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
87	2226752	2	PART	4	FOAM GASKET, BRIDGE SUPPORT SPACER
88	2239902–3	2	FOAM SEAL	1	CONFOR FOAM
93	2227395	2	PART	1	CABLE/HOSE CLAMP, TOP
94	2227396	2	PART	1	CABLE/HOSE CLAMP, BASE
101	2227708	2	SPACER	1	WideOpen RF TUBE LOCATING SPACER
103	2222353	2	PART	4	BRIDGE SUPPORT SPACER
105	2227841	2	PART	2	FOAM GASKET, HOSE/CABLE CLAMP
106	2228286	2	PART	2	FOAM SEAL, CABLE/HOSE CLAMP
107	2228287	2	PART	2	FOAM BLOCK, CABLE/HOSE CLAMP
112	46–220191P1	2	CLAMP	14	CLAMP FOR FLAT CABLE
119	2179397–3	2	BRM ISOLATIO	2	OPEN FOAM CAULKING STRIP – .500” THK
120	2235604	2	SPACER	1	ENDBELL PLATE MOUNTING SPACER
121	2235605	2	PLATE	1	ENDBELL ANCHOR PLATE
122	2235606	2	BLOCK	1	WASHER BLOCK
226	46–294151P11	2	ADHES SPEC	AR	LOCKTITE 271, LOCKTITE NO. 27121,
227	2232030	2	HARDWARE KIT	1	KIT,WO ENCLOSURE UPGRADE HARDWARE
229	2239902–2	2	FOAM SEAL	1	CONFOR FOAM
230	2239904–2	2	PIPE INSULAT	12	PIPE INSULATION
231	2239904	2	PIPE INSULAT	9	PIPE INSULATION
233	2237254	2	TROUGH	1	TROUGH WITH SPONGE
234	2179397–3	2	BRM ISOLATIO	2	OPEN FOAM CAULKING STRIP – .500” THK
300	2214162	2	CLAMP	1	GRADIENT CABLE CLAMP
301	2239148–2	2	BLOCK	1	SPACER BLOCK AND NUT STRIP BLOCK
400	2226534	2	TUBING	1	ZIPPER TUBING FOR GAS LINES 1–1/2 DI

BASE TRIM REAR RIGHT ASM 2225982 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2217312	N	BASE TRIM RE	1	WideOpen BASE TRIM REAR COVER
3	2219358–2	N	REAR ACCESS	1	WideOpen RER ACCESS PANEL–RIGHT
4	46–136246P11	N	TAPE	0.840	5/8" WIDE, BLK ADH BACK, VELCRO HOOK
5	46–136246P12	N	TAPE	0.840	0.625 WIDE X 25 YD,BLK,ADH BACK,LOOP

BASE TRIM REAR LEFT ASM 2225982–2 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2217311	N	BASE TRIM FR	1	WideOpen BASE TRIM FRONT COVER
3	2219358	N	REAR ACCESS	1	WideOpen REAR ACCESS PANEL–LEFT
4	46–136246P11	N	TAPE	0.840	5/8" WIDE, BLK ADH BACK, VELCRO HOOK
5	46–136246P12	N	TAPE	0.840	0.625 WIDE X 25 YD,BLK,ADH BACK,LOOP

WideOpen DOCK BRACKET ASM 2226264 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	46–282757G7	2	DOCK CONTROL	1	WideOpen DOCK CONTROL BOX/BRACKET
3	2218120	2	DOCK MTG SUP	1	WideOpen DOCK MOUNTING SUPPORT
4	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC

REMOTE PAC INTERFACE ASM 2227142 REV 0

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
2	2221024	2	PAC_HANGER_C	1	WideOpen PAC HANGER CABLE INTERFACE
3	2222710	2	CLAMP BRCKT	1	WideOpen PAC INTERFACE CLAMP BRCKT
4	2219096	2	CABLE	1	ECG EXTENSION CABLE
5	2219097	2	CABLE	1	PERIPHERAL PULSE EXTENSION CABLE
6	46–265651P1	2	FITTING	2	COUPLING, PANEL MTG
7	2221023	2	PAC_HANGER_B	1	WideOpen PAC HANGER BAR
8	2238216	N	SCREW	2	SCREW HEXAGON SOCKET 6 MM 70 MM
9	46–328420P502	N	SCREW	4	SCREW BUTTON HEAD 3 MM 8 MM
10	46–317758P18	N	ALERT SYSTEM	2	TUBING
11	46–294151P2	N	ADHES SPEC	AR	LOCKTITE #242 TYPE 11
12	46–265652P1	N	FITTING	2	COUPLING INSERT
13	2109874–17	2	SCREW	1	SCREW COUNTERSUNK HEA 4 MM 40 MM

WideOpen DOCK COVER ASM 2230405 REV 3

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
6	46–271406G2	N	COVER ASM.	1	LCC CASTING W/BEARINGS
31	46–243341G3	N	BUMPER	1	LCC DOCK BUMPER
35	2230001	N	SPACER PAD	1	RUBBER PAD
39	46–197385P1	N	MECHANICAL/M	1	WHITE LITHIUM, MULTI–PURPOSE GREASE.
57	46–271265P1	N	BAR	1	CRADLE LATCH ACTUATOR
72	46–271935P1	N	TUBE SPRING	1	CAST POLYURETHANE
14080	46–220196P4	N	WASHER	1	1/4" SCREW SIZE (0.250), SST
14619	46–208737P19	N	SCREW	4	SCREW
18828	12–241	N	WASHER	2	ID 0.266, OD 0.75, THK .051, BRASS
34356	46–208948P5	N	NUT	3	1/4–20 .438 HEX X .226 THK, 18–8 SST

CARDIAC LCC WideOpen UPGRADE 2235245 REV 1

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2217702	2	WRAP COVER R	1	WideOpen WRAP COVER RIGHT
2	2217703	2	WRAP COVER L	1	WideOpen WRAP COVER LEFT
5	2218146–3	2	LOWER CLAMP	2	WideOpen LOWER CLAMPING BRKT UPGRADE
6	2218146–4	2	LOWER CLAMP	2	WideOpen LOWER CLAMPING BRKT UPGRADE
11	2218898	2	TOP RIGHT FR	1	WideOpen TOP RIGHT FRAME
12	2218897	2	TOP LEFT FRA	1	WideOpen TOP LEFT FRAME
15	2233034	2	FEB MIC & BA	1	CARDIAC_FRONT END BELL ASM
16	2233036	2	REB SPEAKER	1	CARDIAC_REAR END BELL ASM
17	2217461	2	ARC FRONT RI	1	WideOpen ARC COVER FRONT, RIGHT
18	2217462	2	ARC FRONT LE	1	WideOpen ARC FRONT LEFT COVER
19	2217463	2	ARC REAR RIG	1	WideOpen ARC REAR RIGHT COVER
20	2217464	2	ARC REAR LEF	1	WideOpen ARC COVER REAR, LEFT
21	2218899	2	BORE LIGHT E	2	WideOpen BORE LIGHT ENDCAP
26	46–282668P18	2	HOSE, NYLON	1	2 INCH DIAMETER, 132 INCH LENGTH
27	46–208765P28	2	CLAMP	1	DIA RANGE 1–13/16 – 2–3/4, 302/305
28	2222549–2	2	LIGHT PANEL	2	CARDIAC WideOpen LUMITEX LIGHT PNL
30	2162554	2	LASER DIODE	4	LASER FOR CONQUEST MAG FRT COVER ASM
35	46–170617P1	2	ADHESIVE	1	RTV SILICON RUBBER ADHESIVE SEALANT.
39	2221945	2	CABLE	1	MG2–A4–J5 TO MG2–A11–J4 PAC/DISPLAY
40	2221948	2	CABLE	1	MG2–A4–J3 TO MG2–A33–J4 DISPLAY/SRI
41	2221947	2	CABLE	1	MG2A4J4 TO MG2A33–5,6,7,8,9 DSPL/SRI
42	2221946	2	CABLE	2	MG2A4–J1/J2 TO MG2A5–R/L SW PNL/DSPL
44	2215716–2	2	OPR BP/D KIT	1	WideOpen MAGNET HOUSING RH DISPLAY
45	2215716–3	2	OPR BP/D KIT	1	WideOpen MAGNET HOUSING LH DISPLAY
46	2215716–4	1	OPR BP/D KIT	1	WideOpen MAGNET HOUSING CNTR DISPLAY
48	2222329	2	PART	1	RF COIL FEED THRU COVER
73	2224551	2	LATCH	4	WideOpen END BELL LATCH
80	2222806–2	2	PART	1	FOAM BLOCK, WideOpen REAR ENDBELL
86	46–170686P2	2	ADHESIVE	AR	LOCTITE 242, 10CC
88	2239902–3	2	FOAM SEAL	1	CONFOR FOAM
93	2227395	2	PART	1	CABLE/HOSE CLAMP, TOP
94	2227396	2	PART	1	CABLE/HOSE CLAMP, BASE
105	2227841	2	PART	2	FOAM GASKET, HOSE/CABLE CLAMP
106	2228286	2	PART	2	FOAM SEAL, CABLE/HOSE CLAMP
107	2228287	2	PART	2	FOAM BLOCK, CABLE/HOSE CLAMP
110	46–170212P1	2	ADHESIVE	AR	ADHESIVE
112	46–220191P1	2	CLAMP	14	CLAMP FOR FLAT CABLE
120	2235604	2	SPACER	1	ENDBELL PLATE MOUNTING SPACER
121	2235605	2	PLATE	1	ENDBELL ANCHOR PLATE
122	2235606	2	BLOCK	1	WASHER BLOCK
227	2232030	2	HARDWARE KIT	1	KIT,WO ENCLOSURE UPGRADE HARDWARE
229	2239902–2	2	FOAM SEAL	1	CONFOR FOAM
231	2239904	2	PIPE INSULAT	6	PIPE INSULATION
232	2239904–4	2	PIPE INSULAT	6	PIPE INSULATION
233	2237254	2	TROUGH	1	TROUGH WITH SPONGE
300	2214162	2	CLAMP	3	GRADIENT CABLE CLAMP
301	2239148–2	2	BLOCK	3	SPACER BLOCK AND NUT STRIP BLOCK
400	2226534	2	TUBING	1	ZIPPER TUBING FOR GAS LINES 1–1/2 DI

SECTION 2 – POWER OFF PROCEDURE

2–1 SYSTEM POWER SHUTDOWN

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

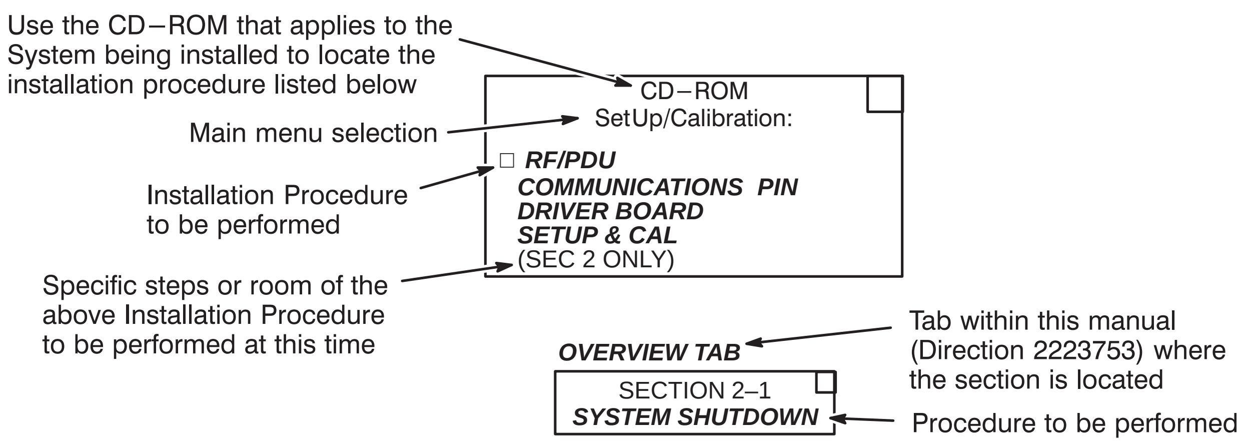
1. Verify that System performance baseline measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.
4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 3 – UPGRADE INSTALLATION FLOWCHART

This section contains information on:

3–1 SIGNA HORIZON MR/*i* & CV/*i* UPGRADE FLOWCHART EXPLANATION

Shown below are three examples of the flow chart Installation Prodcedures and an explanation of the areas within the procedures.



LEGEND FOR FOLLOWING INSTALLATION FLOWCHART

MX INDICATES MILESTONE COMPLETION GOAL IN DAYS

OR INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION

INDICATES GE PROPRIETARY PROCEDURE

1 INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.

NOTE 1: (NOT USED)
NOTE 2: IF TROUBLESHOOTING NECESSARY.
NOTE 3: (NOT USED)
NOTE 4: (NOT USED)
NOTE 5: (NOT USED)
NOTE 6: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

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SIGNA HORIZON 8.x LCC TO SIGNA 8.x LCC WIDE OPEN (MR/*i* & CV/*i*) UPGRADE INSTALLATION FLOW CHART (FIXED–SITES ONLY)

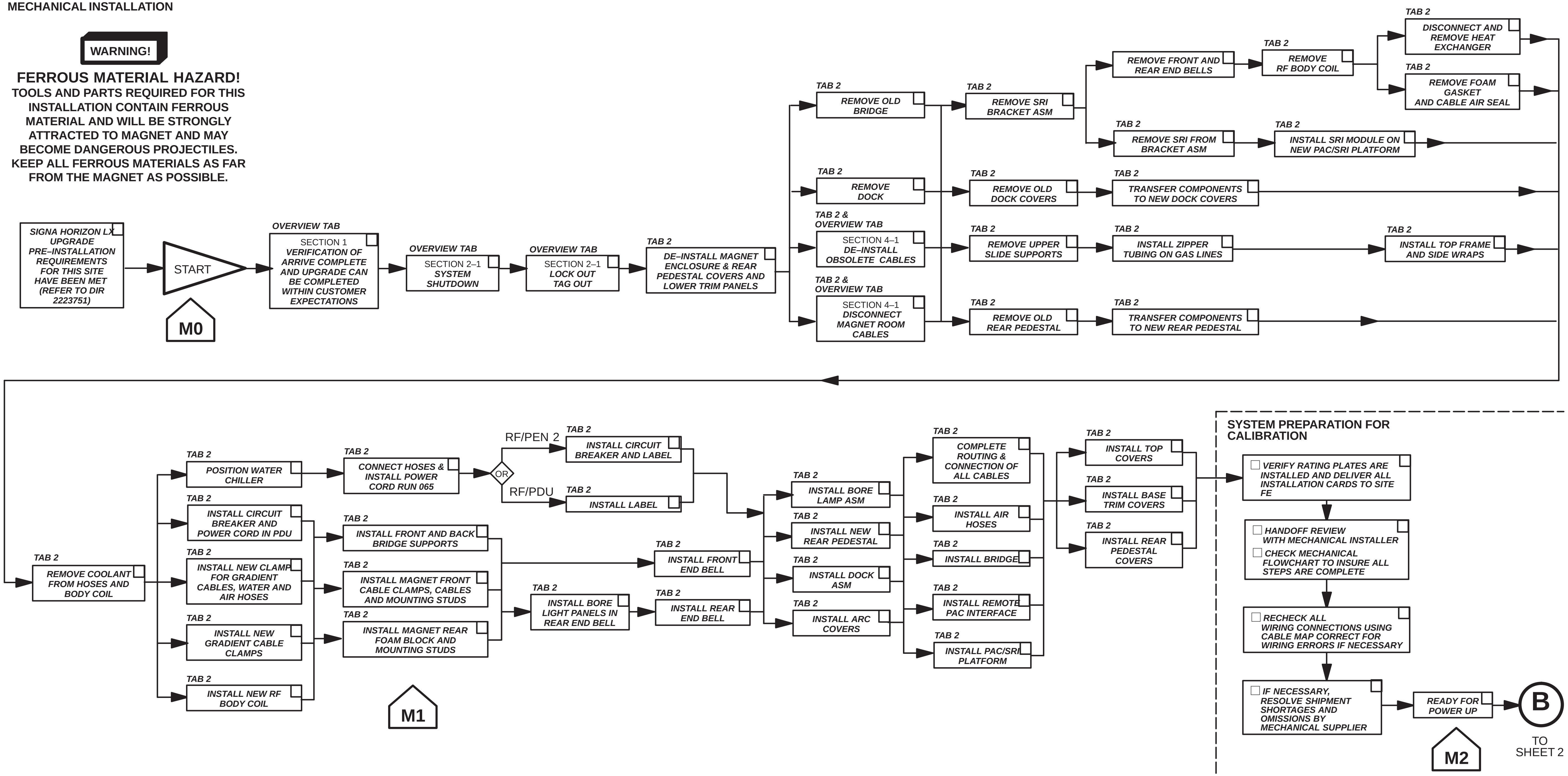
MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS

MECHANICAL INSTALLATION

WARNING!

FERROUS MATERIAL HAZARD!
TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.



SIGNA MR/*i* UPGRADE INSTALLATION FLOW CHART
ILLUSTRATION 1–1 (SHEET 1 OF 2)

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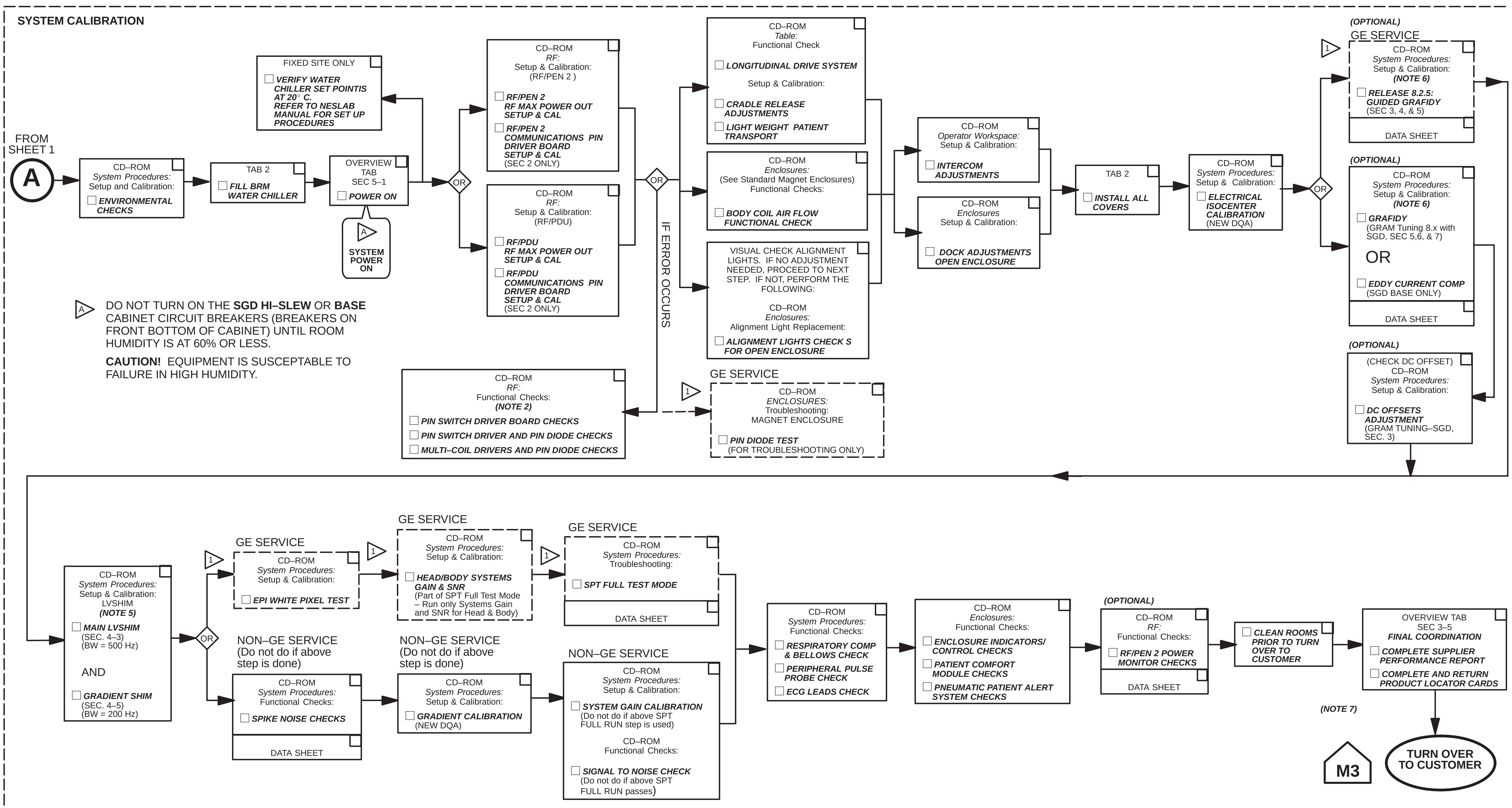
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS



SIGNA MR/*i* UPGRADE INSTALLATION FLOW CHART
ILLUSTRATION 1–1 (SHEET 2 OF 2)

SECTION 4 – SYSTEM CABLES

4–1 CABLES REMOVED FOR UPGRADING TO HORIZON WIDE-OPEN ENCLOSURE

As indicated on the Affected Cables Interconnect Maps that are found on pages 4–3 thru 4–6, remove the following cables which are to be returned for recycling:

TABLE 4–1
HORIZON 8.x CABLES TO BE REMOVED

RUN	“FROM”	“TO”	DESCRIPTION	REMARKS
046	PD1–A2	HE1–J1	Heat Exchanger Power Cord	Power Cable
414	MG2–A35	MG3–A2–J9	Front Cover Microphone Cable	30 ft. 9–pin Sub–D
421	MG2–A33–J3	MG2–A31–J1	Dock Limit Cable	30 inch 15 pin Sub–D
749	MG2–A33–J10	MG2–A36	Bore Temperature Sensor Cable	35 ft 3–cond w/ 9–Sub D
	MG2–A33–J4	MG2–A4–J3	SRI to Right Operator Display Cable	Part of Front Cover
	MG2–A33–J5,J6,J7,J8,J9	MG2–A4–J4	SRI to Right Operator Display Cable	Part of Front Cover
	MG3–A2–J7	MG2–A14–A1	Microphone Cable	Part of Rear End Bell
	MG3–A2–J8	MG2–A14–A2	Speaker Cable	Part of Rear End Bell
	MG2 A15	MG2–A14	Fiber Optic Bundles	Part of Bore Lamp Assembly

4–2 CABLES TO DISCONNECT AND SAVE FOR UPGRADE TO NEW WIDE-OPEN ENCLOSURE

The Affected Cables Interconnect Maps also indicate the cables to be disconnected and reused when installing the new magnet enclosure and water chiller. The upgrade instructions in Tab 2, Magnet Enclosures, designate which ends of the cables to disconnect.

Fold–out Affected Cables for Upgrading to Signa MR/*i* (CV/*i*) from Signa Horizon 8.x cable maps are provided on the following pages. They are:

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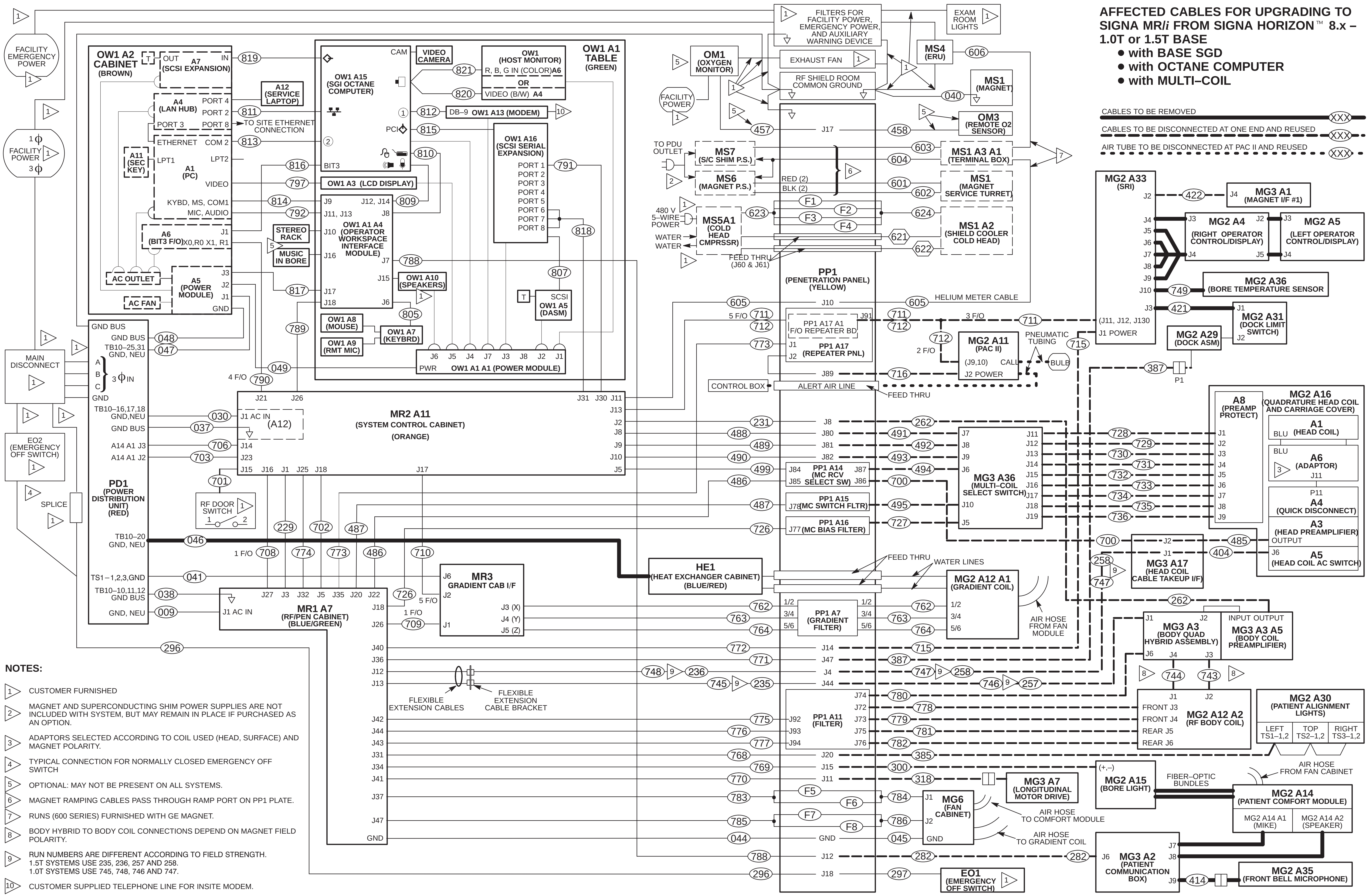
AFFECTED CABLES FOR UPGRADING TO SIGNA MR/*i* FROM SIGNA HORIZON™ 8.x – 1.0T or 1.5T BASE

- with BASE SGD
- with OCTANE COMPUTER
- with MULTI–COIL

CABLES TO BE REMOVED (XXX)

CABLES TO BE DISCONNECTED AT ONE END AND REUSED (XXX)

AIR TUBE TO BE DISCONNECTED AT PAC II AND REUSED (XXX)



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

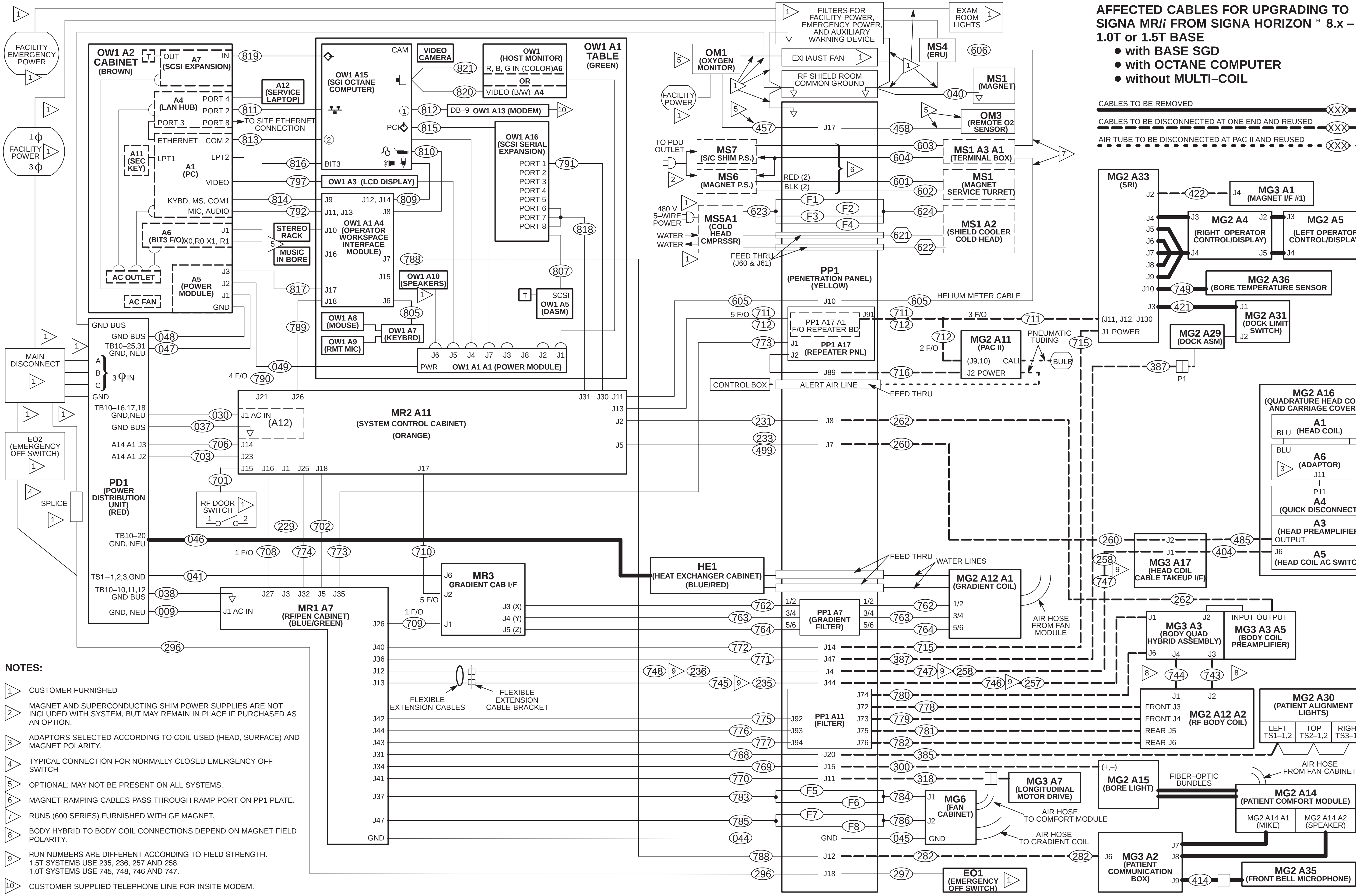
AFFECTED CABLES FOR UPGRADING TO SIGNA MR/*i* FROM SIGNA HORIZON™ 8.x – 1.0T or 1.5T BASE

- with BASE SGD
- with OCTANE COMPUTER
- without MULTI–COIL

CABLES TO BE REMOVED XXX

CABLES TO BE DISCONNECTED AT ONE END AND REUSED XXX

AIR TUBE TO BE DISCONNECTED AT PAC II AND REUSED XXX



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

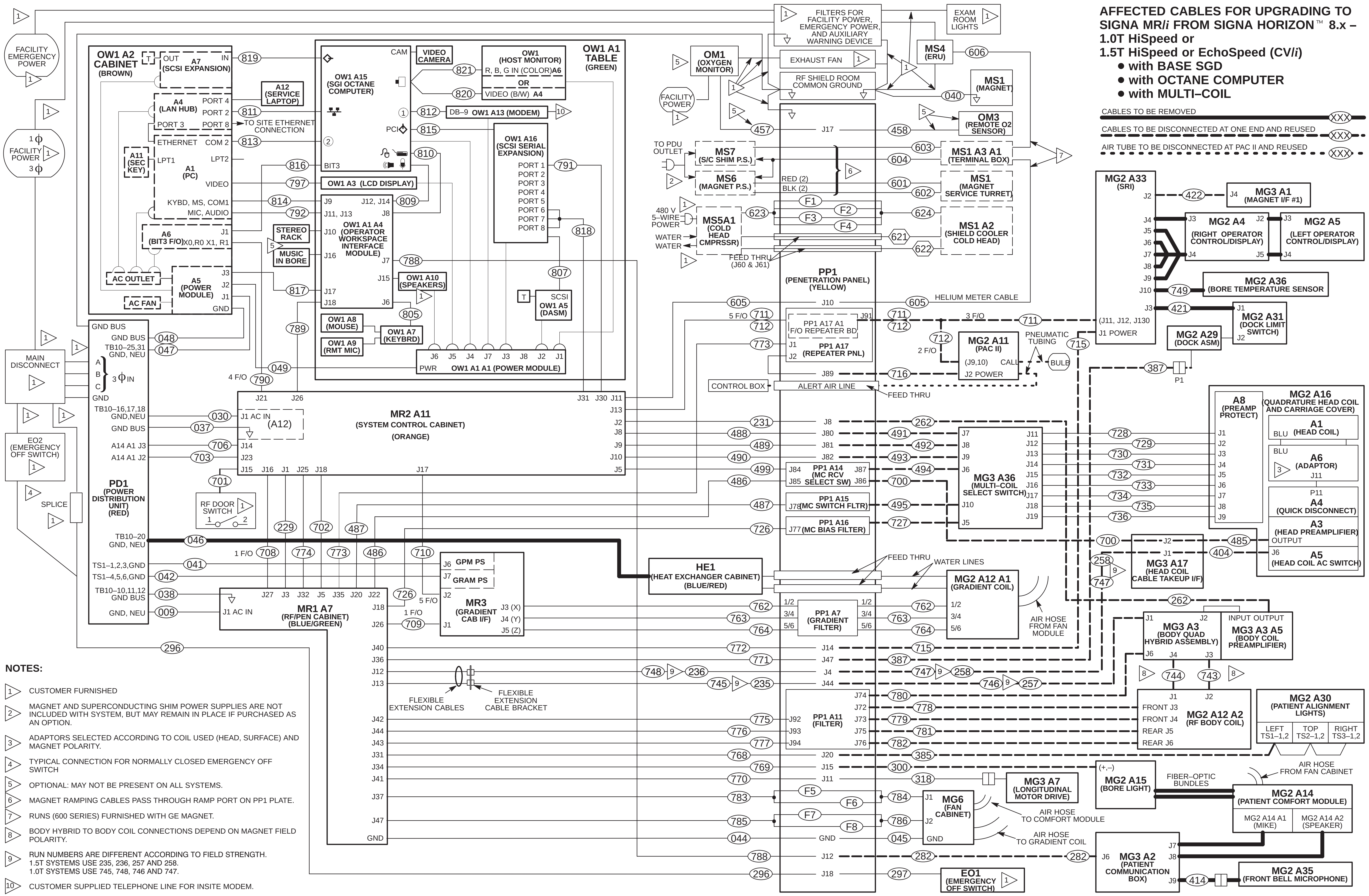
AFFECTED CABLES FOR UPGRADING TO SIGNA MR/*i* FROM SIGNA HORIZON™ 8.x – 1.0T HiSpeed or 1.5T HiSpeed or EchoSpeed (CV/*i*)

- with BASE SGD
- with OCTANE COMPUTER
- with MULTI–COIL

CABLES TO BE REMOVED (XXX)

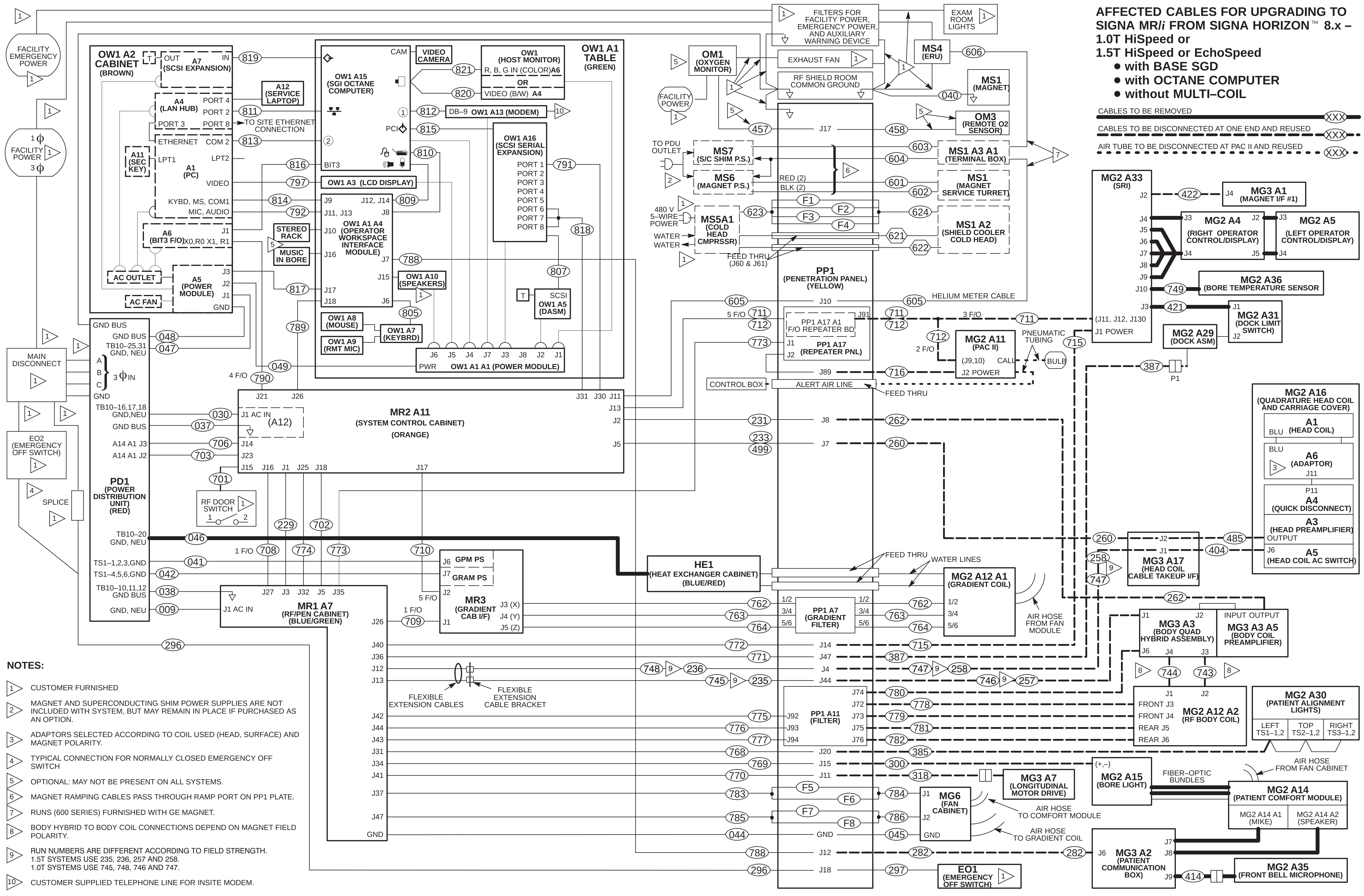
CABLES TO BE DISCONNECTED AT ONE END AND REUSED (XXX)

AIR TUBE TO BE DISCONNECTED AT PAC II AND REUSED (XXX)



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.



AFFECTED CABLES FOR UPGRADING TO SIGNA MR/*i* FROM SIGNA HORIZON™ 8.x – 1.0T HiSpeed or 1.5T HiSpeed or EchoSpeed

- **with BASE SGD**
- **with OCTANE COMPUTER**
- **without MULTI–COIL**

CABLES TO BE REMOVED XXX

CABLES TO BE DISCONNECTED AT ONE END AND REUSED XXX

AIR TUBE TO BE DISCONNECTED AT PAC II AND REUSED XXX

NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

4–3 NEW CABLES TO INSTALL

The following are new cables to be installed for this upgrade. Refer to the applicable Signa MR/*i* (CV/*i*) Interconnect Map as to routing of cables.

TABLE 4–4
NEW CABLES TO BE INSTALLED

RUN	"FROM"	"TO"	DESCRIPTION	REMARKS
065	PD1–26,27,GND	WC1	Water Chiller Power Cord	Power Cable
414	MG2–A35	MG3–A2–J9	Front End Bell Microphone Cable	Part of Front End Bell
421	MG2–A33–J3	MG2–A31–J1	Dock Limit Cable	10 ft 15 pin Sub–D
749	MG2–A33–J10	MG2–A36	Bore Temperature Sensor Cable	Part of Front End Bell
	MG2–A11–J4	MG2–A4–J5	PAC to Operator Display Cable	2221945
	MG2–A5–J1	MG2–A4–J1	Right Control to Oper Display Cable	2221946 Part of Frnt End Bell
	MG2–A5–J2	MG2–A4–J2	Left Control to Oper Display Cable	2221946 Part of Frnt End Bell
	MG2–A33–5,6,7,8,9	MG2–A4–J4	SRI to Operator Display Cable	2221947
	MG2–A33–J4	MG2–A4–J3	SRI to Operator Display Cable	2221945
	MG3–A2–J8	MG2–A14–A2	Speaker Cable	Part of Rear End Bell
	MG2 A15	MG2–A14	Fiber Optic Bundles	Part of Bore Lamp Assembly

4–4 NEW CABLES NOT TO BE INSTALLED

Two new cables are provided with the upgrade that do not need to be installed if existing cables were not removed:

- Run 300 is part of the new Bore Lamp Assembly. If existing Run 300 is present and can be connected to the terminal strip on to the new Bore Lamp Assembly, remove new Run 300 and return for recycling.
- Run 385 is part of the new Front End Bell Assembly. If existing Run 385 is present and can be connected to JP1 of the Voltage Regulator Board on the back of the Front End Bell, remove new Run 385 and return for recycling.

Also:

- If present, the microphone cable on the Rear End Bell should not be connected to J7 of the Patient Communications Box located on the Rear Pedestal.

4–5 EXISTING CABLES TO CONNECT FOR UPGRADE TO NEW WIDE-OPEN ENCLOSURE

The Signa MR/*i* (CV/*i*) Interconnect Maps indicate the cables to be connected when installing the new enclosure and water chiller. The upgrade instructions in Tab 2, Magnet Enclosures, provide information on connecting existing cables.

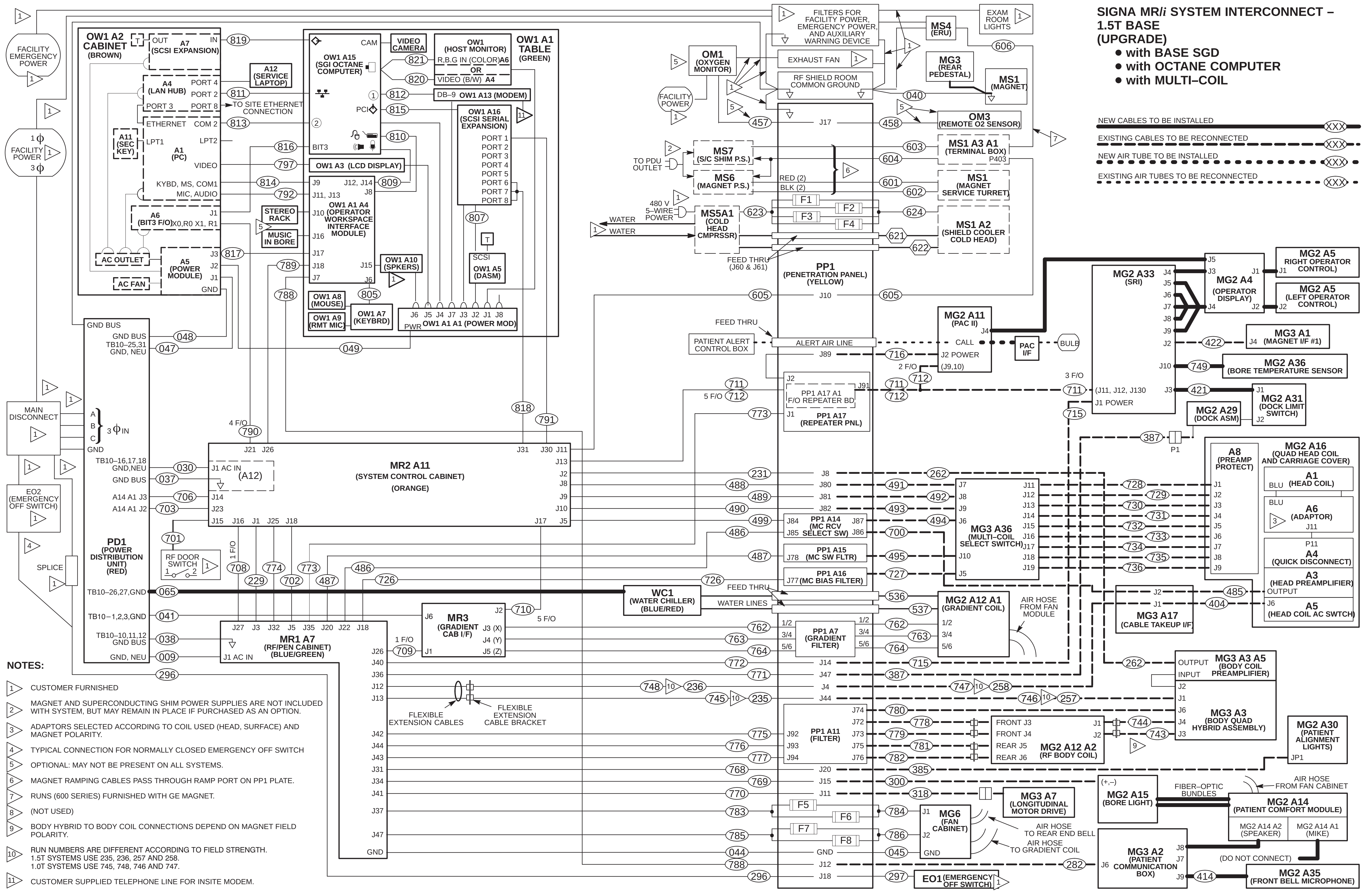
4–6 CABLE INTERCONNECTION DOCUMENTATION

Fold–out MR/*i* (CR/*i*) System Interconnect Cable Maps are provided at the end of this section. They are:

Select the map that applies to the Signa system being installed. Tape map in a convenient location to check off routed cable. Upon completion of connecting cables, return fold–out map to the binder for future reference.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.5T BASE (UPGRADE)

- with BASE SGD
- with OCTANE COMPUTER
- with MULTI–COIL



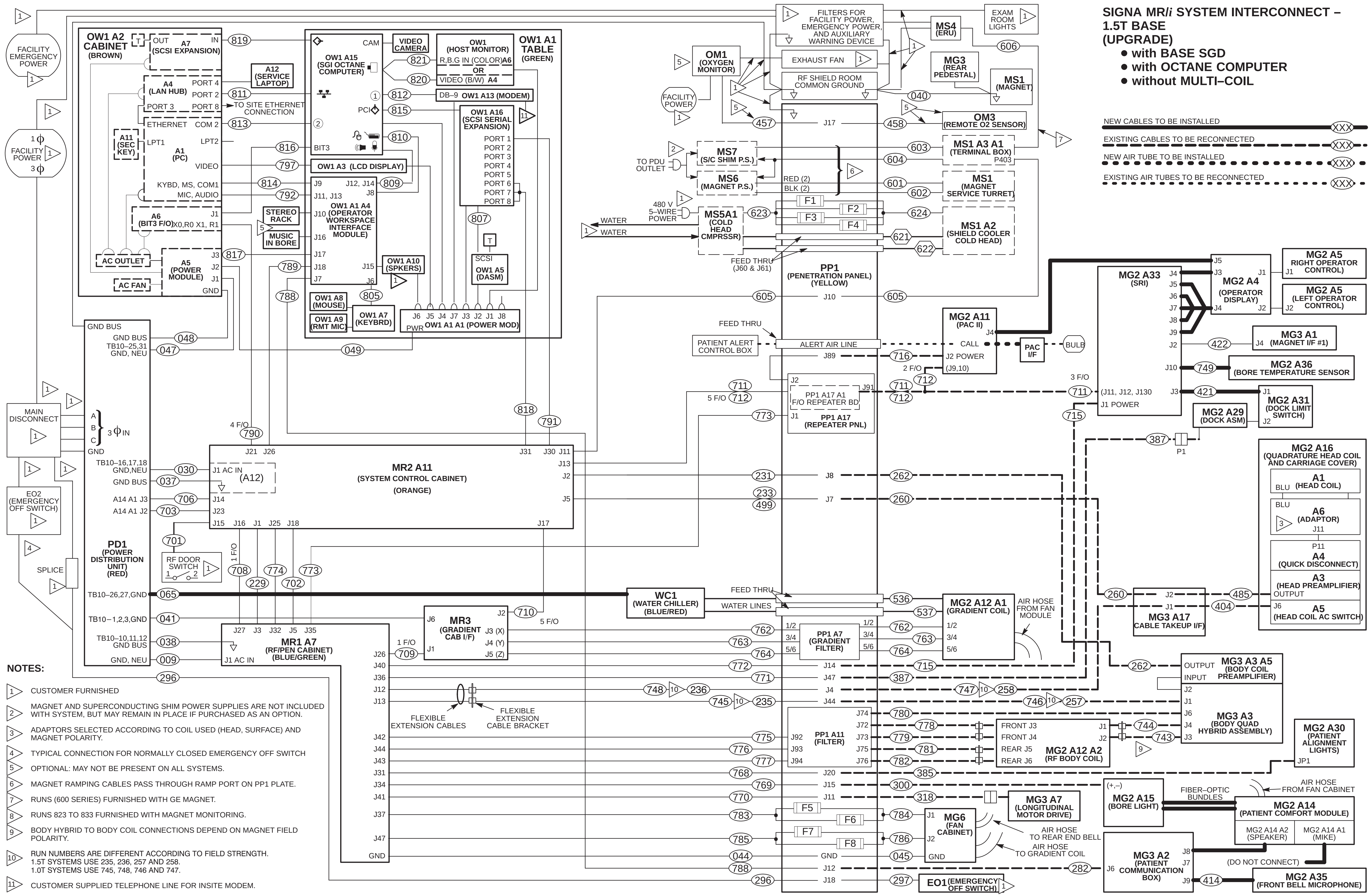
NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. (NOT USED)
9. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
10. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.5T BASE (UPGRADE)

- with BASE SGD
- with OCTANE COMPUTER
- without MULTI–COIL

NEW CABLES TO BE INSTALLED
EXISTING CABLES TO BE RECONNECTED
NEW AIR TUBE TO BE INSTALLED
EXISTING AIR TUBES TO BE RECONNECTED



NOTES:

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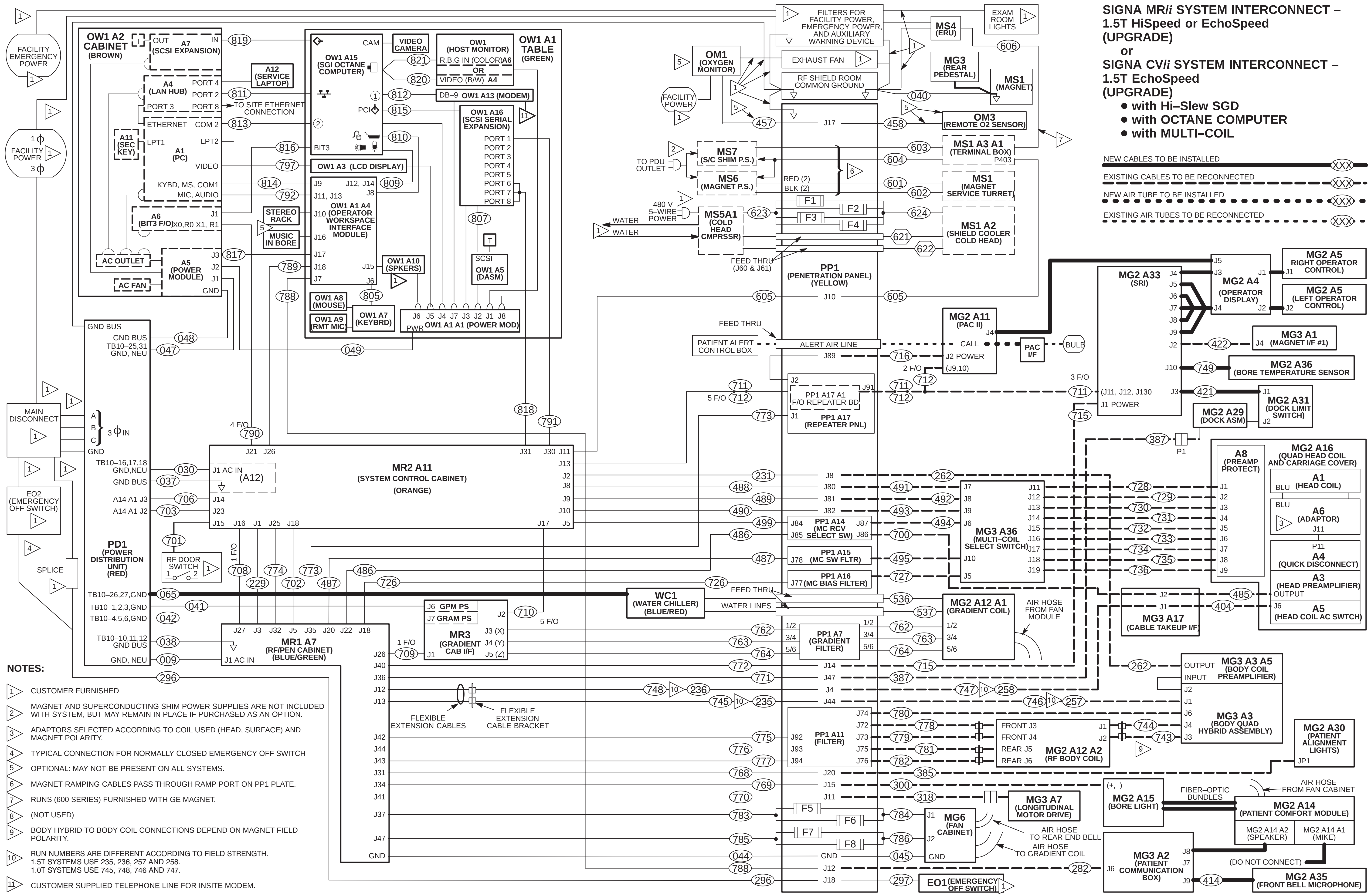
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.5T HiSpeed or EchoSpeed (UPGRADE)
or
SIGNA CV/*i* SYSTEM INTERCONNECT – 1.5T EchoSpeed (UPGRADE)

- with Hi–Slew SGD
- with OCTANE COMPUTER
- with MULTI–COIL

NEW CABLES TO BE INSTALLED (XXX)
EXISTING CABLES TO BE RECONNECTED (XXX)
NEW AIR TUBE TO BE INSTALLED (XXX)
EXISTING AIR TUBES TO BE RECONNECTED (XXX)



NOTES:

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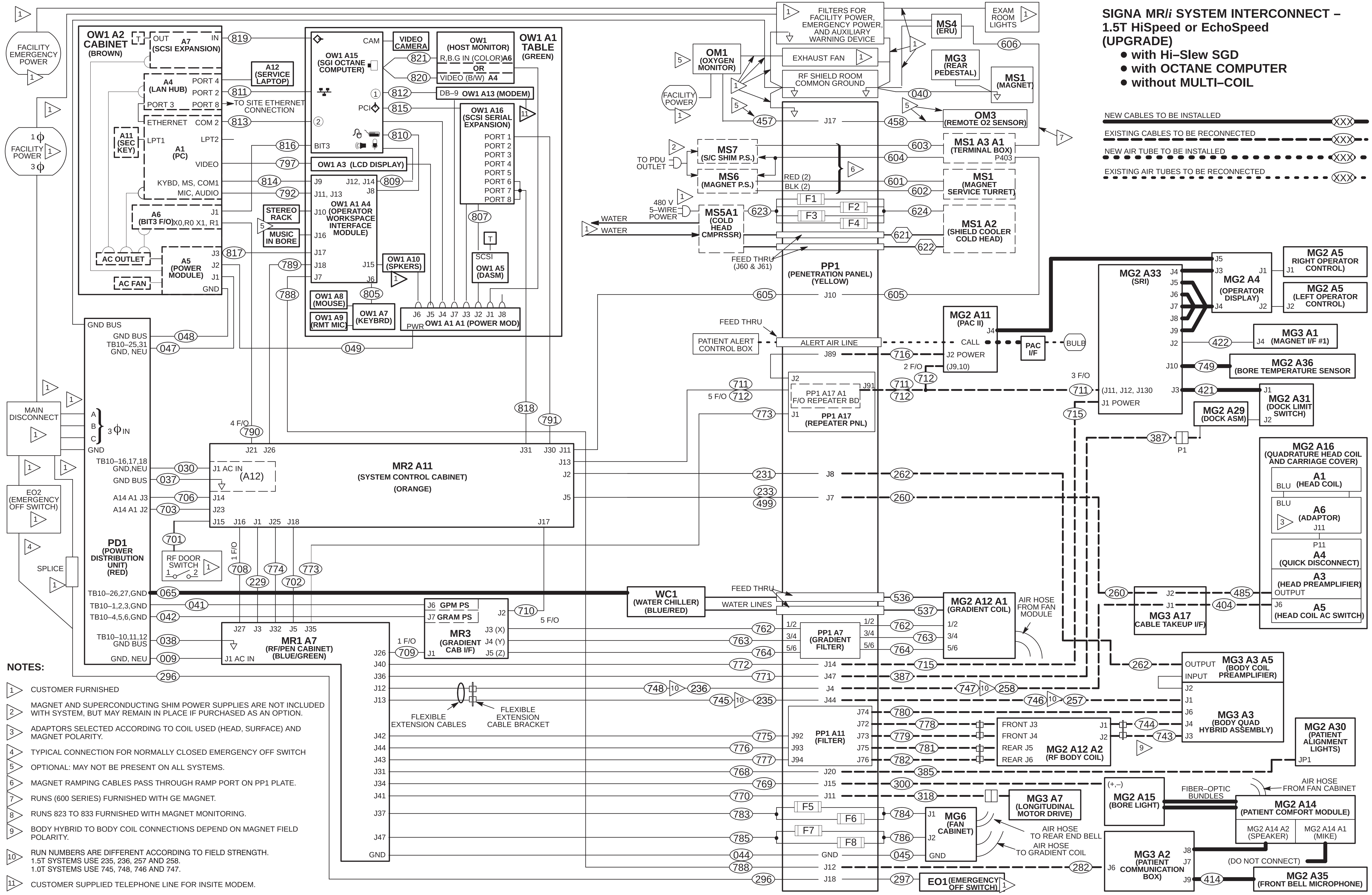
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.5T HiSpeed or EchoSpeed (UPGRADE)

- with Hi–Slew SGD
- with OCTANE COMPUTER
- without MULTI–COIL

NEW CABLES TO BE INSTALLED XXX
EXISTING CABLES TO BE RECONNECTED XXX
NEW AIR TUBE TO BE INSTALLED XXX
EXISTING AIR TUBES TO BE RECONNECTED XXX



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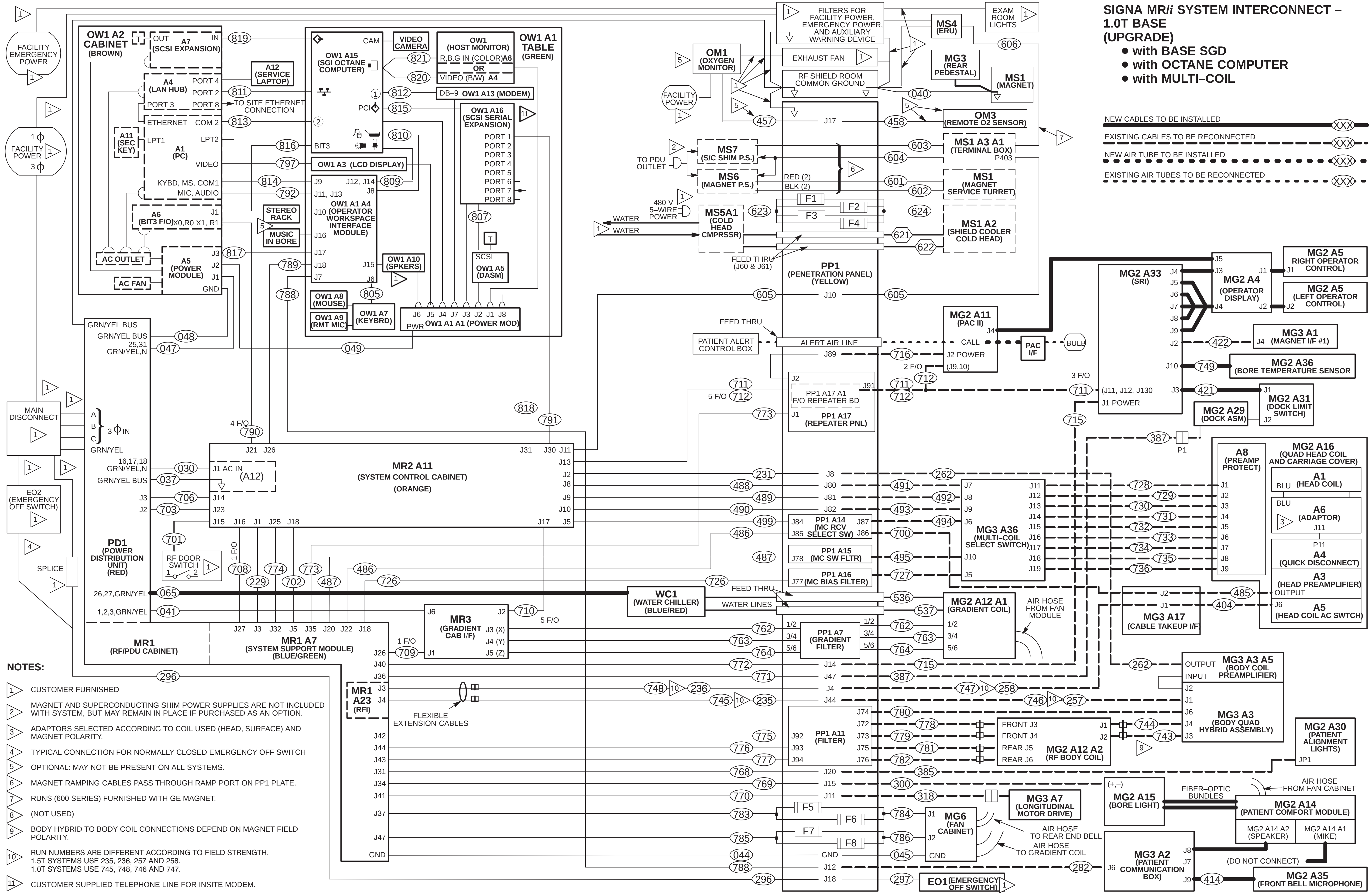
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.0T BASE (UPGRADE)

- with BASE SGD
- with OCTANE COMPUTER
- with MULTI–COIL

NEW CABLES TO BE INSTALLED
EXISTING CABLES TO BE RECONNECTED
NEW AIR TUBE TO BE INSTALLED
EXISTING AIR TUBES TO BE RECONNECTED



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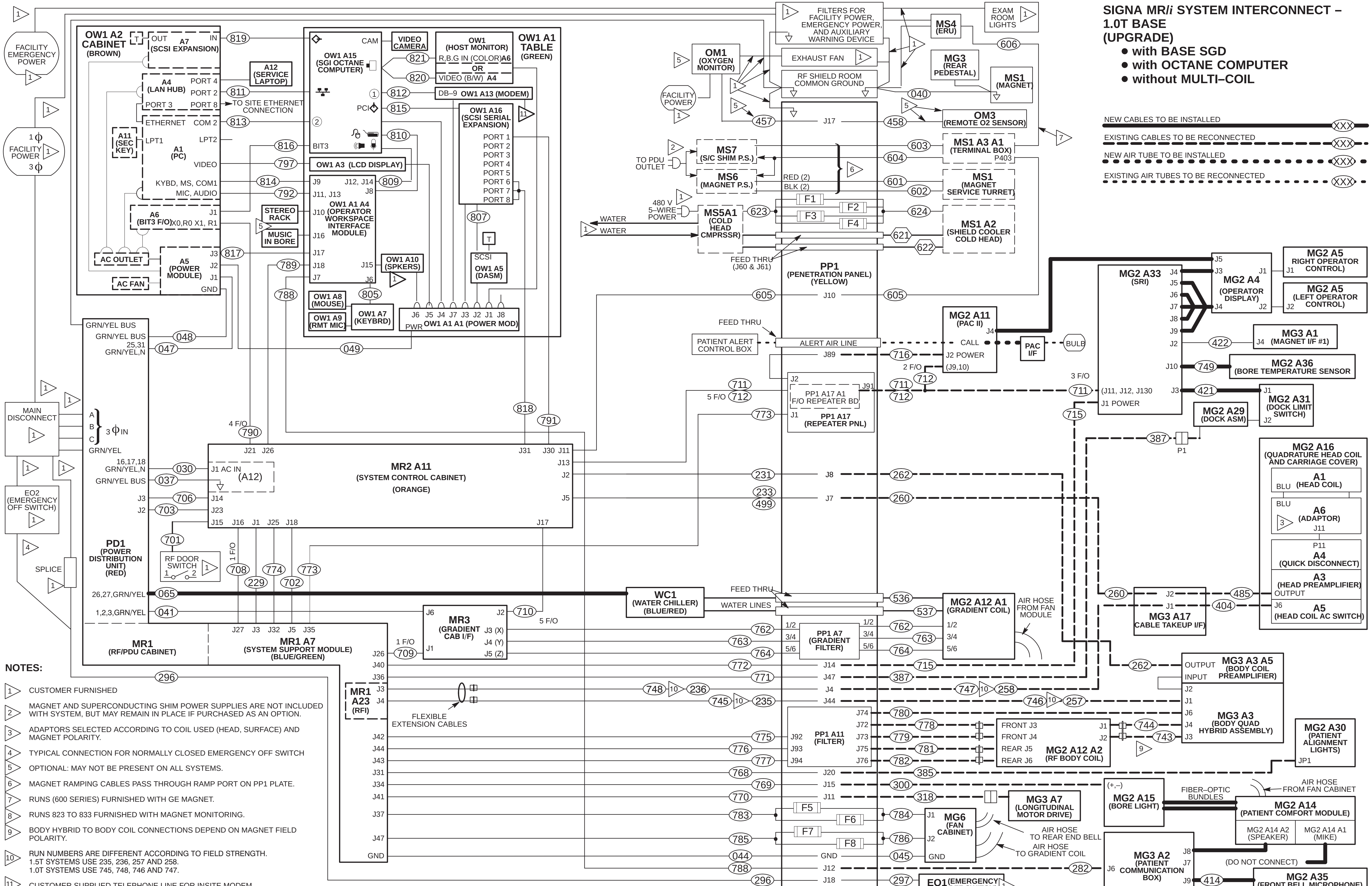
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.0T BASE (UPGRADE)

- with BASE SGD
- with OCTANE COMPUTER
- without MULTI–COIL

NEW CABLES TO BE INSTALLED
EXISTING CABLES TO BE RECONNECTED
NEW AIR TUBE TO BE INSTALLED
EXISTING AIR TUBES TO BE RECONNECTED



NOTES:

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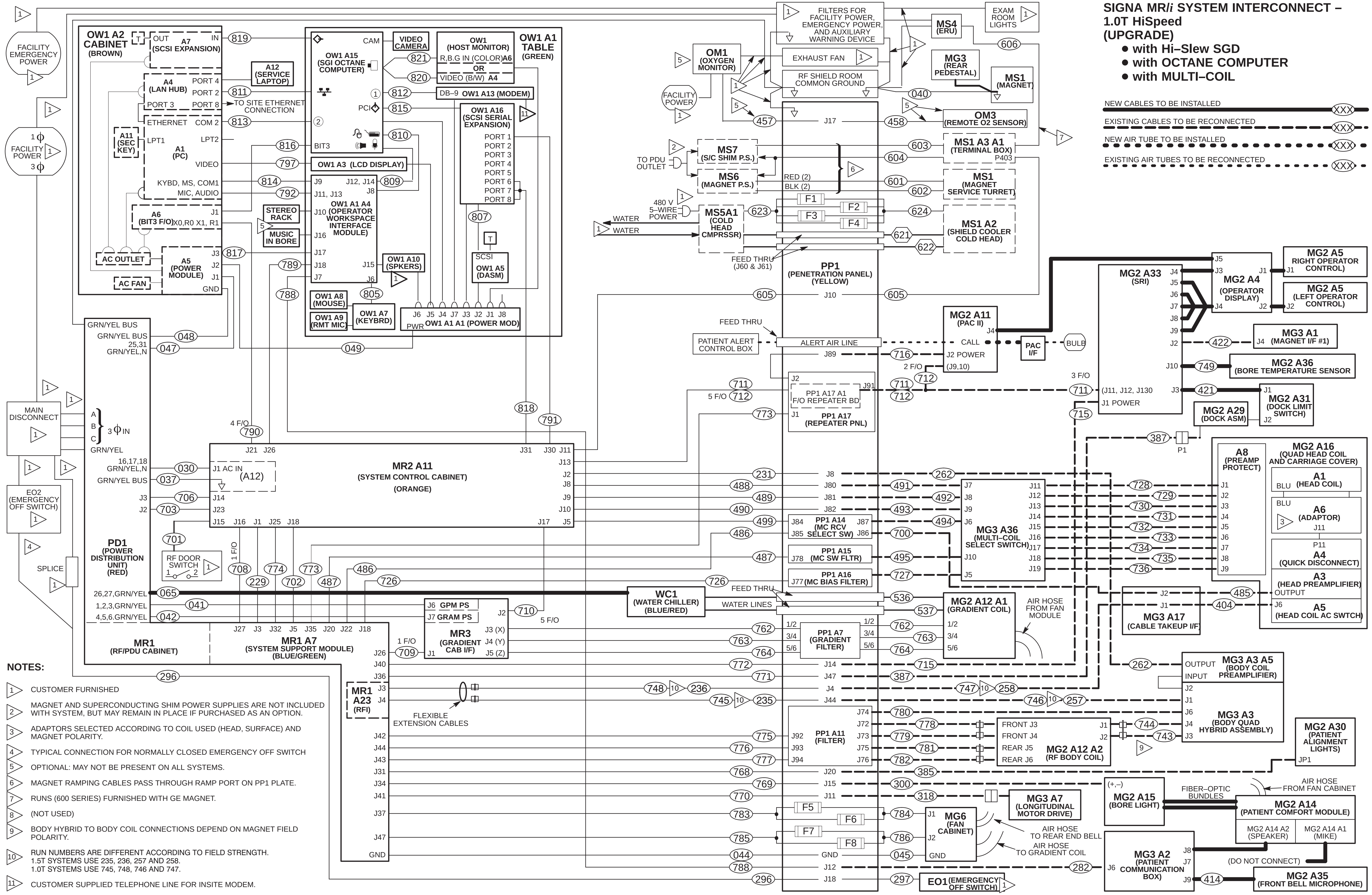
11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.0T HiSpeed (UPGRADE)

- with Hi–Slew SGD
- with OCTANE COMPUTER
- with MULTI–COIL

NEW CABLES TO BE INSTALLED
EXISTING CABLES TO BE RECONNECTED
NEW AIR TUBE TO BE INSTALLED
EXISTING AIR TUBES TO BE RECONNECTED



NOTES:

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11. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

SIGNA MR/*i* SYSTEM INTERCONNECT – 1.0T HiSpeed (UPGRADE)

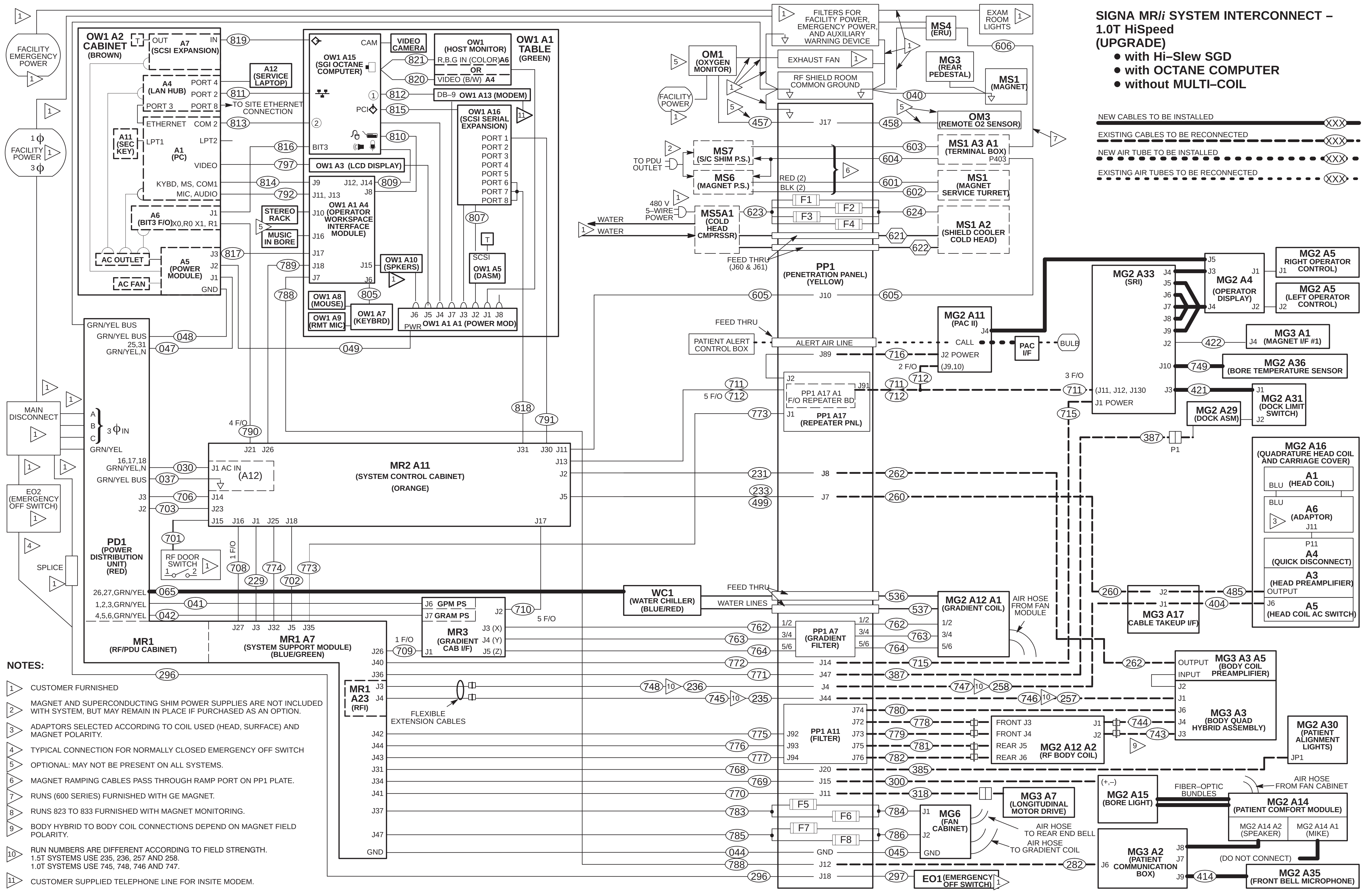
- with Hi–Slew SGD
- with OCTANE COMPUTER
- without MULTI–COIL

NEW CABLES TO BE INSTALLED XXX

EXISTING CABLES TO BE RECONNECTED XXX

NEW AIR TUBE TO BE INSTALLED XXX

EXISTING AIR TUBES TO BE RECONNECTED XXX



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SECTION 5 – COMPLETION

5–1 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Press the FULL ON button on the PDU front control panel.
4. Except for the Gradient Cabinet 8645 power modules, switch all cabinet circuit breakers to ON. The 8645 power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.

5–2 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the “Return pile”, Recycling Operation will route them to appropriate location.

5–3 MATERIAL DISPOSITION GUIDANCE

The “discard” or “return pile” consists of items removed by the Signa MR/*i* Upgrade or traded in for new options and have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the return pile is per GE Medical Systems Policy and Procedures.

Note

The return pile includes items in Sections NO TAG through 5–9, RETURN PILE CHECKLIST.

1. For coordination of return pile, contact GEMS Recycling Center: Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414–747–6855); WizMail “RECYCLING”; Field ASPEN 1–800–525–1516, Box #26041.

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
- Site name, address, and phone number
- Date of completed Horizon upgrade.
- Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.

2. Prepare the return pile for shipment as follows:

   a. Package other removed modules and boards in a cardboard box and secure inside System Cabinet (or any other cabinet where they fit).

   b. Use ramp to move cabinets on to shipping pallets. Three installers are required for this heavy lifting task. Fasten with shipping brackets that were removed previously from new cabinets.

3. Process product locater cards for all serialized components removed from the system according to policy.

4. Regulations governing the disposal of this material in the recycle pile vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the “recycle pile” per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

5–4 RETURN PILE CHECKLIST, SIGNA HORIZON 8.x LCC TO SIGNA WIDE-OPEN UPGRADE

The following items are designated to be returned after the upgrade. All materials should be able to be packaged for return on the two provided cover dollies. Refer to Section 5–3 , MATERIAL DISPOSITION GUIDANCE.

☐ 1. Magnet Enclosure Front Cover
☐ 2. Magnet Enclosure Back Cover
☐ 3. Magnet Enclosure Side Covers (4)
☐ 4. Front End Bell
☐ 5. Rear End Bell
☐ 6. Left and Right Side Slide Support assemblies
☐ 7. Service Platform
☐ 8. Rear Pedestal Frame
☐ 9. Rear Pedestal Covers
☐ 10. RF Coil in shipping container
☐ 11. Bore Lamp Asm with Fiber Optic bundle
☐ 12. Fiberglass Bridge
☐ 13. Obsolete Cables
☐ 14. Heat Exchanger
☐ 15. Coolant removed from Heat Exchanger and Body Coil

5–5 FINAL COORDINATION

1. Record and enter applicable data into applicable site configuration files and records.
2. Process product locater installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

Product Locator File
P.O. Box 414, W–523
Milwaukee, WI 53201–0414

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

3. Process the **Supplier Performance Report** located in Common Forms. A reference copy of the form is provided at the end of this section.
4. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals. Since considerable hardware remains from initial installation, (electronics cabinets, operator workspace, etc.) retain *Direction 2138247 Signa Renewal Parts Release 5.x & 8.x*
5. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
6. Verify that coordination of application support for instruction of site users on Software Release 8.x has been made before returning system to the users. Turn site over to applications who will instruct users.

GE Medical Systems Field Service

SUPPLIER PERFORMANCE REPORT

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: ______________________________ Date: ______________

Work Performed / Evaluated: ______________________________

Commodity / Item Purchased: ______________________________

GE Customer Name: ______________________________

FDO Number: ______________ Test Equipment Bar−code No; ______________

GEMS Evaluator Name: ______________________________

Please rate suppliers as 1 to 5 on each of the 15 questions below:
1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

	Supplier schedules well and:					
1	starts on time?	1	2	3	4	5
2	finishes on time?	1	2	3	4	5
3	avoids disruptions?	1	2	3	4	5
	Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4	5
5	cooperates with changes in schedule?	1	2	3	4	5
6	deals with unexpected problems?	1	2	3	4	5
	Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4	5
8	adequate for contracted work?	1	2	3	4	5
9	calibrated and in good working order?	1	2	3	4	5
	Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4	5
11	always keep job sites clean and free of debris?	1	2	3	4	5
12	ensure both function and appearance are as expected?	1	2	3	4	5
	Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4	5
14	behavior?	1	2	3	4	5
15	attitude?	1	2	3	4	5

Examples / Details: ______________________________

____

____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

5. Senior Operations Specialist for your LCT
6. The contractor who performed the work
7. ISS (already in default distribution)
8. Sourcing Supplier Quality (already in default distribution)

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

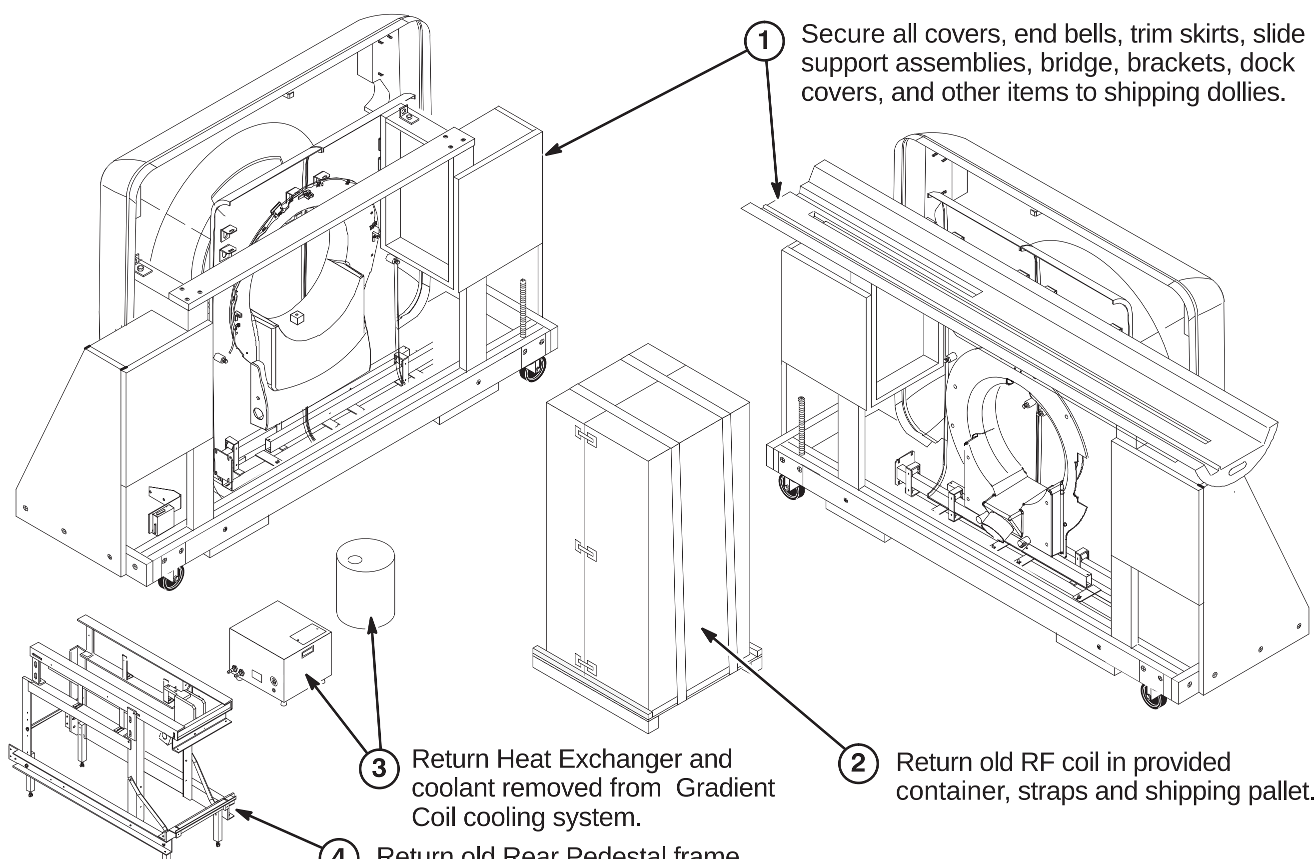
IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM MAGNET AS POSSIBLE.

INSTALLATION PROCEDURES SUMMARY

- ❑ **A1 – Items to be Returned for Recycling at Completion of Upgrade**
- ❑ **A2 – Enclosure Covers and Dock Removal**
- ❑ **A3 – Disconnect Rear Pedestal Cables**

❑ A1 – ITEMS TO BE RETURNED FOR RECYCLING AT COMPLETION OF UPGRADE

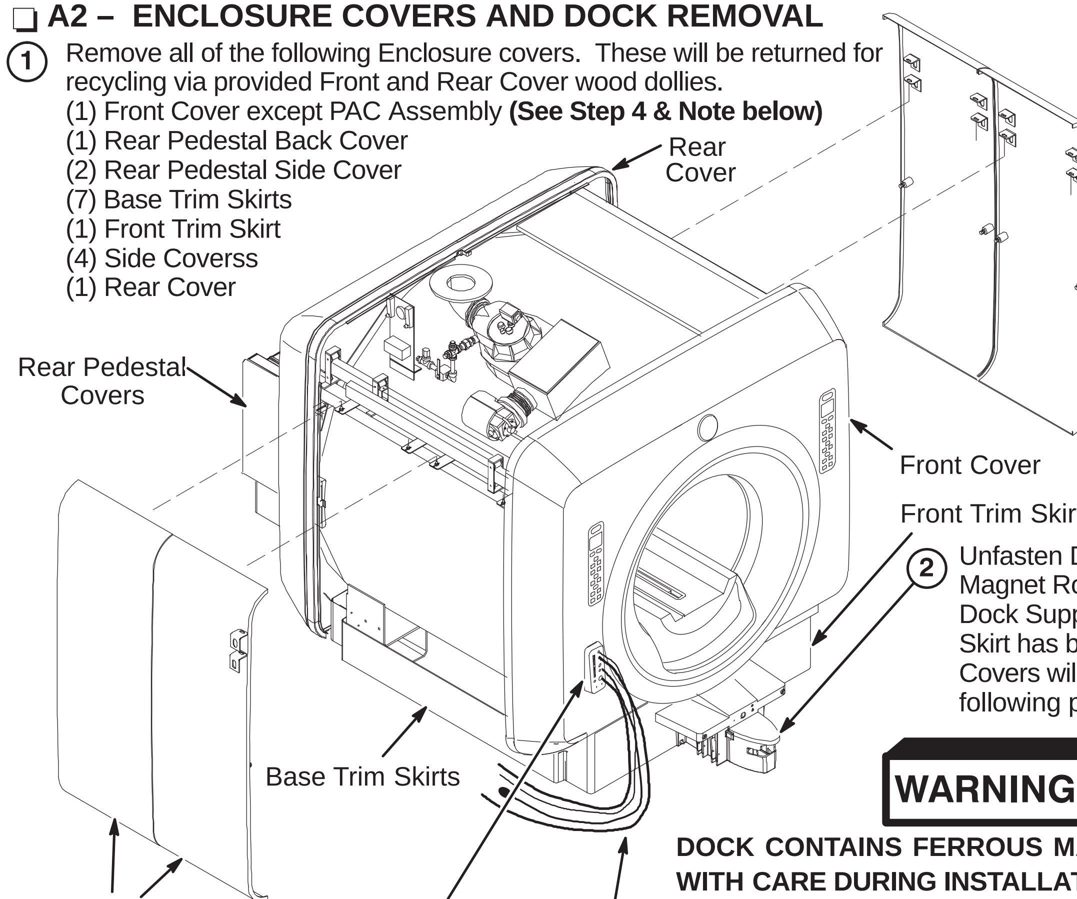
Note: All items de–installed during this upgrade are to be returned for recycling. Secure as many of the items as possible to the two shipping dollies provided for this purpose. Shown below is an example of how the dollies might look when loaded. (not all items are shown)



1. Secure all covers, end bells, trim skirts, slide support assemblies, bridge, brackets, dock covers, and other items to shipping dollies.
2. Return old RF coil in provided container, straps and shipping pallet.
3. Return Heat Exchanger and coolant removed from Gradient Coil cooling system.
4. Return old Rear Pedestal frame.

❑ A2 – ENCLOSURE COVERS AND DOCK REMOVAL

1. Remove all of the following Enclosure covers. These will be returned for recycling via provided Front and Rear Cover wood dollies.
   (1) Front Cover except PAC Assembly **(See Step 4 & Note below)**
   (1) Rear Pedestal Back Cover
   (2) Rear Pedestal Side Cover
   (7) Base Trim Skirts
   (1) Front Trim Skirt
   (4) Side Coverss
   (1) Rear Cover



2. Unfasten Dock and remove from Magnet Room. Also remove Dock Support after Front Trim Skirt has been removed. Dock Covers will be replaced in a following procedure.

WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY

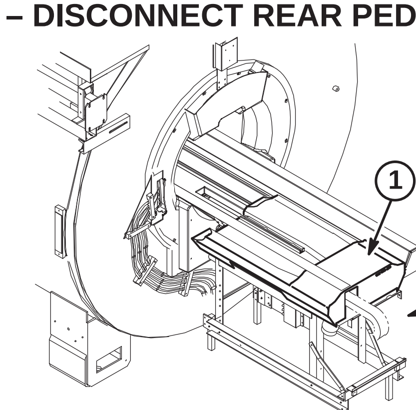
4. Unfasten and save PAC Assembly. It will be upgraded and transferred to new SRI/PAC Platform.
3. Disconnect and save ECG & Peripheral Gating cables, Respiration Bellows, and Patient Alert Pneumatic Air Tube and Bulb. These will be reused in new Wide–Open enclosure.

Note: All cables attached to Front Cover to be returned except for Run 385. Disconnect and save. It will be reused in new Wide–Open enclosure.

WARNING!

DO NOT LIFT THE DOCK MORE THAN 2 INCHES (5 CM) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installling the dock as there is significant risk of the magnet attracting the dock if it needs to be raised more that 2 inches (5 cm).

❑ A3 – DISCONNECT REAR PEDESTAL CABLES

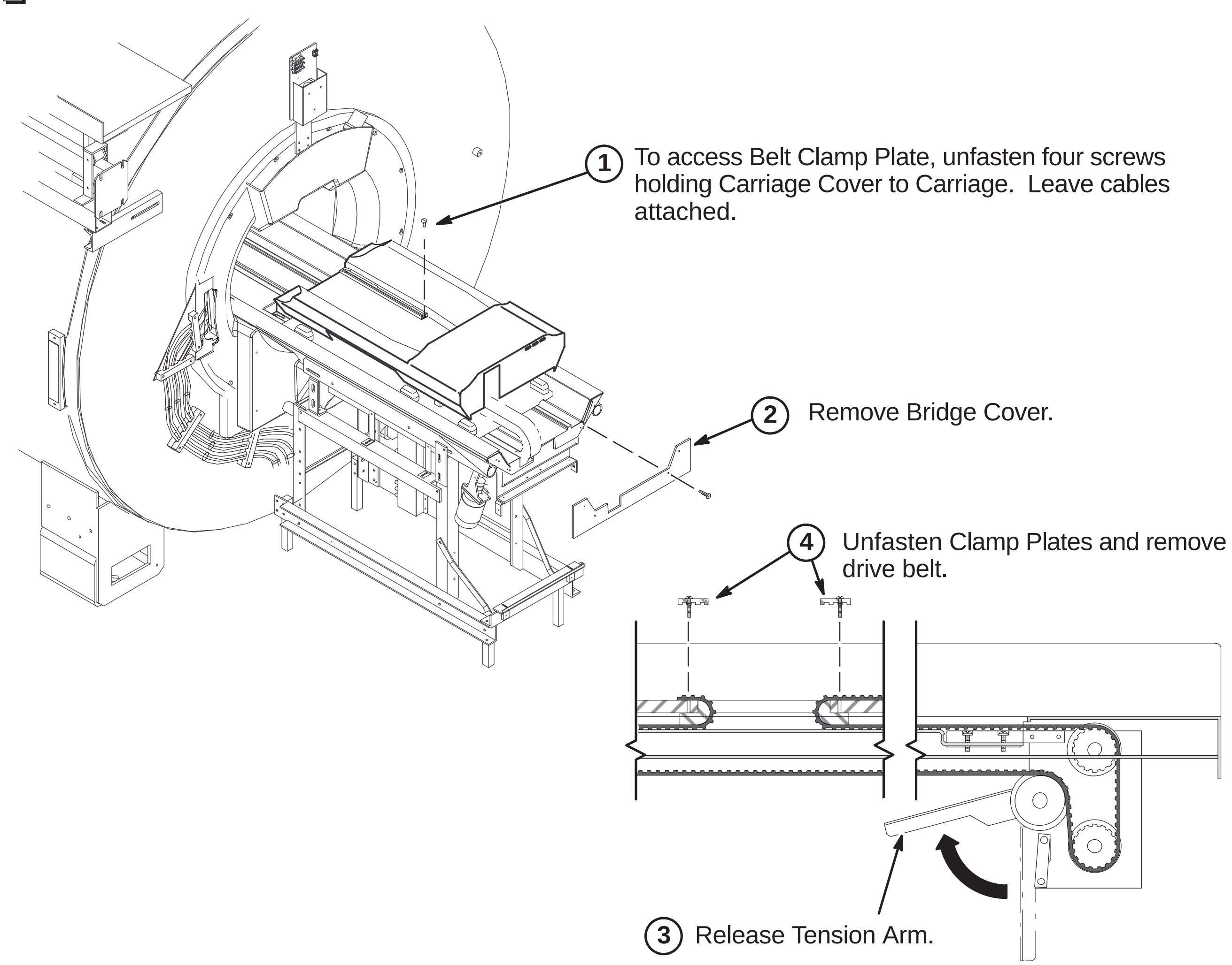


1. Move Carriage to limit of travel at rear of Bridge.
2. Disconnect all cables connected to Rear Pedestal components.

INSTALLATION PROCEDURES SUMMARY

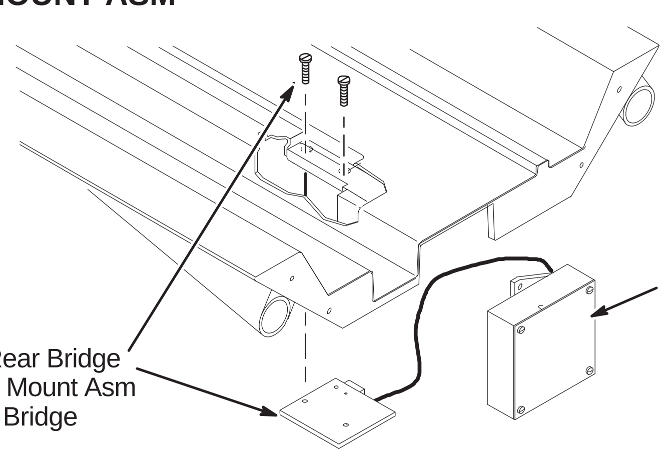
- ❑ **B1 – Remove Drive Belt**
- ❑ **B2 – Unfsten Bridge Mount Assembly**
- ❑ **B3 – Remove Bridge**
- ❑ **B4 – Removing Service Platform and Slide Support Assemblies**

❑ B1 – REMOVE DRIVE BELT



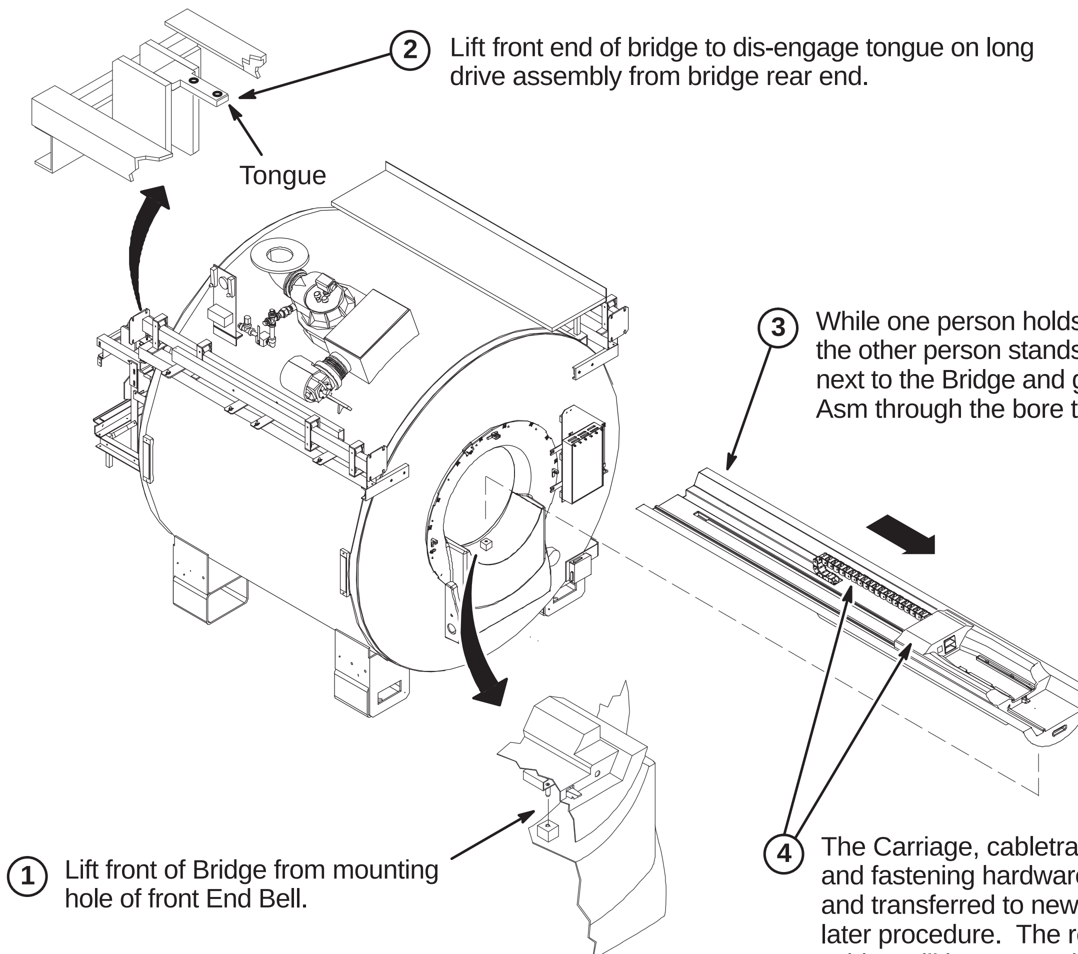
1. To access Belt Clamp Plate, unfasten four screws holding Carriage Cover to Carriage. Leave cables attached.
2. Remove Bridge Cover.
3. Release Tension Arm.
4. Unfasten Clamp Plates and remove drive belt.

❑ B2 – UNFASTEN BRIDGE MOUNT ASM



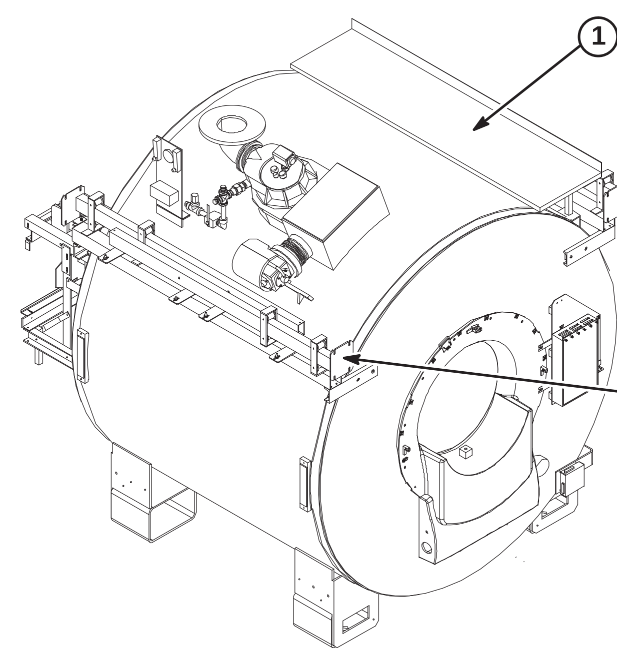
1. Unfasten Bridge, Tongue and Rear Bridge Mount Asm. Save Rear Bridge Mount Asm and fasteners for installing new Bridge

❑ B3 – REMOVE BRIDGE



1. Lift front of Bridge from mounting hole of front End Bell.
2. Lift front end of bridge to dis-engage tongue on long drive assembly from bridge rear end.
3. While one person holds holds the front end, the other person stands in back of the bore next to the Bridge and guides the Bridge Asm through the bore towards the front.
4. The Carriage, cabletrack with cables, belt, and fastening hardware are to be removed and transferred to new fiberglass Bridge in later procedure. The removed fiberglass Bridge will be returned for recycling.

❑ B4 – REMOVING SERVICE PLATFORM AND SLIDE SUPPORT ASSEMBLIES



1. Remove Service Platform and return for recylcing.
2. Remove left and right side Slide Support assemblies. Place on bottom of provided cover dollies for return for recycling.

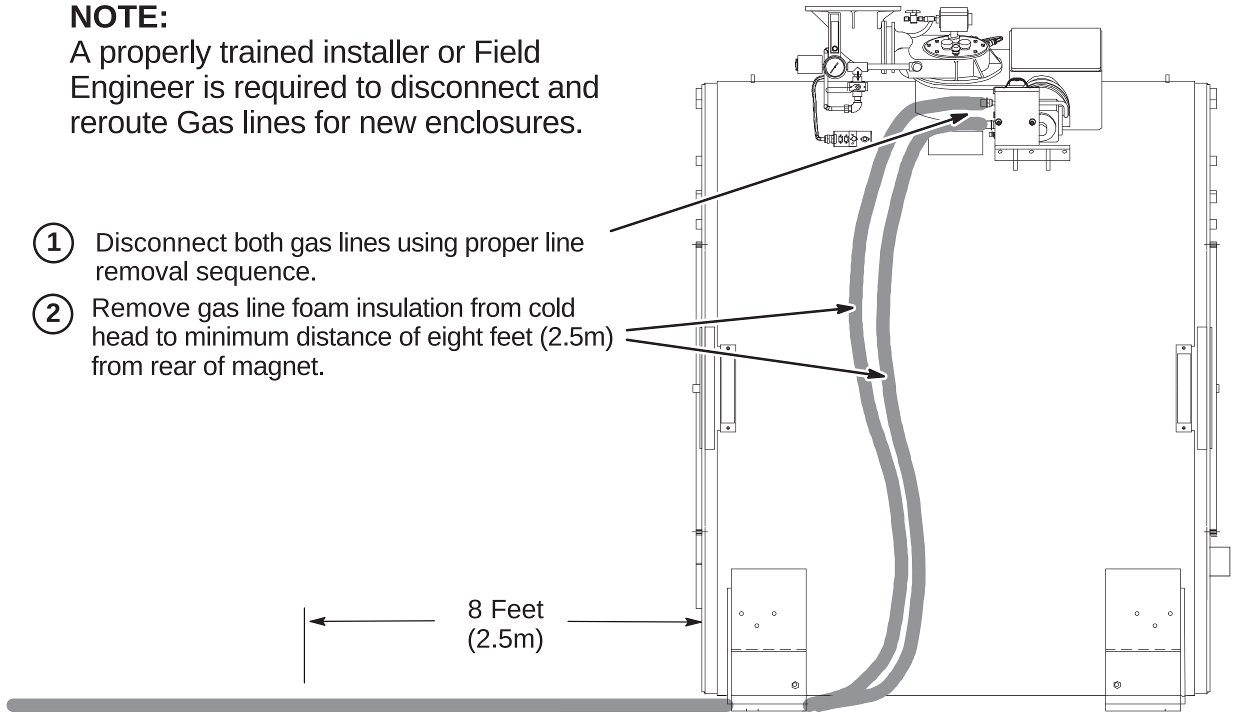
INSTALLATION PROCEDURES SUMMARY

- ❑ C1 – Remove Gas Line Foam Insulation and Disconnect Gas Lines
- ❑ C2 – Zipper Tubing Installation Preparation
- ❑ C3 – Install Zipper Tubing on Gas Lines
- ❑ C4 – Connect and Route Gas Lines

❑ C1 – REMOVE GAS LINE FOAM INSULATION AND DISCONNECT GAS LINES

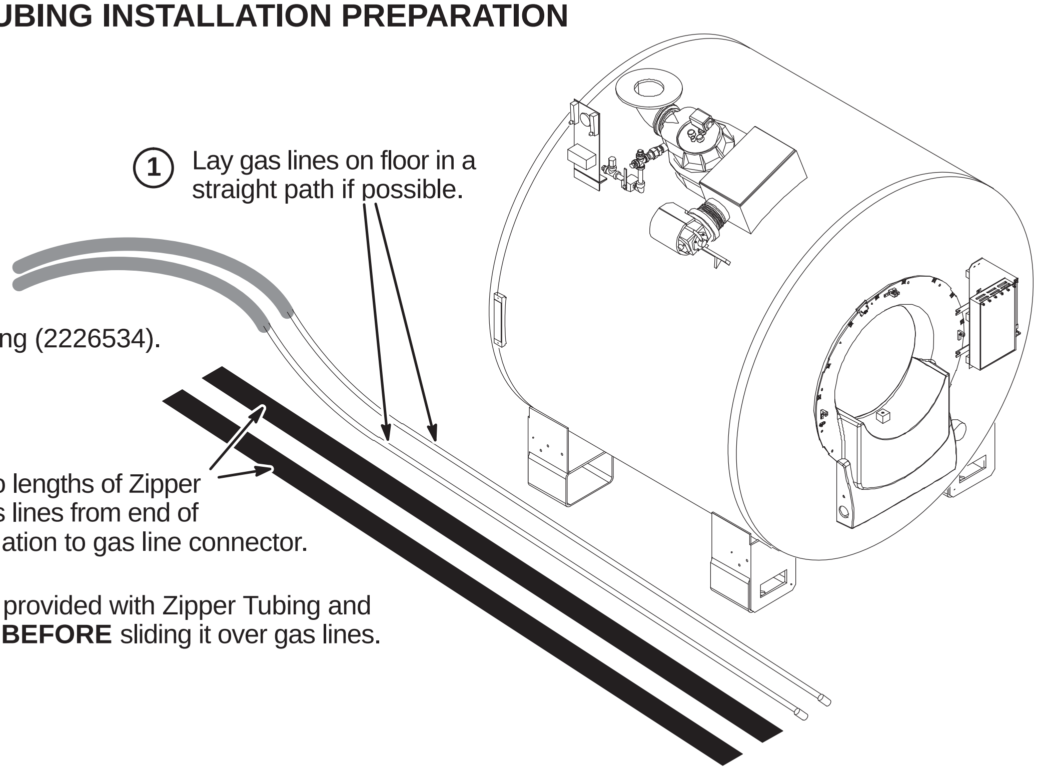
NOTE:
A properly trained installer or Field Engineer is required to disconnect and reroute Gas lines for new enclosures.

1. Disconnect both gas lines using proper line removal sequence.
2. Remove gas line foam insulation from cold head to minimum distance of eight feet (2.5m) from rear of magnet.

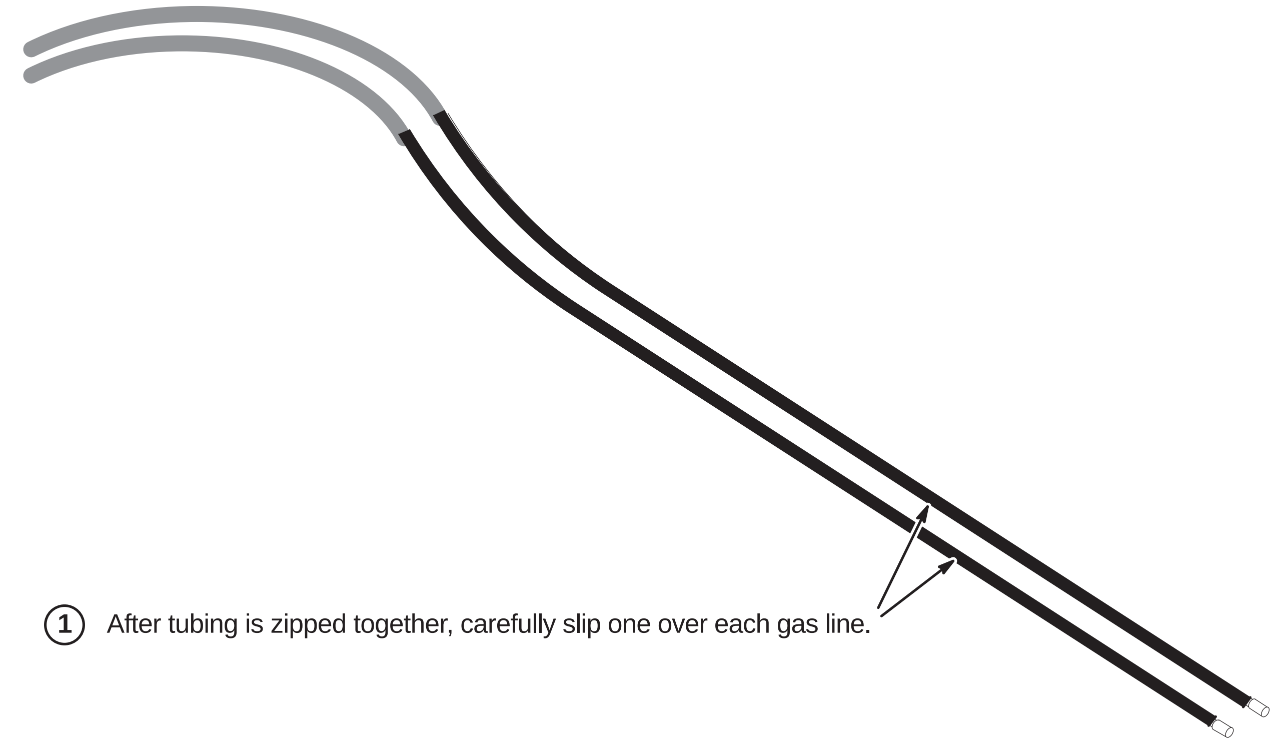


❑ C2 – ZIPPER TUBING INSTALLATION PREPARATION

1. Lay gas lines on floor in a straight path if possible.
2. Locate Zipper Tubing (2226534).
3. Roll out and cut two lengths of Zipper Tubing to cover gas lines from end of trimmed foam insulation to gas line connector.
4. Follow instructions provided with Zipper Tubing and zip tubing together **BEFORE** sliding it over gas lines.



❑ C3 – INSTALL ZIPPER TUBING ON GAS LINES

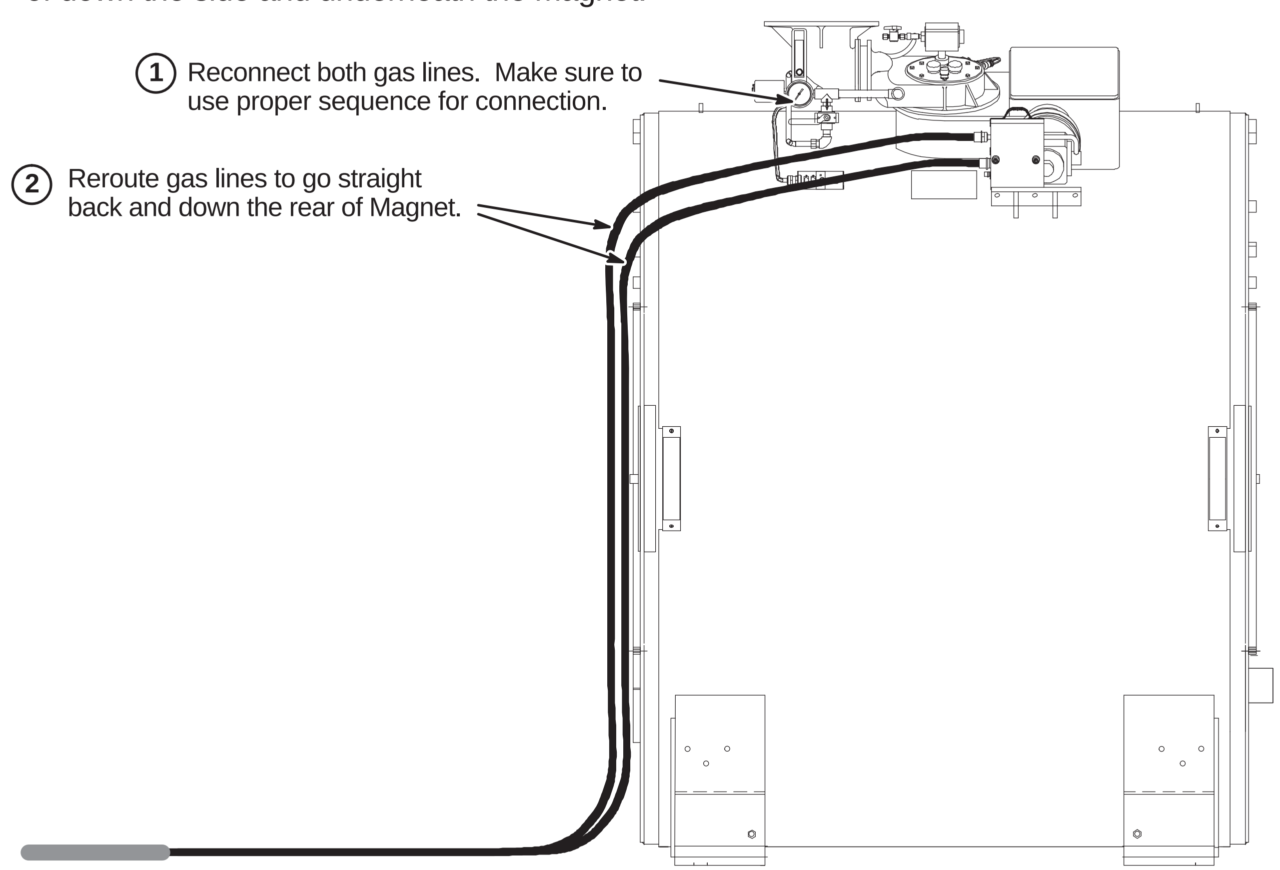


1. After tubing is zipped together, carefully slip one over each gas line.

❑ C4 – CONNECT AND ROUTE GAS LINES

Note: Gas lines will be relocated from former route of down the side and underneath the magnet.

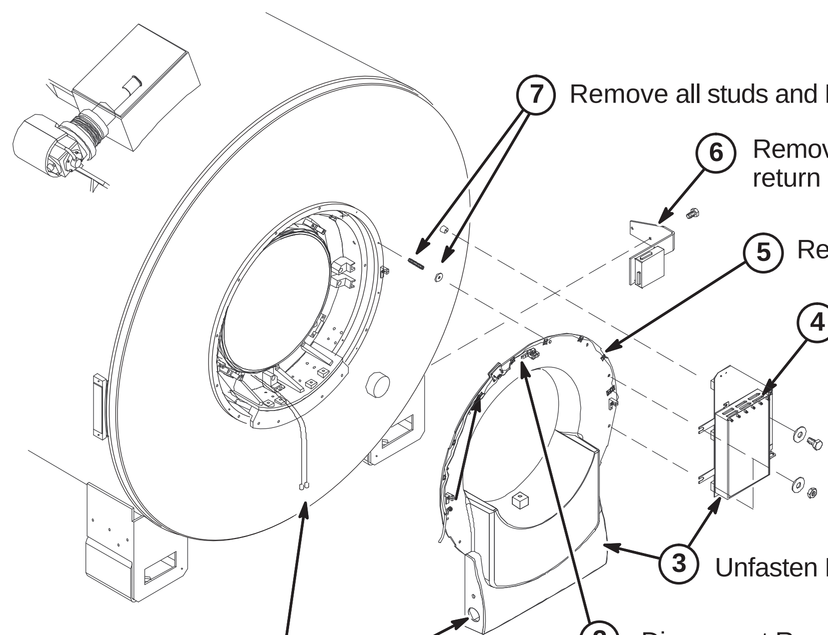
1. Reconnect both gas lines. Make sure to use proper sequence for connection.
2. Reroute gas lines to go straight back and down the rear of Magnet.



INSTALLATION PROCEDURES SUMMARY

- ❑ **D1 – Remove Front End Bell Asm, SRI, And Dock Control Box**
- ❑ **D2 – Disconnect Air Hoses & Remove Rear Pedestal**
- ❑ **D3 – Remove Bore Light Assembly and Run 749**
- ❑ **D4 – Remove Rear End Bell**
- ❑ **D5 – Remove RF Body Coil**

❑ D1 – REMOVE FRONT END BELL ASM, SRI, AND DOCK CONTROL BOX

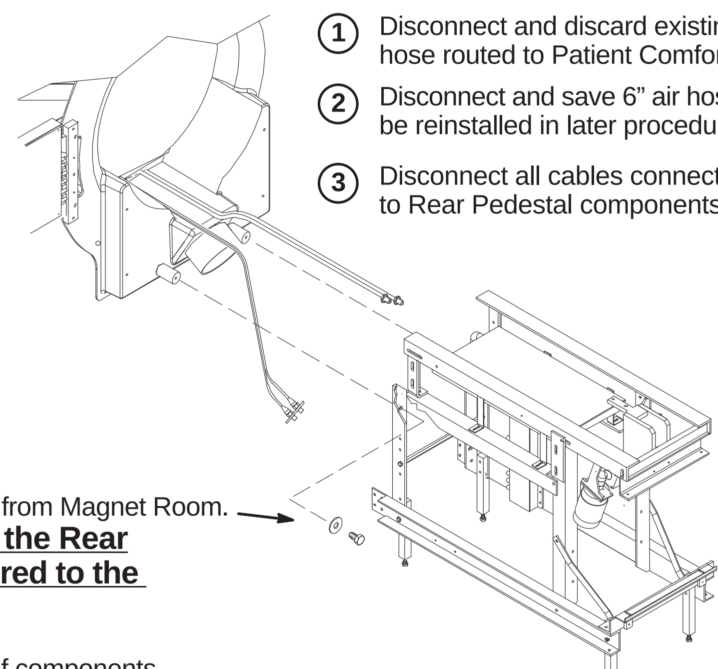


7. Remove all studs and Neoprene Rubber Washers.
6. Remove Dock Control Box and bracket for return to manufacturer.
5. Remove Front End Bell Asm
4. Remove SRI module from bracket. It will be transferred to PAC/SRI platform in later procedure. Bracket will be returned.
   Note: Run 421 (46–287908G2), which connects J3 on the SRI, will be replaced with a longer cable (46–287908G1) for this upgrade.
3. Unfasten Front End Bell Asm and SRI Asm from studs.
2. Disconnect Run 385 from circuit board on Front End Bell. Run 385 will be reconnected to new End Bell.
1. Disconnect Runs 778 and 779 with Bulkhead asm from RF Coil Cables .

❑ D2 – DISCONNECT AIR HOSES & CABLES AND REMOVE REAR PEDESTAL

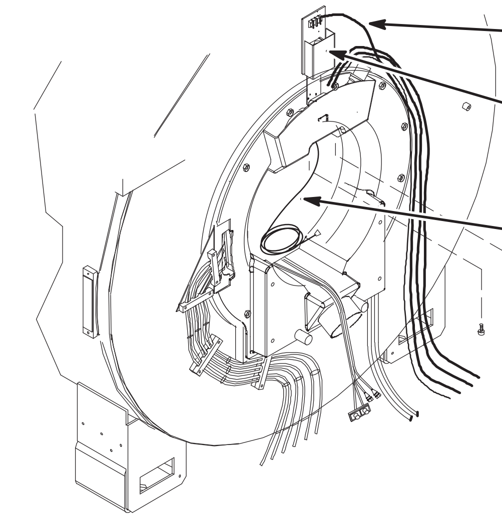
WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AMD MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.



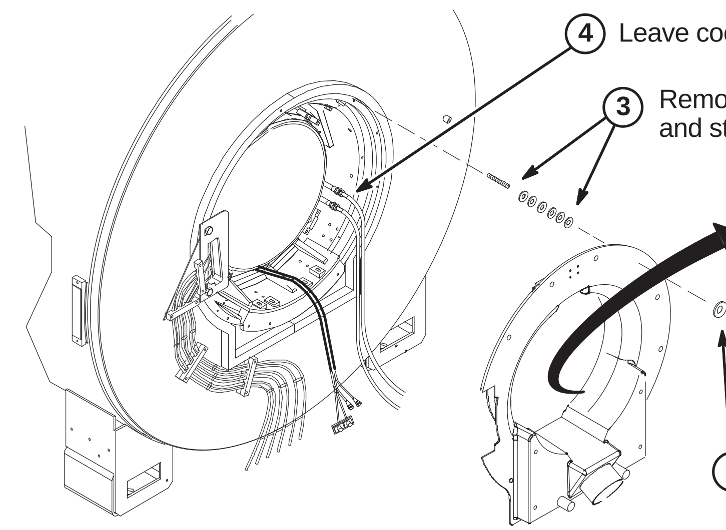
1. Disconnect and discard existing 2” air hose routed to Patient Comfort Module.
2. Disconnect and save 6” air hose. It will be reinstalled in later procedure.
3. Disconnect all cables connected to Rear Pedestal components.
4. Unfasten Rear Pedestal and remove from Magnet Room.
   <u>Many of the componenets of the Rear Pedestal HAVE to be transferred to the new Rear Pedestal.</u>
5. Refer to later procedure for transfer of components.

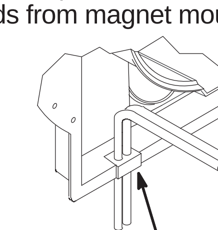
❑ D3 – REMOVE BORE LIGHT ASSEMBLY AND RUN 749



1. Disconnect Run 300 from Bore Light. Save cable. It can be reused and reconnected to new Bore Light.
2. Remove fiber optic lines from light tubes in bore. Unfasten Bore Light Asm with fiber optic lines and place on cover dolly to return for recycling.
3. Disconnect and remove Run 749, Bore Temperature Sensor, and return with other parts.

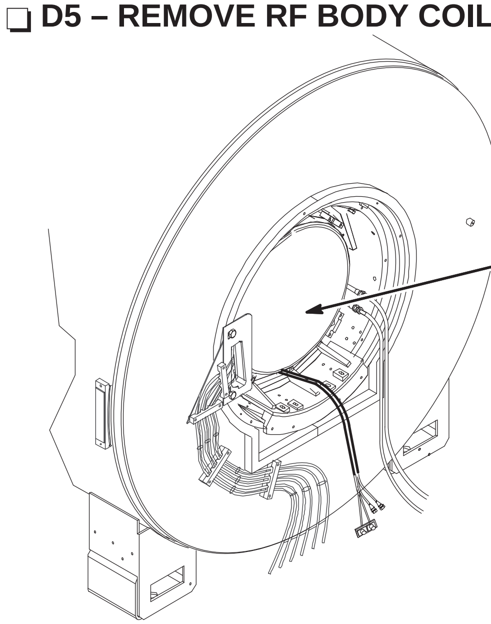
❑ D4 – REMOVE REAR END BELL




Rear View of End Bell

4. Leave coolant hoses attached to Body Coil.
3. Remove Neoprene Rubber Washers and studs from magnet mounting ring.
2. Remove rubber water hoses from opening in End Bell.
1. Unfasten and remove Rear End Bell Asm.

❑ D5 – REMOVE RF BODY COIL



1. Remove RF Body Coil and return to manufacturer using same container that new RF Body Coil was delivered in.

Note: Make sure to strap container to pallet.

FIXED–SITE LCC MAGNET ENCLOSURE UPGRADE INSTRUCTIONS

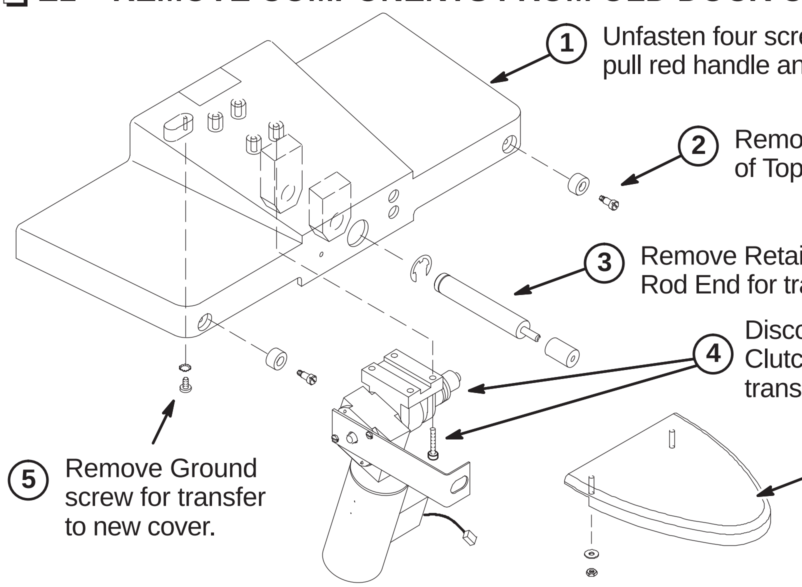
INSTALLATION PROCEDURES SUMMARY

- ❑ **E1 – Remove Components from Old Dock Cover**
- ❑ **E2 – Install Components on New Dock Cover**
- ❑ **E3 – Remove Components from Old Rear Pedestal**
- ❑ **E4 – Install Components on New Rear Pedestal**

WARNING!

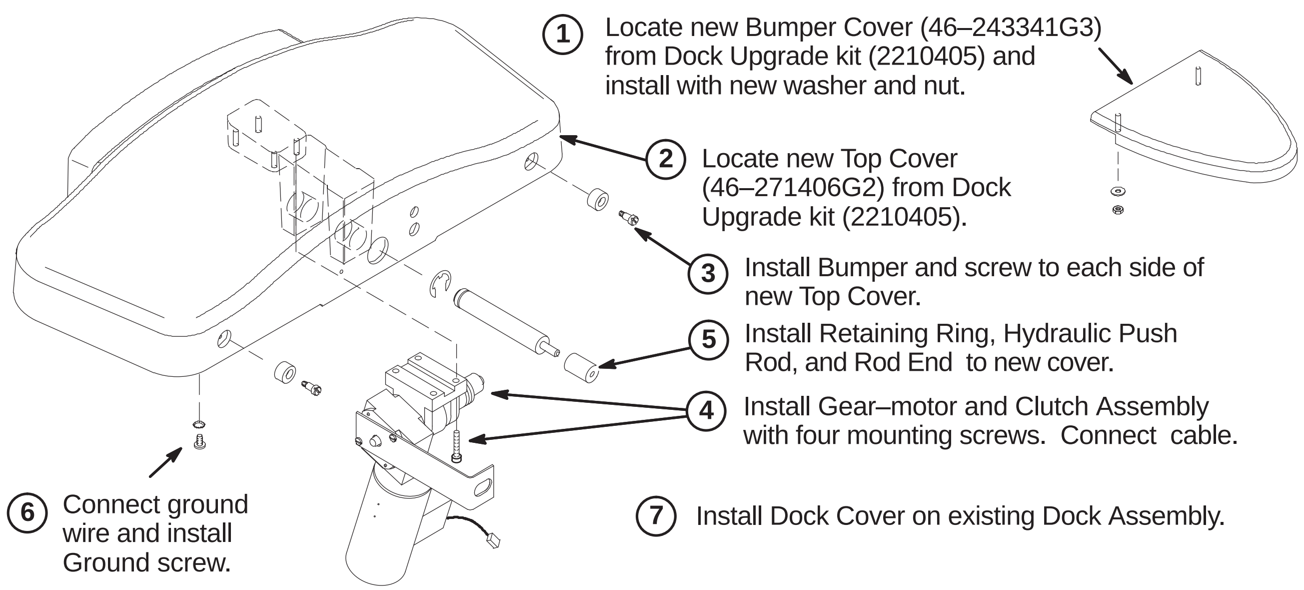
THE DOCK, ESPECIALLY THE GEAR–MOTOR ASSEMBLY, IS HIGHLY FERROUS! Before executing the following two procedures, confirm that the Dock is outside the magnet room.

❑ E1 – REMOVE COMPONENTS FROM OLD DOCK COVER



1. Unfasten four screws holding Top Cover to Dock Assembly. pull red handle and remove cover from base.
2. Remove Bumper and screw from each side of Top Cover for transfer to new cover.
3. Remove Retaining Ring, Hydraulic Push Rod, and Rod End for transfer to new cover.
4. Disconnect cable. Remove Gear–motor and Clutch Assembly and four mounting screws for transfer to new cover.
5. Remove Ground screw for transfer to new cover.
6. Remove Bumper cover.
7. Return old Top Cover and Bumper Cover to manufacturer.

❑ E2 – INSTALL COMPONENTS ON NEW DOCK COVER

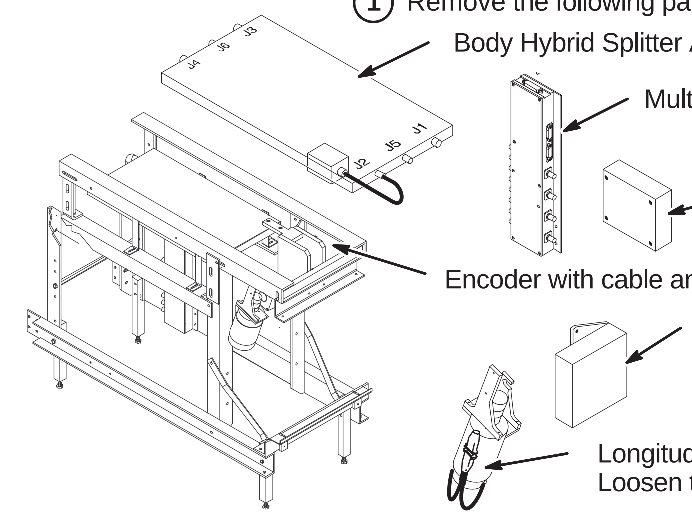


1. Locate new Bumper Cover (46–243341G3) from Dock Upgrade kit (2210405) and install with new washer and nut.
2. Locate new Top Cover (46–271406G2) from Dock Upgrade kit (2210405).
3. Install Bumper and screw to each side of new Top Cover.
5. Install Retaining Ring, Hydraulic Push Rod, and Rod End to new cover.
4. Install Gear–motor and Clutch Assembly with four mounting screws. Connect cable.
6. Connect ground wire and install Ground screw.
7. Install Dock Cover on existing Dock Assembly.

WARNING!

THE LONGITUDINAL DRIVE MOTOR IS HIGHLY FERROUS! Before executing the following two procedures, confirm that the Rear Pedestal is outside of the magnet room.

❑ E3 – REMOVE COMPONENTS FROM OLD REAR PEDESTAL



1. Remove the following parts for transfer to new Rear Pedestal.

Body Hybrid Splitter Assembly with Preamp

Multi–coil Select Switch Module

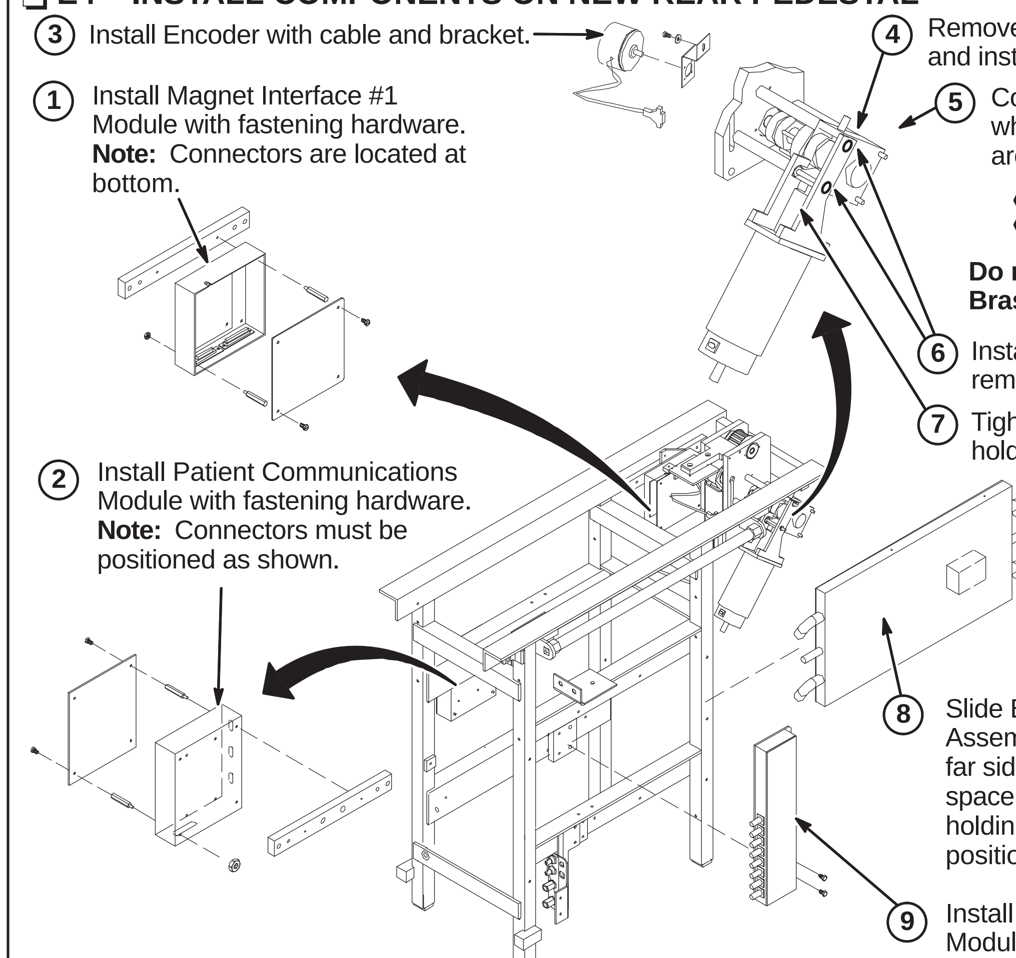
Patient Communications Module with fastening hardware

Encoder with cable and mounting bracket.

Magnet Interface #1 Module with fastening hardware

Longitudinal Drive Motor with mounting bracket and coupling. Loosen two set screws holding coupling to shaft.

❑ E4 – INSTALL COMPONENTS ON NEW REAR PEDESTAL



3. Install Encoder with cable and bracket.
1. Install Magnet Interface #1 Module with fastening hardware. **Note:** Connectors are located at bottom.
2. Install Patient Communications Module with fastening hardware. **Note:** Connectors must be positioned as shown.
4. Remove two (2) flat head screws and install Longitudinal Drive Asm.
5. Connect coupling to shaft while positioning bracket around brass rods.

CAUTION

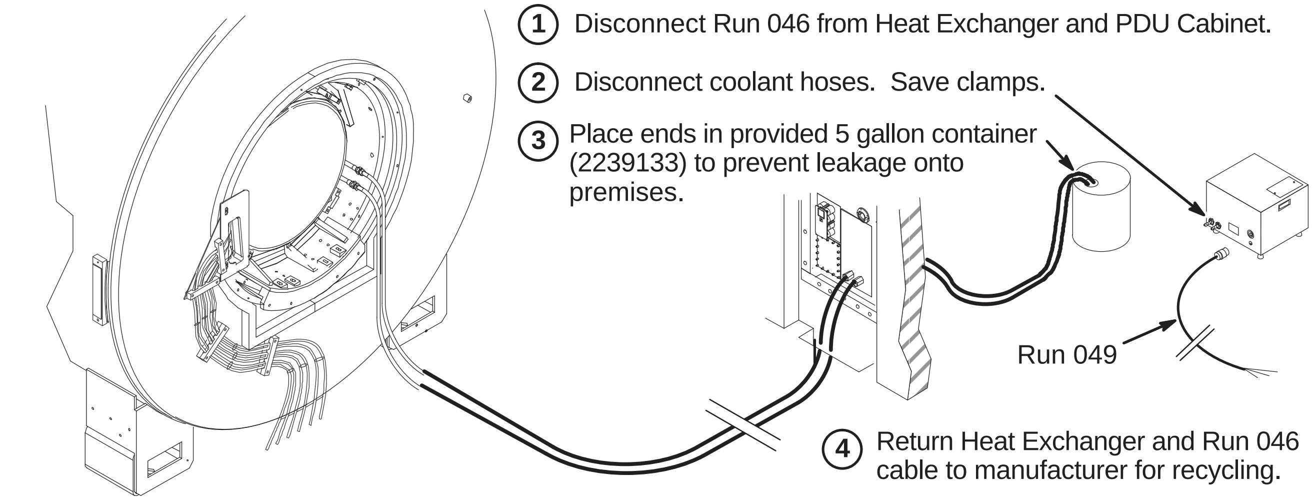
Do not overtighten nuts. Brass threads will strip easily.

6. Install two (2) screws removed in above step.
7. Tighten two (2) Set Screws holding coupling to shaft.
8. Slide Body Hybrid Splitter Assembly with Preamp (on far side) between insulated spacers. Tighten screws holding spacers after positioning Splitter Assembly.
9. Install Multi–coil Select Switch Module with provided screws.

INSTALLATION PROCEDURES SUMMARY

- ❑ **F1 – Disconnect Heat Exchanger**
- ❑ **F2 – Drain Coolant from Coolant Hoses (see Note in procedure)**
- ❑ **F3 – Connect Coolant Hose**
- ❑ **F4 – Install Power Cord**
- ❑ **F5 – Filling the System with Coolant**
- ❑ **F6 – Starting Water Chiller**

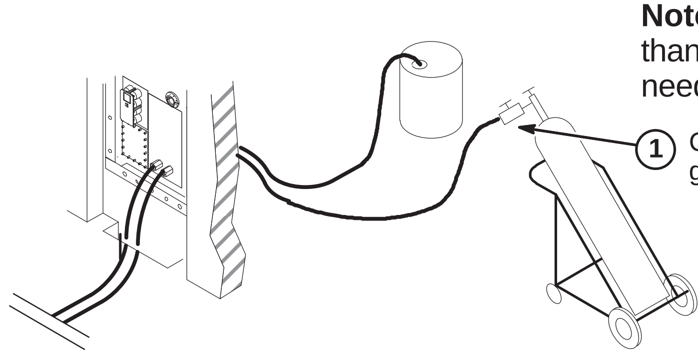
❑ F1 – DISCONNECT HEAT EXCHANGER



1. Disconnect Run 046 from Heat Exchanger and PDU Cabinet.
2. Disconnect coolant hoses. Save clamps.
3. Place ends in provided 5 gallon container (2239133) to prevent leakage onto premises.
4. Return Heat Exchanger and Run 046 cable to manufacturer for recycling.

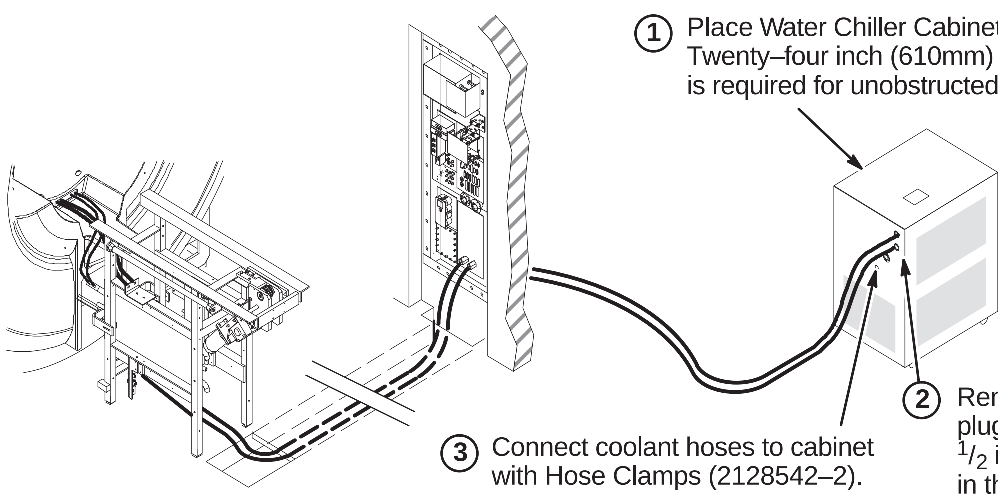
❑ F2 – DRAIN COOLANT FROM COOLANT HOSES

Note: If system has been in operation for less than 6 months, this Procedure (**E2**), does not need to be completed.



1. Connect one of the hoses to locally obtained nitrogen gas cylinder.
2. Carefully open valve on cylinder and purge coolant from hoses into 5 gallon container.
3. When complete, close and seal 5 gallon container for return for recycling,

❑ F3 – CONNECT COOLANT HOSE

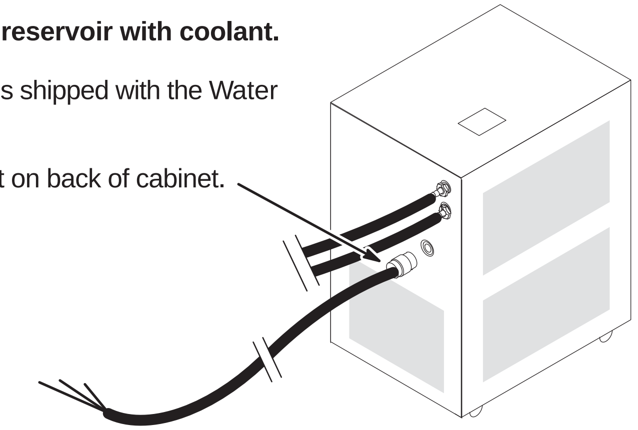


1. Place Water Chiller Cabinet in Equipment Room. Twenty–four inch (610mm) clearance on all sides is required for unobstructed flow of cooling air.
2. Remove two protective plugs and install provided $^{1}/_{2}$ inch hose barbs fittings in these connections.
3. Connect coolant hoses to cabinet with Hose Clamps (2128542–2).

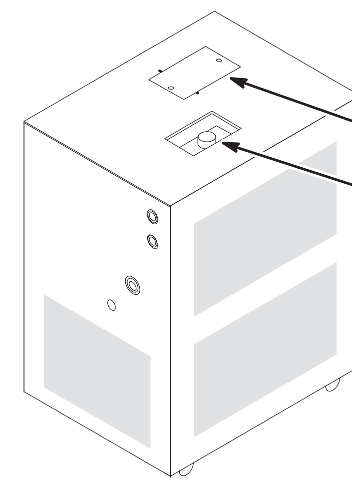
❑ F4 – INSTALL POWER CORD

CAUTION:
Do not start Water Chiller pump before filling reservoir with coolant.
Consult vendor Water Chiller manual –
Installation, Operation, and Service Manual that is shipped with the Water Chiller.

1. Connect power cable Run 065 (2223355) to outlet on back of cabinet.

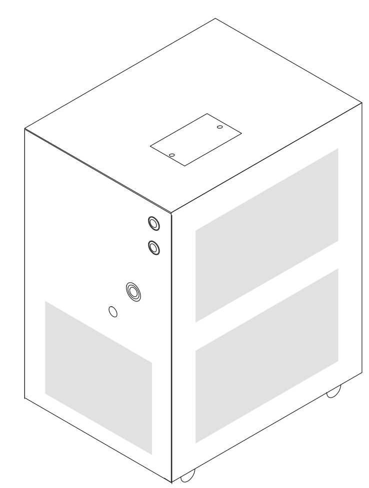


❑ F5 – FILLING THE SYSTEM WITH COOLANT



1. Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System
2. Remove cover.
3. Consult vendor Water Chiller manual – *Installation: Filling Requirements*.
Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.

❑ F6 – STARTING WATER CHILLER



1. Consult vendor Water Chiller manual – *Operation: Start Up.* Follow instructions in manual for starting water chiller operation.
2. Verify unit is set to control at 20° C.
3. If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller or store it with other service manuals on site.

INSTALLATION PROCEDURES SUMMARY

- ❑ G1 – Compact PDU Circuit Breaker Installation Preparation } Compact PDU Only
- ❑ G2 – Install Circuit Breaker in Compact PDU } Compact PDU Only
- ❑ G3 – Route/Connect Run 065 In Compact PDU Cabinet } Compact PDU Only

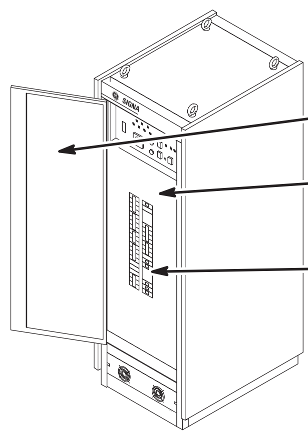
OR

- ❑ G4 – Route/Connect Run 065 In RF/PDU Cabinet } RF/PDU Only
- ❑ G5 – Apply Gradient Water Chiller Label To PDU Module in RF/PDU } RF/PDU Only

Note:
Circuit Breaker (46–320337P4) is only used in upgading the Compact PDU Cabinet, Procedures **F1** thru **F3**.

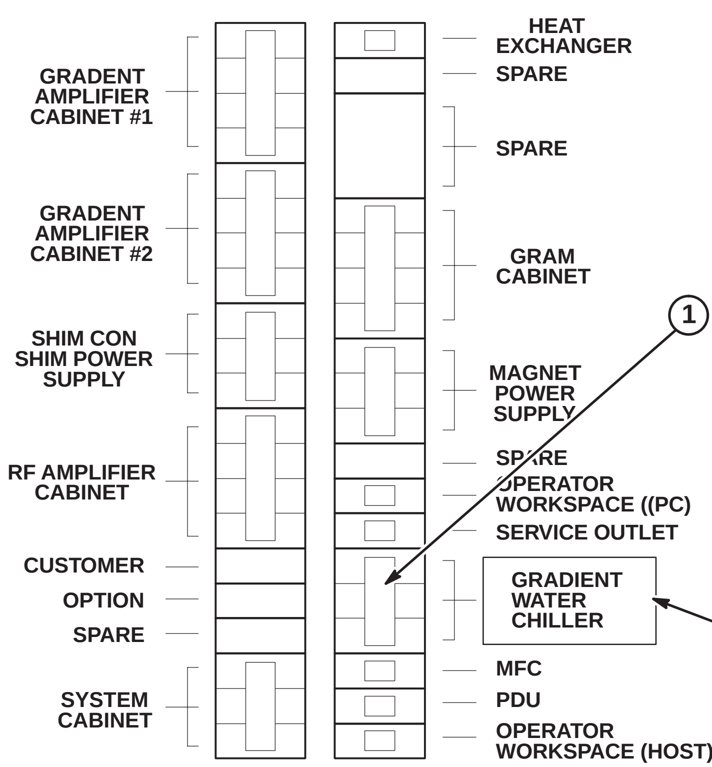
It is not used on the RF/PDU Cabinet upgrade, Procedures **F4** and **F5**.

❑ G1 – COMPACT PDU CIRCUIT BREAKER INSTALLATION PREPARATION



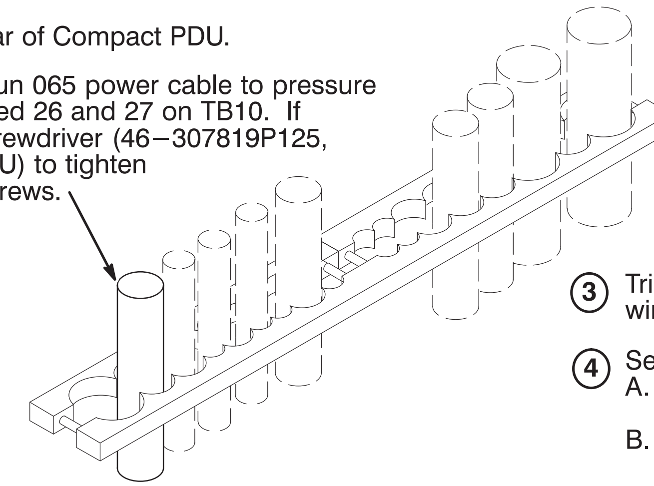
1. Make certain that power at the main disconnect is OFF, locked and tagged before proceeding.
2. Open front door of Compact PDU.
3. Remove the panel that holds the row of rows of circuit breakers.
4. Pry out the three blanks located on the front panel for the circuit breaker position marked RESISTIVE SHIM POWER SUPPLY.

❑ G2 – INSTALL CIRCUIT BREAKER IN COMPACT PDU



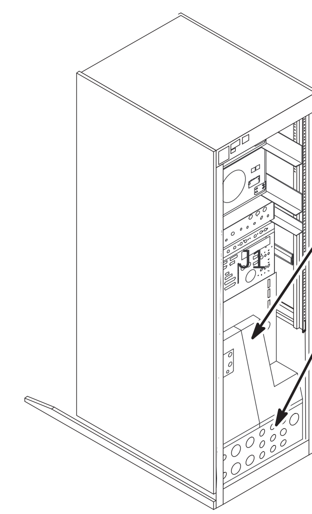
1. Install the Gradient Water Chiller circuit breaker (46–320337P4) into the position originally marked for Resistive Shim Power Supply.
   – Connect circuit breaker to existing wire harness as marked.
   – Plug sensor wire on the circuit breaker into mating connector on the PDU harness.
2. Replace front panel and make sure circuit breaker is set to OFF.
3. Place GRADIENT WATER CHILLER label (2230962) over Resistive Shim Power Supply.

❑ G3 – ROUTE/CONNECT RUN 065 IN COMPACT PDU CABINET

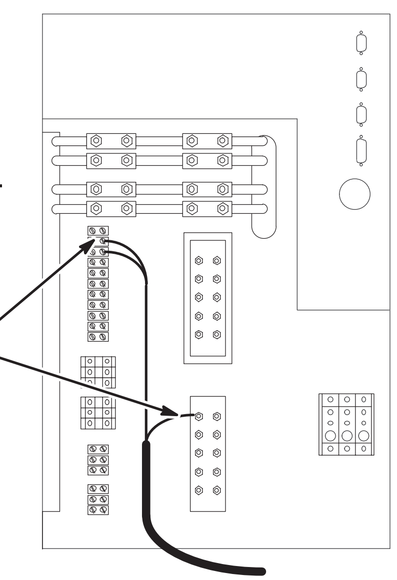


1. Gain acces to rear of Compact PDU.
2. Route/connect Run 065 power cable to pressure connectors marked 26 and 27 on TB10. If available, use screwdriver (46–307819P125, provided with PDU) to tighten terminal block screws.
3. Trim and connect GRN/YEL ground wire to ground bus connectors.
4. Secure cable in place with either:
   A. Cable Ties.
   OR
   B. Strain relief clamp. (shown)
   Tighten screws.

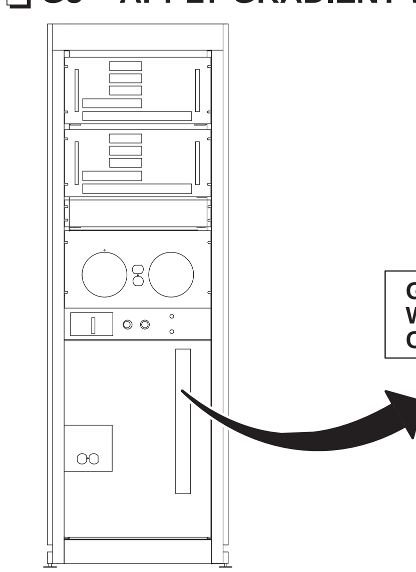
❑ G4 – ROUTE/CONNECT RUN 065 IN RF/PDU CABINET

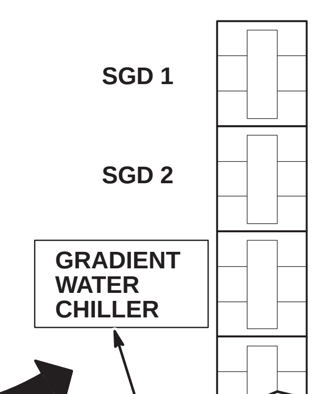


1. Make sertain that power at the main disconnect is OFF, locked and tagged before proceeding.
2. Remove covers from rear of PDU Module.
3. Route Run 065 thru Heat Exchanger opening in Interface Panel.
4. Route and connect as marked Run 065 to terminal block marked 26 and 27. Connect GRN/YEL wire to stud on Ground Plate.
5. Install covers on PDU Module.



❑ G5 – APPLY GRADIENT WATER CHILLER LABEL TO PDU MODULE IN RF/PDU





Note: Circuit Breaker (46–320337P4) is not used on the RF/PDU cabinet upgrade.

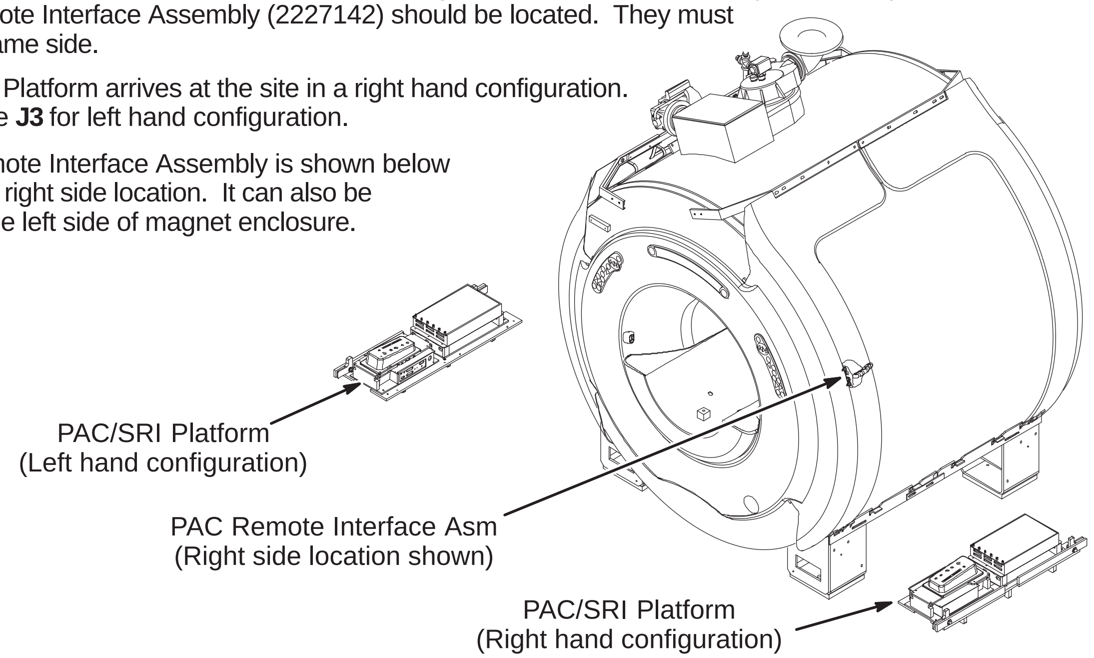
1. Place GRADIENT WATER CHILLER label (2230962) over Resistive Shim (Option).

INSTALLATION PROCEDURES SUMMARY

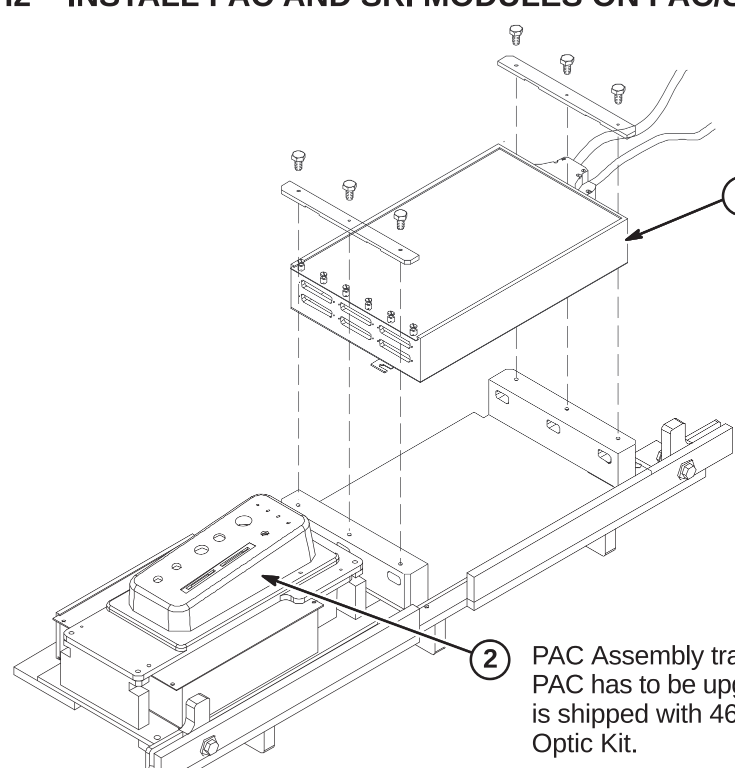
- ❑ **H1 – Determine PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **H2 – Install PAC and SRI Modules on PAC/SRI Platform**
- ❑ **H3 – PAC/SRI Platform Left Hand Configuration (If required)**
- ❑ **H4 – Location of Magnet Ground Cable**

❑ H1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

1. Consult with customer to determine which side of magnet the PAC/SRI Platform (2223440–2) and PAC Remote Interface Assembly (2227142) should be located. They must to be on the same side.
2. The PAC/SRI Platform arrives at the site in a right hand configuration. See Procedure **J3** for left hand configuration.
3. The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.



❑ H2 – INSTALL PAC AND SRI MODULES ON PAC/SRI PLATFORM

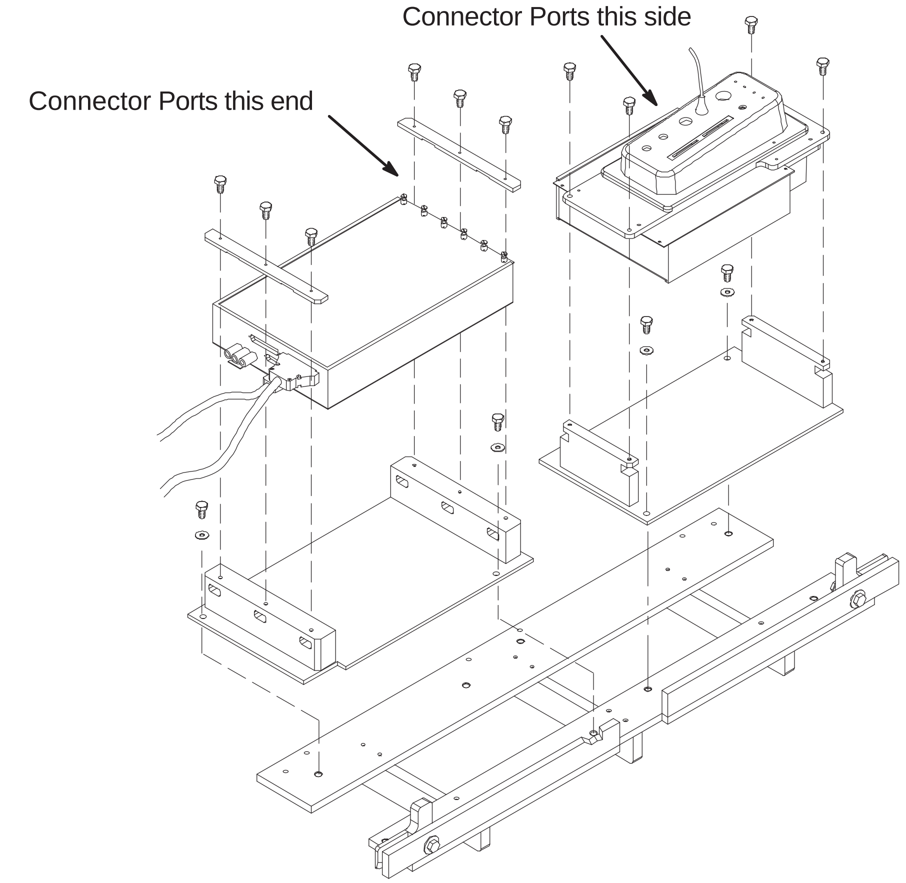


1. Install previously removed SRI Module on PAC/SRI Platform as shown.
2. PAC Assembly transferred from Front Cover. Before installation, PAC has to be upgraded using Direction 2245769. This Direction is shipped with 46–317999G4, Photo–Plethysmograph Fiber Optic Kit.

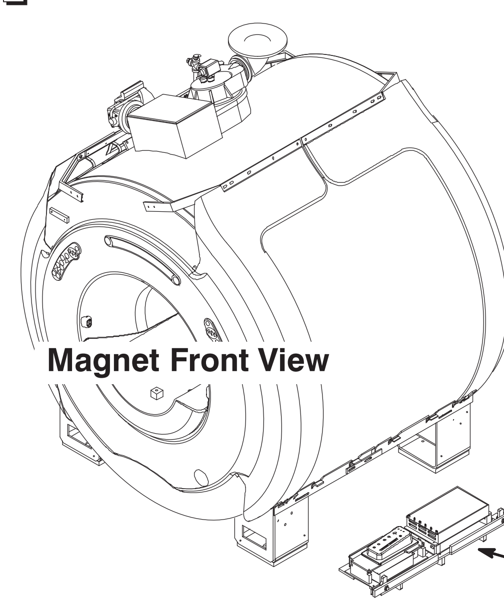
Note: Must use G4 cable because G3 will be too short.

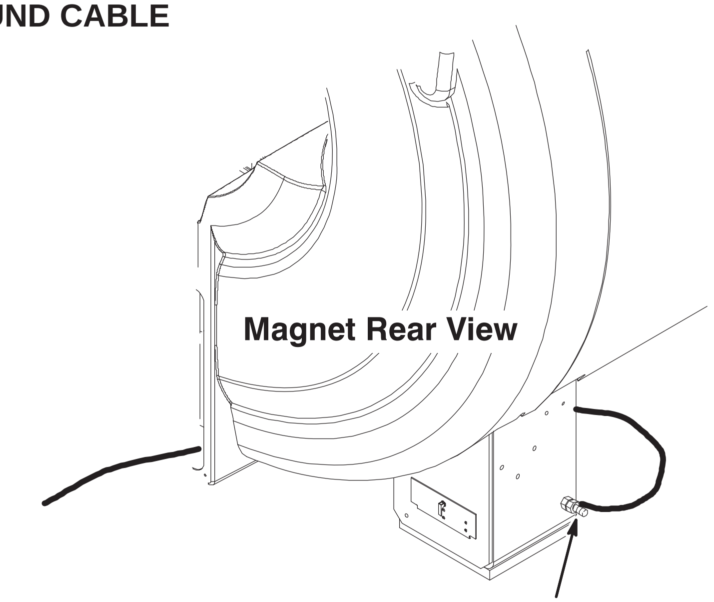
❑ H3 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

1. For right hand configuration, reassemble PAC/SRI as shown.



❑ H4 – LOCATION OF MAGNET GROUND CABLE





1. If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable has to be connected to the opposite side rear left leg as shown above.
2. If PAC/SRI Platform is located on left side of magnet, then the Magnet ground cable has to be connected to the opposite side rear right leg.

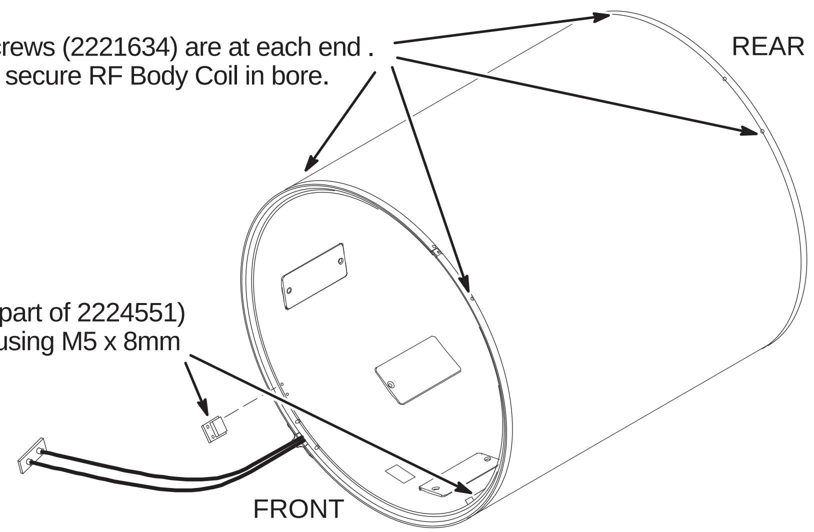
INSTALLATION PROCEDURES SUMMARY

- ❑ **J1 – RF Body Coil Installation Preparation**
- ❑ **J2 – RF Body Coil Installation**
- ❑ **J3 – Position RF Body Coil in Gradient Coil**
- ❑ **J4 – Securing RF Body Coil**

NOTE:
The RF Coil for an MR/*i* system (BRM–D) is shorter than the one used for a CV/*i* system (CRM).

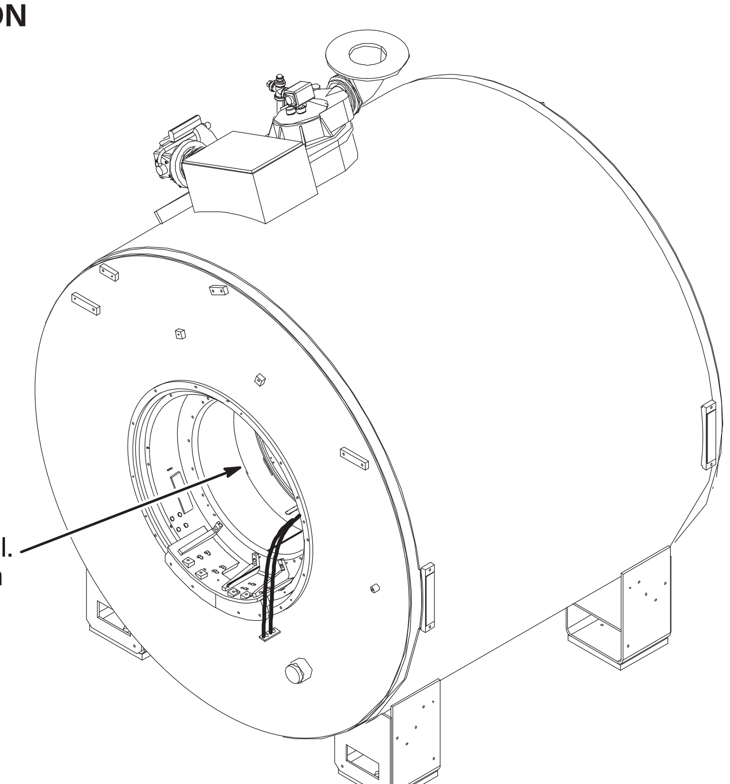
❑ J1 – RF BODY COIL INSTALLATION PREPARATION

1. Make sure two set screws (2221634) are at each end . These will be used to secure RF Body Coil in bore.



2. Make sure that two Latch Catches (part of 2224551) are attached to each end of RF Coil using M5 x 8mm screws (2227318) and Blue Loctite.

❑ J2 – RF BODY COIL INSTALLATION

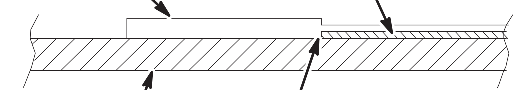


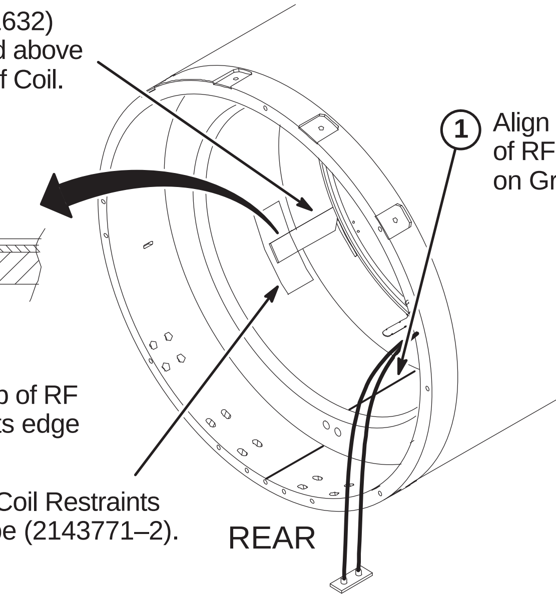
1. Carefully insert RF Coil into Gradient Coil. RF Coil should be of equall distance from the front and rear of the Gradient Coil.

❑ J3 – POSITION RF BODY COIL IN GRADIENT COIL

Note: Steps 2, 3, and 4 are to be performed only if RF Coil is for an MR/*i* upgrade. The RF Coil Constraints are not used for cardiac CV/*i* systems.

2. Slide two RF Coil Constraints (2221632) between RF and Gradient Coils and above black rubber pads on each side of Rf Coil.





1. Align black mark on end of RF Coil with black line on Gradient Coil.

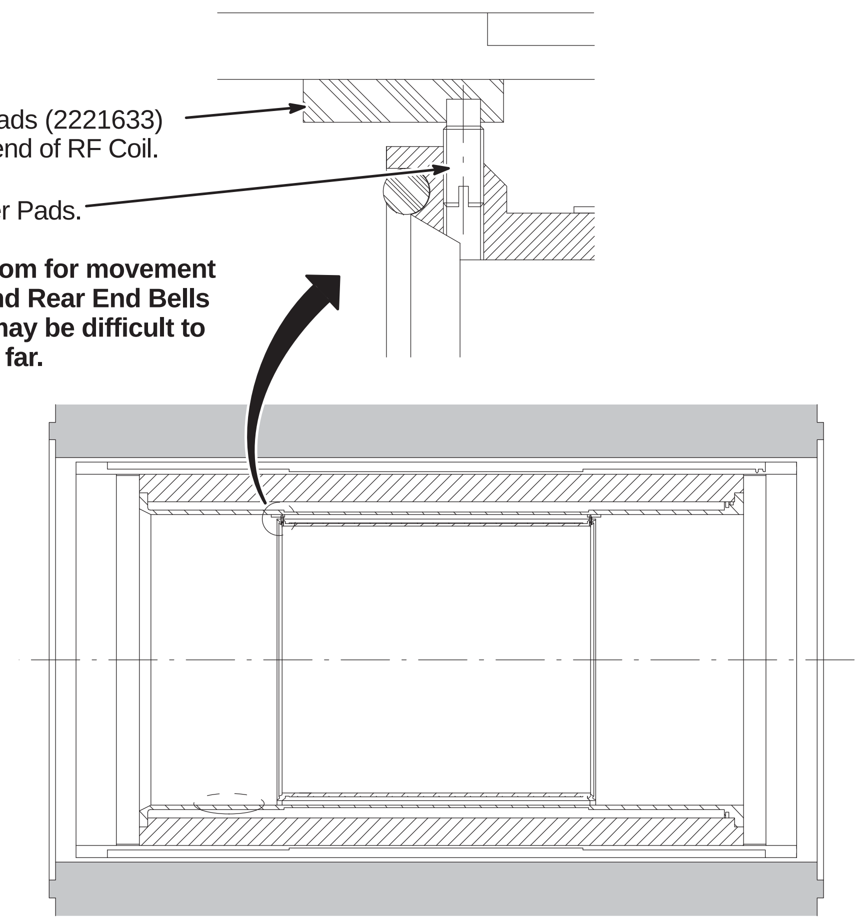
3. Position RF Coil until lip of RF Coil Restraints contacts edge of RF Shield Retainer.

4. Tape both ends of RF Coil Restraints with strips of Teflon tape (2143771–2).

❑ J4 – SECURING RF BODY COIL

1. Position Upper Mounting Pads (2221633) above Set Screws at each end of RF Coil.

2. Turn Set Screws into Rubber Pads.
DO NOT OVER TIGHTEN.
RF Coil will need some room for movement when clamped to Front and Rear End Bells in later procedure. Also may be difficult to back out if screwed in too far.

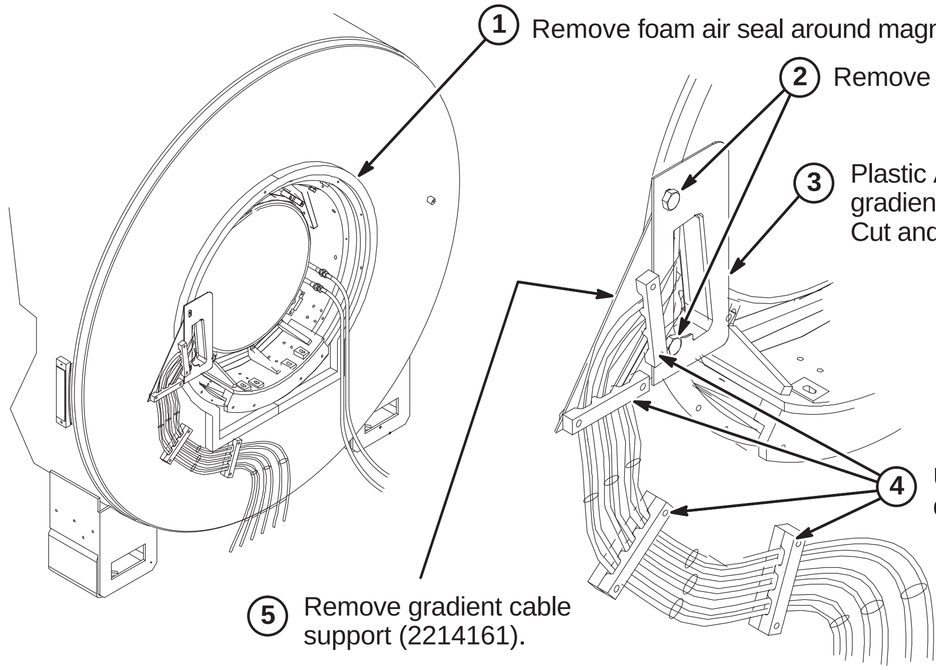


SECTION VIEW OF SIDE OF COIL INSTALLATION

INSTALLATION PROCEDURES SUMMARY

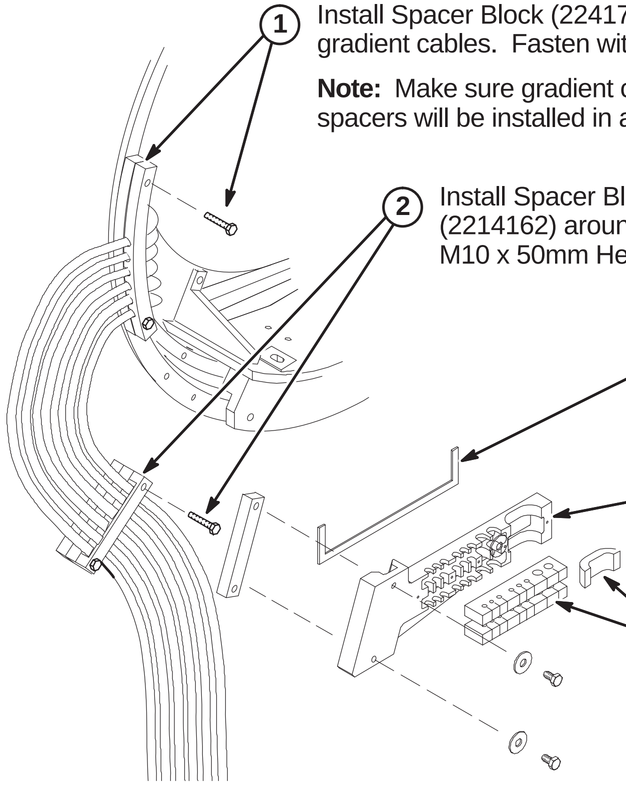
- ❑ **K1 – Remove Foam Gasket and Gradient Cable Air Seal**
- ❑ **K2 – Install Cable Clamps**
- ❑ **K3 – Route Gradient Cables & Coolant Hoses**
- ❑ **K4 – Route Air Hose and Install Cable/Hose Clamp**

❑ K1 – REMOVE FOAM GASKET, GRADIENT CABLE AIR SEAL, CLAMPS & BRACKETS



1. Remove foam air seal around magnet mounting ring.
2. Remove two screws.
3. Plastic Air Seal (2177301) around gradient cables has to be removed. Cut and break Air Seal as necessary.
4. Unfasten screws. Remove and discard outside cable clamps.
5. Remove gradient cable support (2214161).

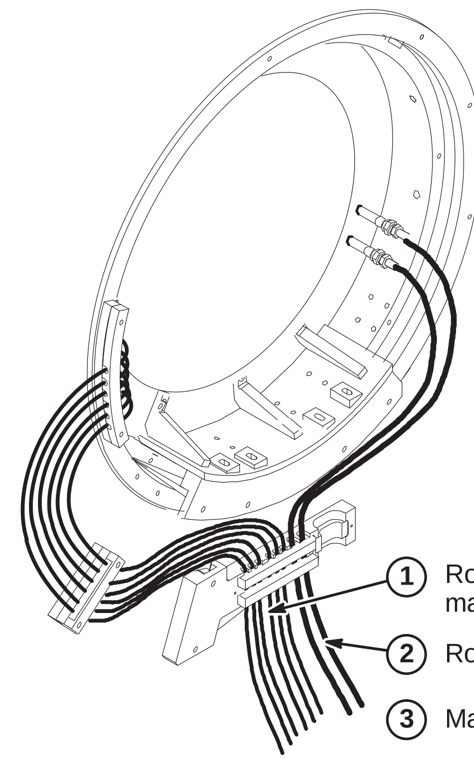
❑ K2 – INSTALL CABLE CLAMPS



1. Install Spacer Block (2241781–2) and Gradient Clamp (2241781) around routed gradient cables. Fasten with two M10 x 50mm Hex Head Screws (2109866–24).

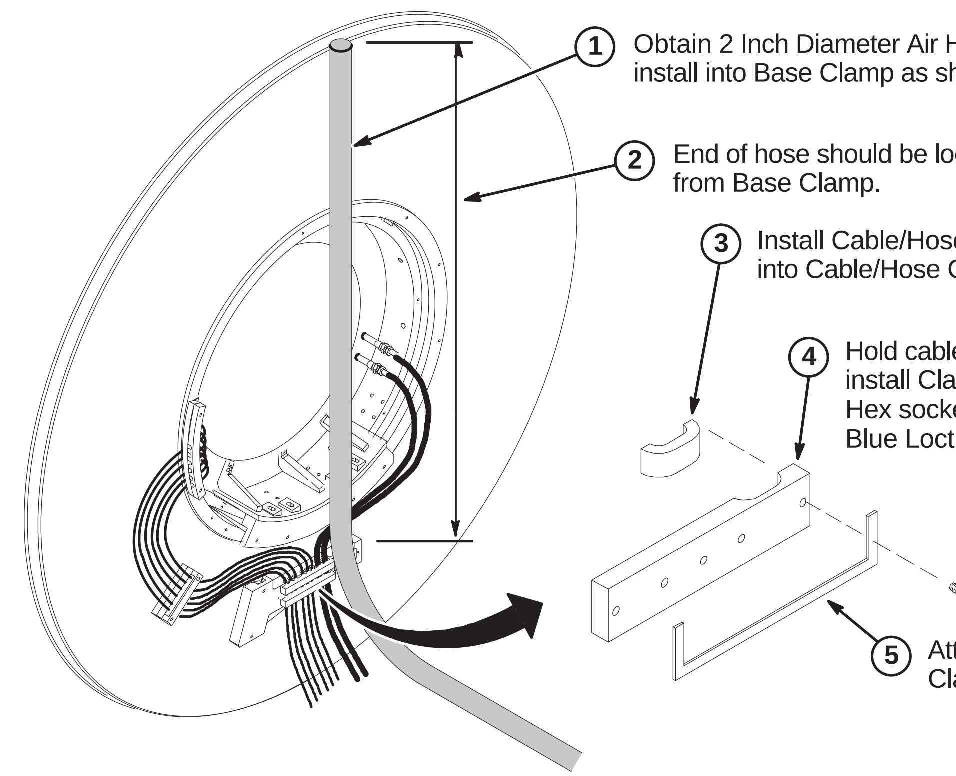
   Note: Make sure gradient cables are eequally spaced as they are routed. Foam spacers will be installed in a later procedure.
2. Install Spacer Block (2239148–2) and Gradient Clamp (2214162) around routed gradient cables. Fasten with two M10 x 50mm Hex Head Screws (2109866–24).
3. Remove protective cover and apply Cable/Hose Clamp Foam Seal (2228286) to Hose/Cable Clamp Base (2227395) as shown.
4. Attach Cable/Hose Clamp Base to magnet using two M10 x 25mm (2109866–20) Hex Head screws and Blue Loctite 242 (46–170686P2).
5. Install two Cable/Hose Clamp Foam Seals (2227841) and Cable/Hose Camp Foam Block (2228287) as shown.

❑ K3 – ROUTE GRADIENT CABLES & COOLANT HOSES



1. Route Gradient cables thru Cable Base. Start with the right side. Do not make sharp bends in the cables.
2. Route coolant hoses.
3. Make sure foam is completely around each cable and hose to form air seal.

❑ K4 – ROUTE AIR HOSE AND INSTALL CABLE/HOSE CLAMP

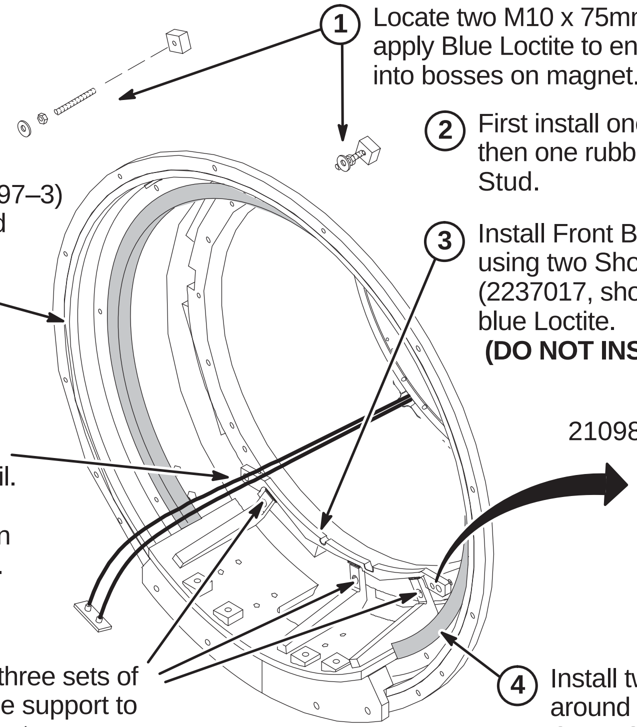


1. Obtain 2 Inch Diameter Air Hose (46–282668P18) and install into Base Clamp as shown.
2. End of hose should be located 64 inches (1625mm) from Base Clamp.
3. Install Cable/Hose Clamp Foam Block (2228287) into Cable/Hose Clamp Top (2227396) as shown.
4. Hold cables and hoses in place and install Clamp Top using five M6 x 30mm Hex socket screws (2109867–16) and Blue Loctite.
5. Attach Foam Seal (2228286) to Clamp Top.

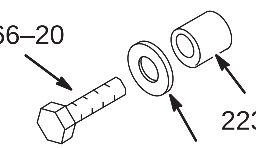
INSTALLATION PROCEDURES SUMMARY

- ❑ **L1 – Install Bore Front Seals, Mounting Bracket, and Studs**
- ❑ **L2 – Install Magnet Front Cable Clamps**
- ❑ **L3 – Attach Operator Display Cable to Front of Magnet**
- ❑ **L4 – Install Gradient Cable Foam Segments**

❑ L1 – INSTALL BORE FRONT SEALS, MOUNTING BRACKET, AND STUDS

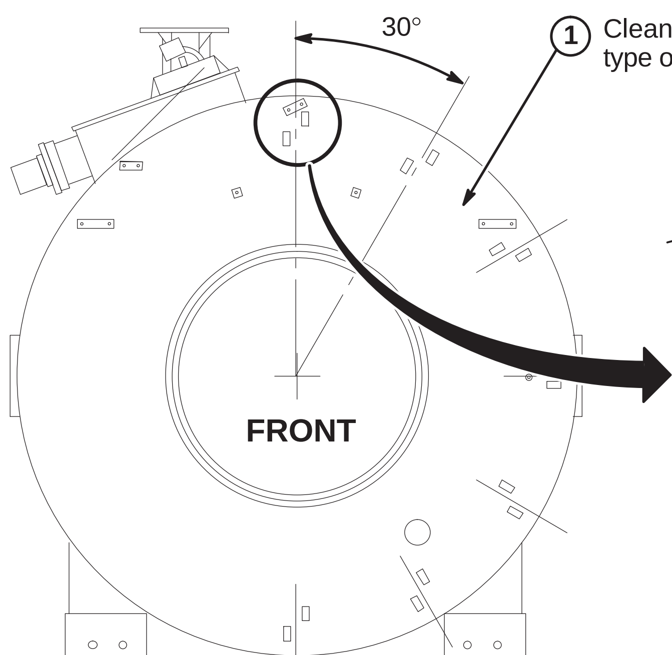


1. Locate two M10 x 75mm Studs (2116325–10) and apply Blue Loctite to end opposite the slot and install into bosses on magnet.
2. First install one M10 Nut (2109875–4) and then one rubber washer (2167321) on each Stud.
3. Install Front Bridge Support (2235935) using two Shoulder Bolt Assemblies (2237017, shown below) at each end and blue Loctite. **(DO NOT INSTALL FOR CV/i UPGRADE)**

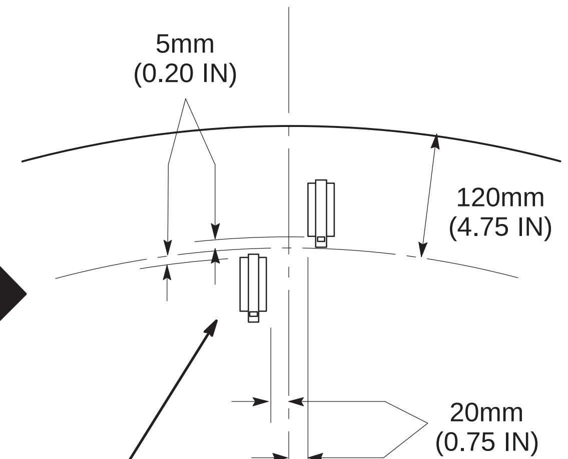


4. Install two Foam Seals (2174929–6) around inside edge of the Gradient Outer Cylinder.
5. If installed, remove the three sets of fasteners connecting the support to the Gradient Coil. Repeat process for other end of magnet.
6. Route Body Coil Cables straight out from Body Coil. **Do not fit into bridge support notch.** Secure in place with fiberglass tape.
7. Install Foam Strip (2179397–3) between magnet bore and Gradient Coil at both front and rear of magnet.

❑ L2 – INSTALL MAGNET FRONT CABLE CLAMPS

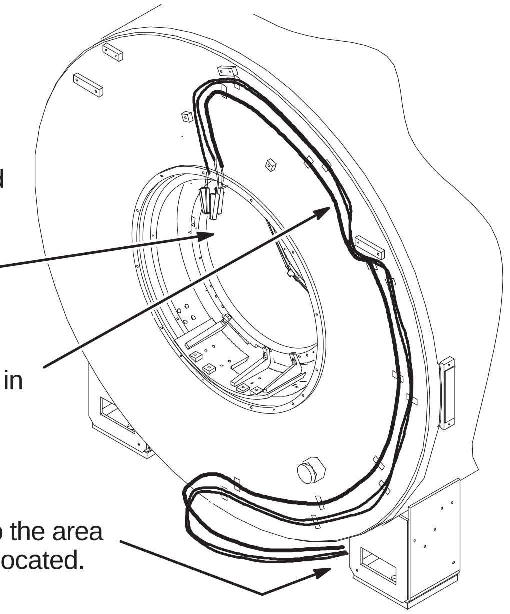


1. Clean Magnet surface with isopropel alcohol or other type of cleaning fluid before attaching cable clamps.



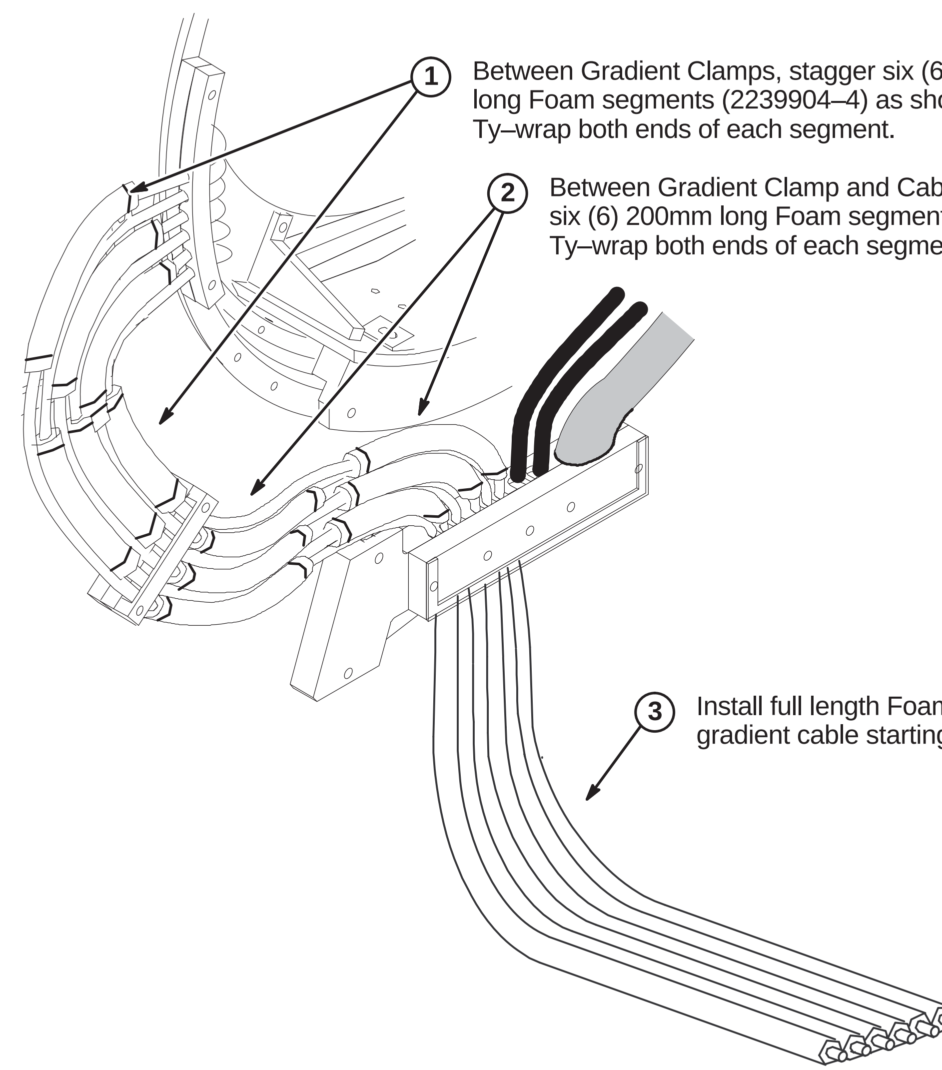
2. Locate and attach fourteen (14) Cable Clamps (46–220191P1) every 30° as shown

❑ L3 – ATTACH OPERATOR DISPLAY CABLES TO FRONT OF MAGNET



1. Locate three cables: 2221945, 2221947, and 2221948 for routing on front of magnet.
2. Position the cable ends with the right angle connector at the top of the Bore just below the magnet end ring as shown.
3. Route the three cables as shown and hold in place with cable clamps.
4. Route the other ends of the cables to the area where the SRI/PAC platform will be located. (right side location shown)

❑ G4 – INSTALL GRADIENT CABLE FOAM SEGMENTS

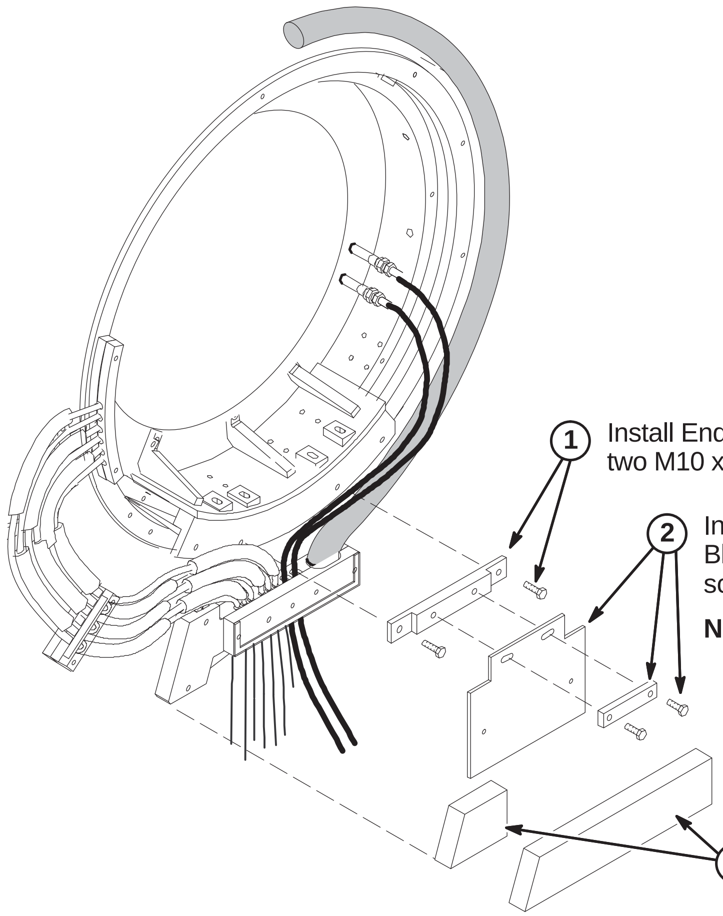


1. Between Gradient Clamps, stagger six (6) 200mm long Foam segments (2239904–4) as shown. Ty–wrap both ends of each segment.
2. Between Gradient Clamp and Cable/Hose Clamp, stagger six (6) 200mm long Foam segments (2239904–4) as shown. Ty–wrap both ends of each segment.
3. Install full length Foam (2239904) around each gradient cable starting at Cable/Hose Clamp.

INSTALLATION PROCEDURES SUMMARY

- ❑ **M1 – Install Foam Gasket and Gradient Cable Air Seal**
- ❑ **M2 – Install REar Studs, Bridge Support, & Secure RF Coil Cables**
- ❑ **M3– Attach Foam Block**

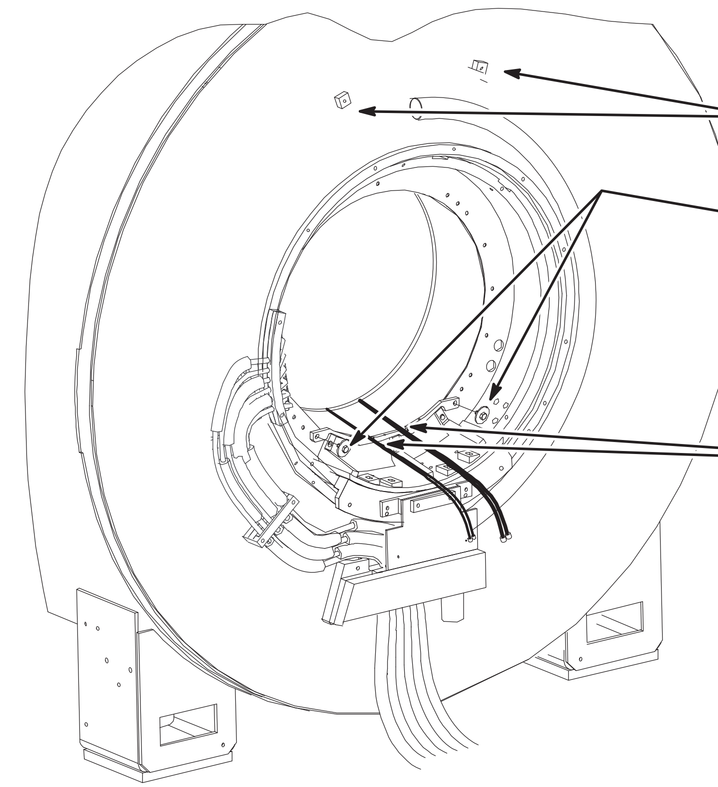
❑ M1 – INSTALL FOAM GASKET AND GRADIENT CABLE AIR SEAL



1. Install End Bell Mounting Spacer (2235604) using two M10 x 40mm Hex head screws (2109866–23).
2. Install End Bell Anchor Plate (2235605) and Washer Block (2235606) using two M10 x 30mm Hex head screws (2109866–21).

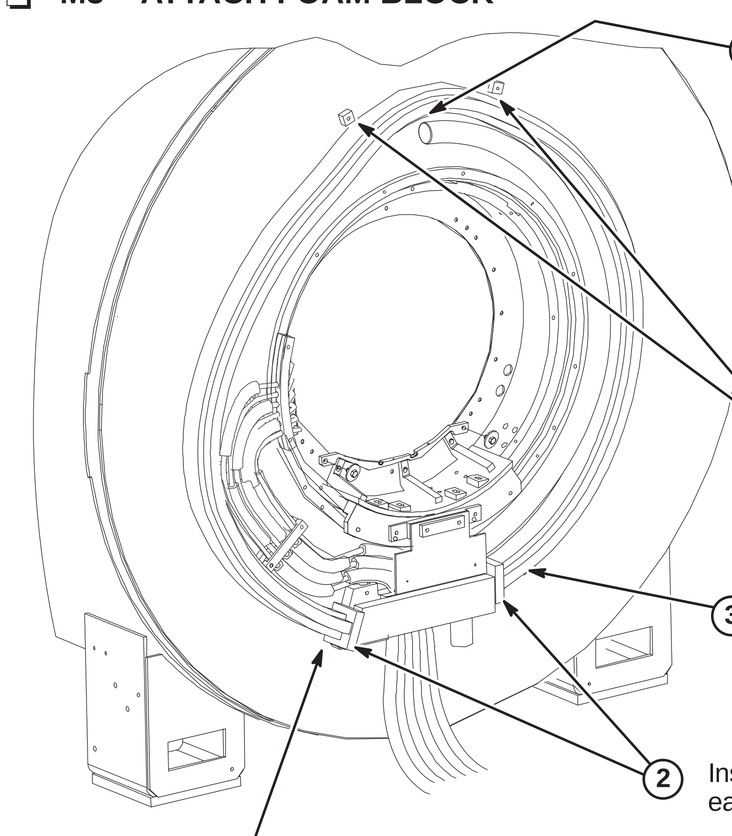
   Note: Center plate on studs before tightening nuts.
3. Obtain Foam Blocks (2239902–2 & 2239902–3) and attach to Rear Pedestal Plate as shown.

❑ M2 – INSTALL REAR STUDS, BRIDGE SUPPORT, & SECURE RF COIL CABLES



1. Locate two M10 x 75mm Studs (2116325–10) and apply Blue Loctite to end opposite the slot and install into bosses on magnet.
2. Install Front and Rear Bridge Supports (2235935) using two Shoulder Bolt Assemblies (2237017) and blue Loctite for each support.
   (MR/*i* ONLY)
3. Route Body Coil cables thru grooves in bridge support and secure in place with ty–wraps.

❑ M3 – ATTACH FOAM BLOCK

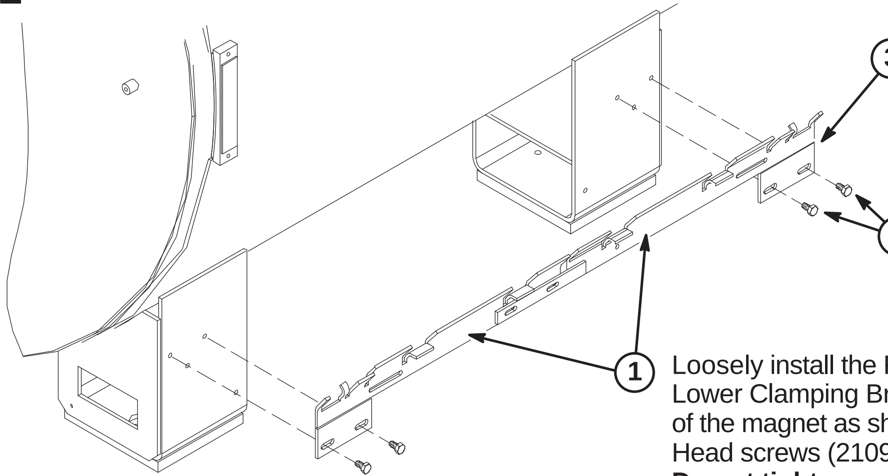


5. Foam placement in this area needs to avoid patient comfort air plenum on Rear End Bell.
4. There should be a 1 inch (25mm) space between the Foam and the mounting bosses.
3. Start routing of Rear End Bell Foam Block (2222806) and attach as shown. Make sure ends of Foam are butted up against both sides of Foam pieces installed in Step 2..
2. Install two (2) Foam pieces (2239902) on each side of Cable/Hose clamp as shown.
6. Cut Foam Block to fit.
1. Clean Magnet surface with isopropel alcohol or other cleaning solvent before attaching Rear End Bell Foam Block in next step.

INSTALLATION PROCEDURES SUMMARY

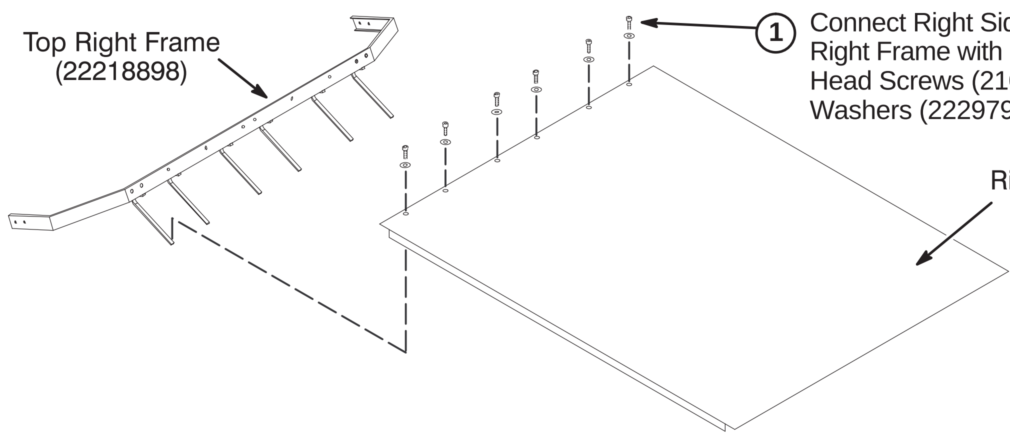
- ❑ N1 – Install Lower Clamping Brackets
- ❑ N2 – Connect Right Side Top Frame and Wrap Cover
- ❑ N3 – Connect Left Side Top and Wrap Cover
- ❑ N4 – Install Right and Left Side Frame and Wrap Covers
- ❑ N5 – Install Condensation Trough

❑ N1 – INSTALL LOWER CLAMPING BRACKETS



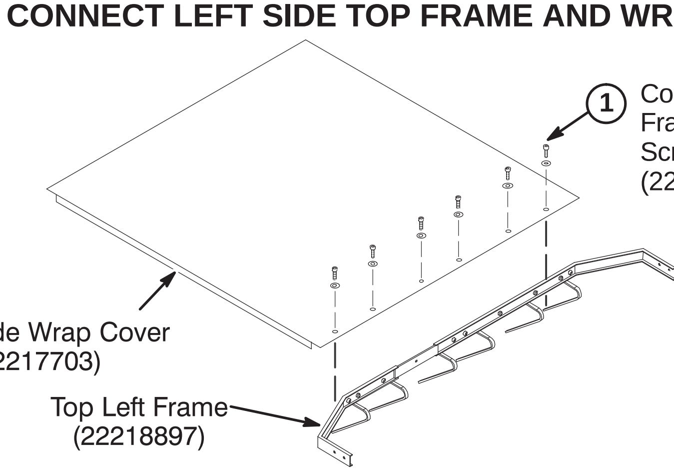
1. Loosely install the Right and Left (2218146–3 & –4) Lower Clamping Bracket Assemblies to each side of the magnet as shown using M10 x 25mm Hex Head screws (2109866–20) and Blue Loctite. **Do not tighten screws.**
2. Loosely install at each end of right and left brackets two (2) 1/2 x 3/4 long Hex Head Screws (46–260950P1).
3. Each end of bracket should be initially located about 2 to 3mm past magnet legs.

❑ N1 – CONNECT RIGHT SIDE TOP FRAME AND WRAP COVER



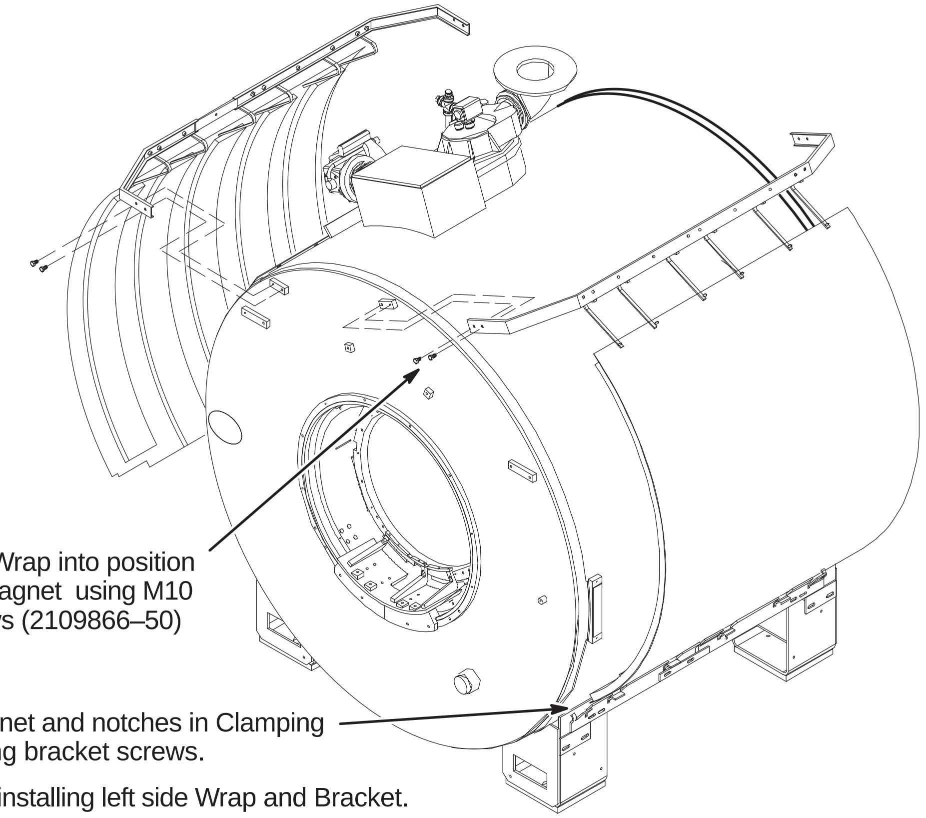
1. Connect Right Side Wrap Cover and Top Right Frame with six (6) M6 x 12mm Button Head Screws (2109872–6), M6 Nylon Washers (2229794), and Blue Loctite.

❑ N3 – CONNECT LEFT SIDE TOP FRAME AND WRAP COVER



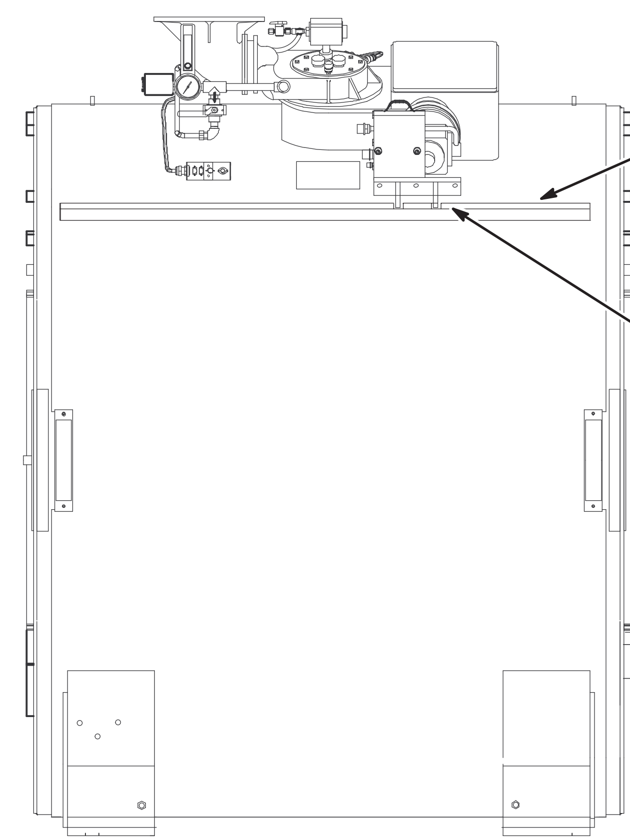
1. Connect Left Side Wrap Cover and Top Leftt Frame with six (6) M6 x 12mm Button Head Screws (2109872–6), M6 Nylon Washers (2229794), and Blue Loctite.

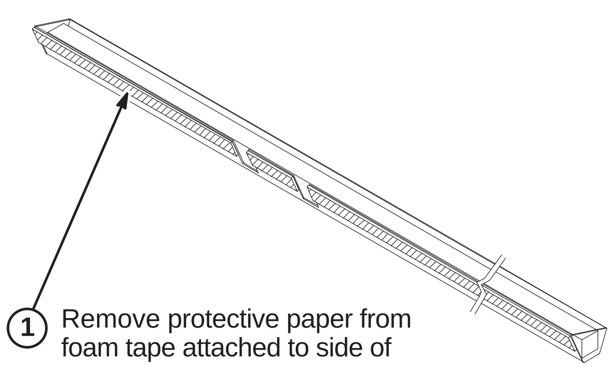
❑ N4 – INSTALL RIGHT AND LEFT SIDE FRAME AND WRAP COVERS



1. Lift right side Frame and Wrap into position as shown and fasten to magnet using M10 x 14mm Hex Head Screws (2109866–50) and Blue Loctite.
2. Slide Wrap between Magnet and notches in Clamping Bracket. Tighten clamping bracket screws.
3. Repeat steps 1 and 2 for installing left side Wrap and Bracket.

❑ N4 – INSTALL CONDENSATION TROUGH





1. Remove protective paper from foam tape attached to side of Condensation Trough (2237254).
2. Install Trough on left side of magnet with slots aligned with cold head support brackets.
3. Along the entire length of the Trough, lay a bead of RTV Silicone Rubber Adhesive (46–170617P1) on top edge nearest the magnet.

INSTALLATION PROCEDURES SUMMARY

- ❑ **O1 – Position Front End Bell on Magnet**
- ❑ **O2 – Install Front End Bell on Magnet**
- ❑ **O3 – Secure Front End Bell to RF Coil**
- ❑ **O4 – Align front End Bell and Secure to Magnet**

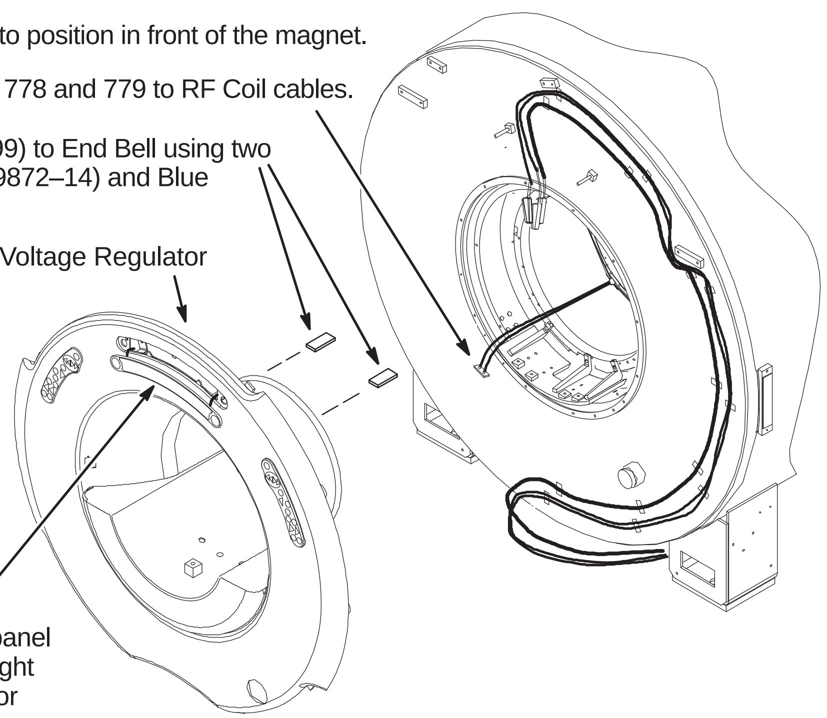
❑ O1 – POSITION FRONT END BELL ON MAGNET

(1) Carefully move the Front End Bell into position in front of the magnet.

(2) Connect existing Runs 778 and 779 to RF Coil cables.

(3) Attach Bore Light End Caps (2218899) to End Bell using two M4 x 10mm Pan Head Screws (2109872–14) and Blue Loctite in each End Cap.

(4) Connect existing Run 385 to JP1 on Voltage Regulator board located behind Display Panel.



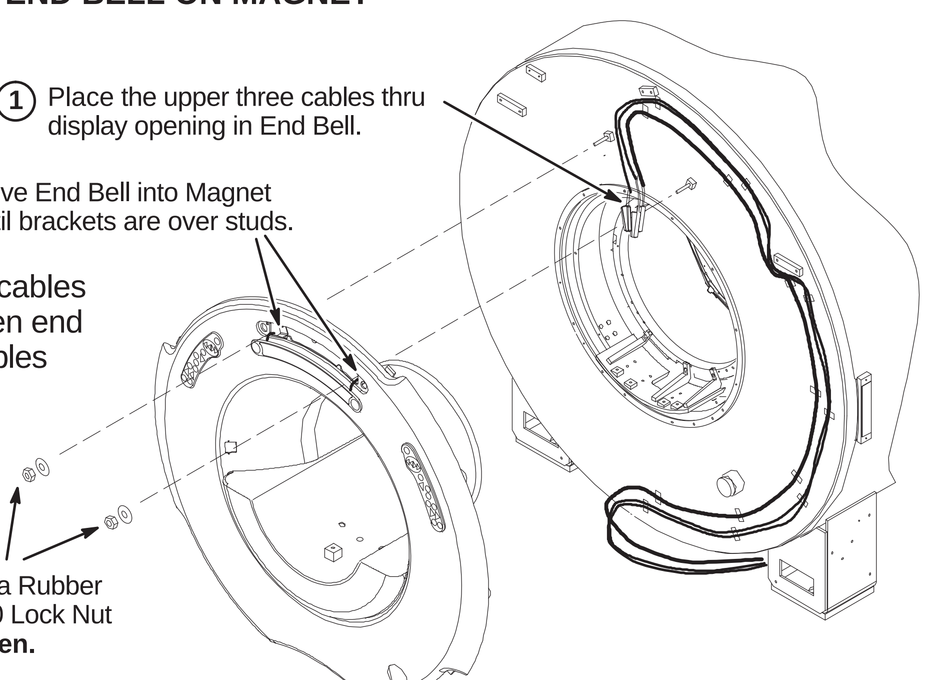
(5) Carefully pull Operator Display Panel away from mounting studs. Display panel is connected to cables from left and right Operator Control panels. Let Operator Display panel hang below opening.

❑ O2 – INSTALL FRONT END BELL ON MAGNET

(1) Place the upper three cables thru display opening in End Bell.

(2) Move End Bell into Magnet until brackets are over studs.

Note: Make sure display cables do not get pinched between end ring and End Bell. Pull cables thru display opening.



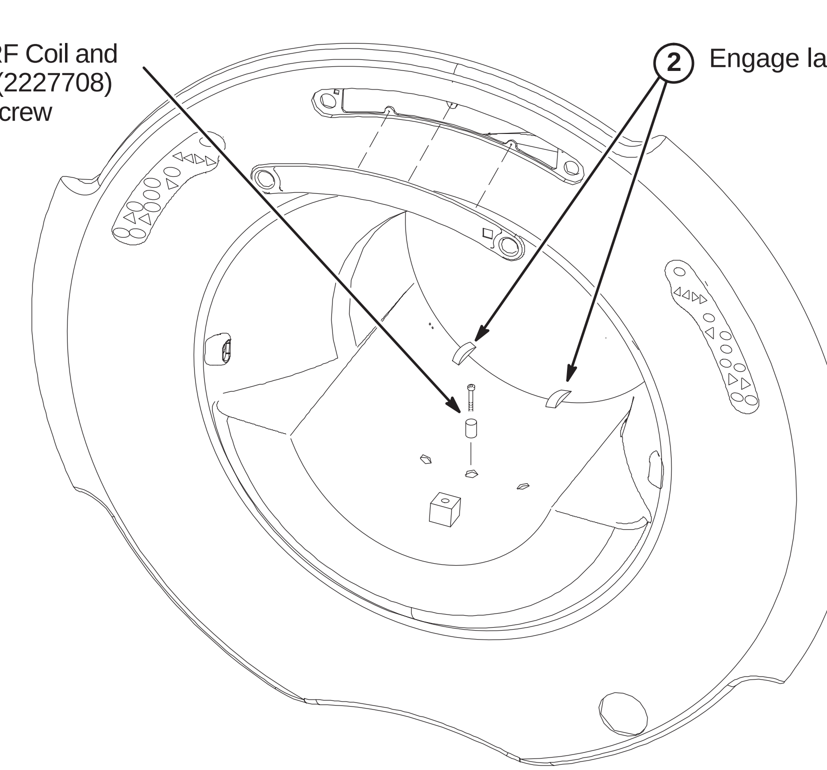
(6) Install loosely on each stud a Rubber Washer (2167321) and M10 Lock Nut (2109877–3). **Do not tighten.**

(7) Make sure microphone cable is routed towards bottom of Magnet.

❑ O3 – SECURE FRONT END BELL TO RF COIL

(1) Press the End Bell towards the RF Coil and install the RF Tube Location pin (2227708) using M6 x 30mm Hex Socket screw (2109867–16) and Blue Loctite.

(2) Engage latches.



❑ O4 – ALIGN FRONT END BELL AND SECURE TO MAGNET

(1) Connect the three cables as marked to J1, J2 and J3 on the back of the Operator Display.

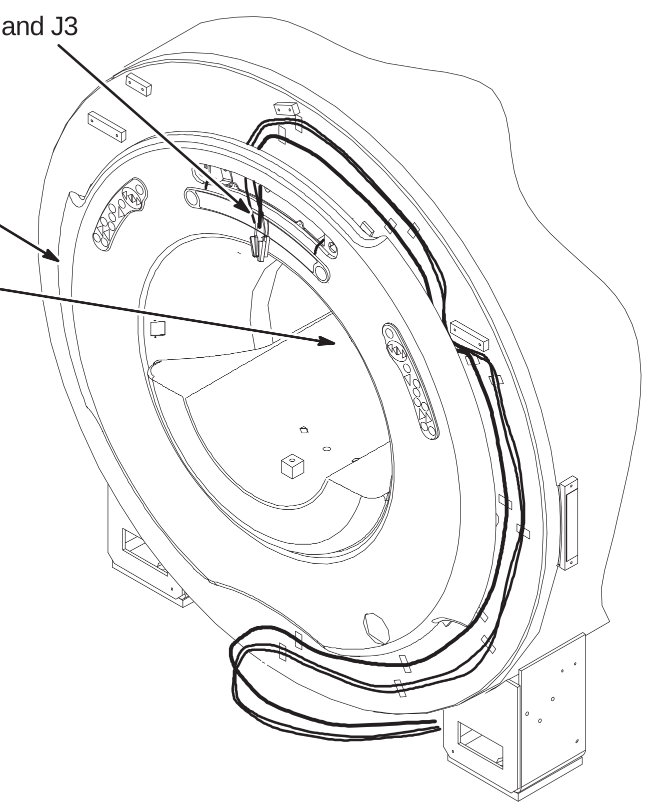
(2) Adjust M10 Lock Nuts so that upper section of End Bell is the same distance, ~43mm (1.7 IN.), from the magnet as the lower, left, and right.

(3) Make sure End Bell and RF Coil are engaged properly.

Note: The gap between the RF coil and the Front End Bell shall not exceed 0.125 inches (3.175mm)

(4) After gap meets specification above, tighten the inside M10 Nut and Rubber Washer against the End Bell bracket.

(5) Install the Operator Display Panel on the End Bell by gently pressing the display in the locations of the ball studs until it snaps in place.

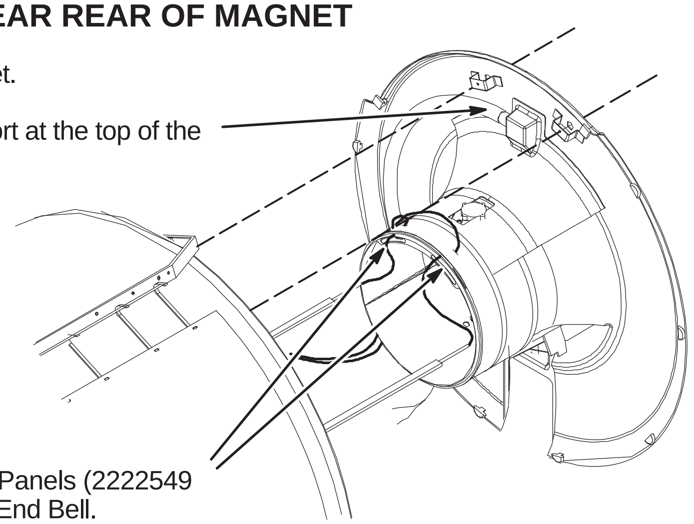


INSTALLATION PROCEDURES SUMMARY

- ❑ **P1 – Position Rear End Bell Near Rear of Magnet**
- ❑ **P2 – Route Bore Light Paenl in End Bell**
- ❑ **P3– Route Body Coil Cables Thru End Bell Opening**
- ❑ **P4 – Secure Rear End Bell to RF Coil and Magnet**
- ❑ **P5 – Install Bridge Support Spacers (MR/*i* Only)**
- ❑ **P6 – Install Bore Light Panels**

❑ P1 – POSITION REAR END BELL NEAR REAR OF MAGNET

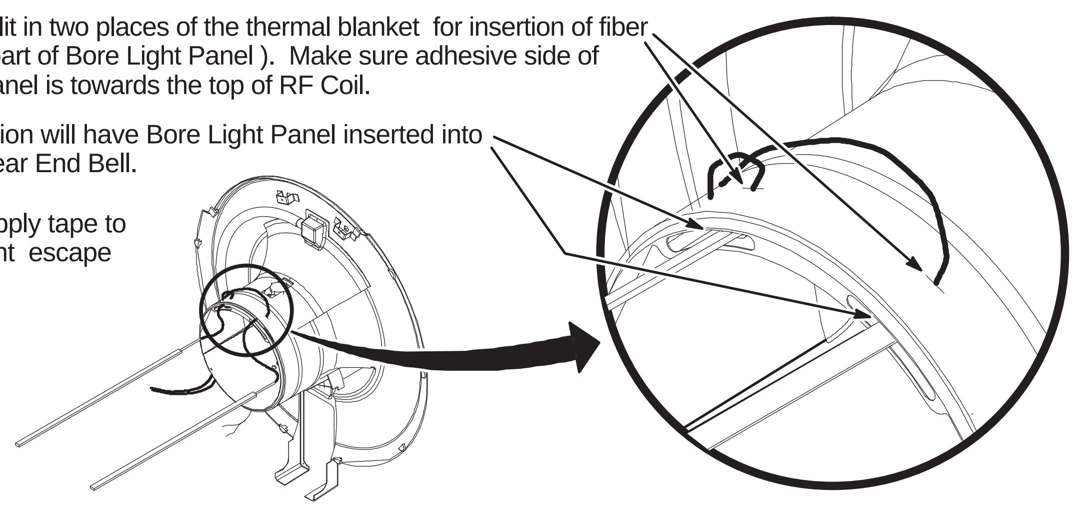
1. Position Rear End Bell near rear of Magnet.
2. Attach Air Hose (46–282668P18) to the port at the top of the End Bell using clamp (46–208765P28).



3. See Procedure **P2,** if Bore Light Panels (2222549 or 2222549–2) are not routed in End Bell.

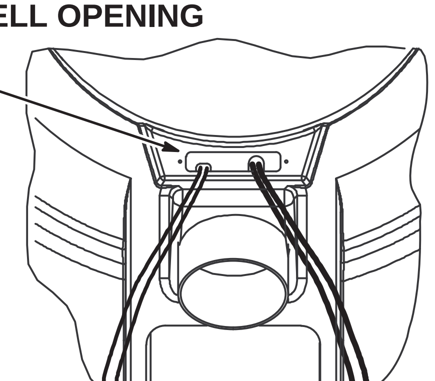
❑ P2 – ROUTE BORE LIGHT PANEL IN END BELL

1. Cut a small slit in two places of the thermal blanket for insertion of fiber optic cable (part of Bore Light Panel). Make sure adhesive side of Bore Light Panel is towards the top of RF Coil.
2. Final installation will have Bore Light Panel inserted into position in Rear End Bell.
3. If possible, apply tape to slits to prevent escape of air flow.



❑ P3 – ROUTE BODY COIL CABLES THRU END BELL OPENING

1. Insert Body Coil cables thru rectangular opening in the bottom of EndBell.
2. Continue to move End Bell into position while aligning hose, cables, and fiber optics.

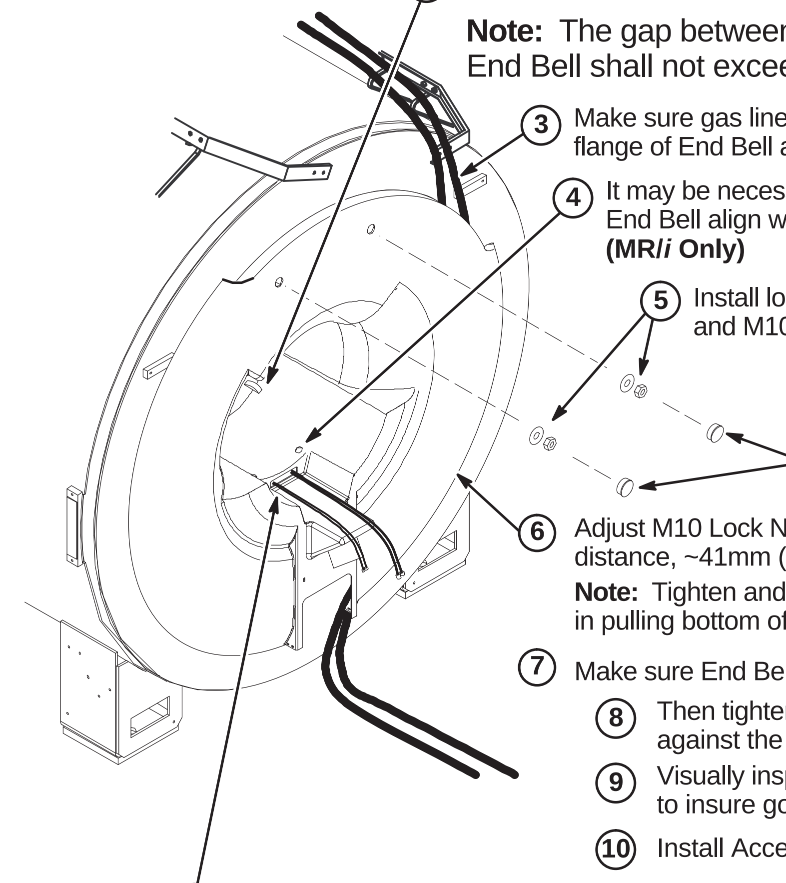


❑ P4 – SECURE REAR END BELL TO RF COIL AND MAGNET

1. Remove and save Access Hole Plugs
2. Position End Bell on two studs and against RF Coil. Egage two latches.

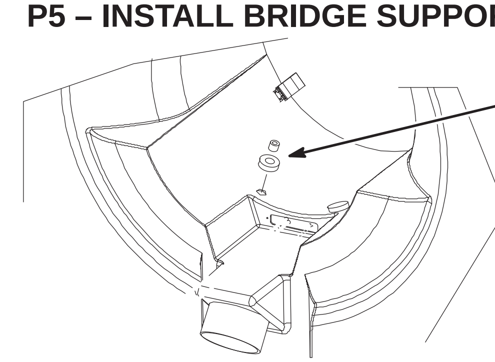
Note: The gap between the RF coil and the Rear End Bell shall not exceed 0.125 inches (3.175mm)

3. Make sure gas lines are routed over top left Frame Bracket behind flange of End Bell as shown.
4. It may be necessary to adjust the latches so that the holes in the End Bell align with the threaded holes in the Bridge Support. **(MR/*i* Only)**
5. Install loosely on each stud a Rubber Washer (2167321) and M10 Lock Nut (2109877–3). **Do not tighten.**



6. Adjust M10 Lock Nuts so that upper section of End Bell is the same distance, ~41mm (1.6 IN.), from the magnet as the lower section.
   Note: Tighten and adjust Nuts when arc covers are installed to aid in pulling bottom of End Bell using lower brackets.
7. Make sure End Bell and RF Coil are engaged properly.
8. Then tighten the inside M10 Nut and Rubber Washer against the End Bell bracket.
9. Visually inspect the fit between the foam seal and End Bell to insure good contact.
10. Install Access Hole Plugs.
11. Install feed thru cover (2222329) and gasket over the Body Coil cables using M4 x 10mm Pan Head screws (2109873–14) and Blue Loctite.

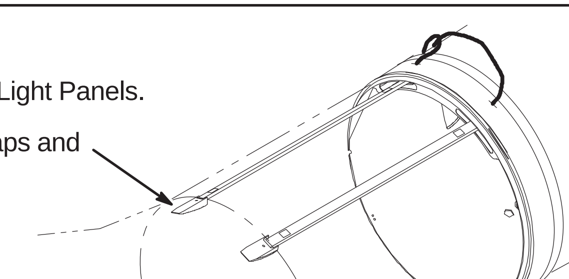
❑ P5 – INSTALL BRIDGE SUPPORT SPACERS (MR/*i* Only)



1. Install four Bridge Support Spacers (2222353) and four Foam Gaskets (2226752) over the two holes at the bottom of each End Bell.

❑ P6 – INSTALL BORE LIGHT PANELS

1. Remove the adhesive protector from the Bore Light Panels.
2. Insert Bore Light Panels into Bore Light End Caps and tighten screws.
3. Press Bore Light Panels into place.



INSTALLATION PROCEDURES SUMMARY

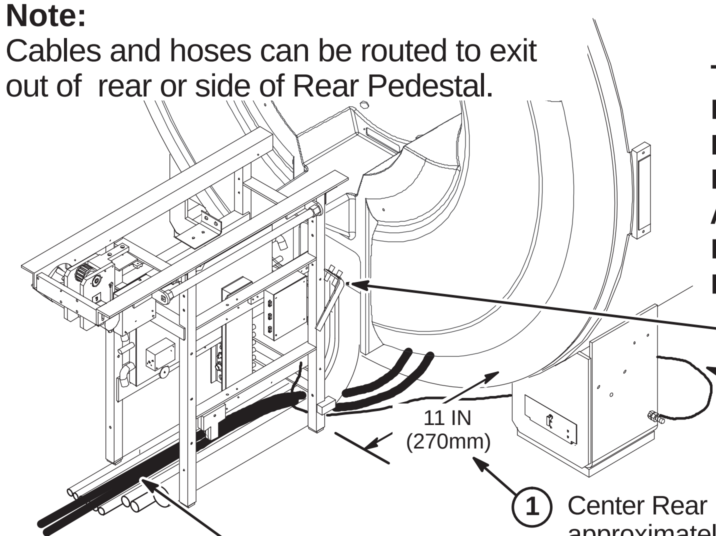
- ❑ Q1 – Location of Rear Pedestal
- ❑ Q2 – Route/Connect Runs 258/747, 318, 700, 781, 782, Bias and Drive Cables
- ❑ Q3 – Route/Connect Runs 257/746, 262, 743, 744, and 780
- ❑ Q4 – Route/Connect Runs 282, 414, 422, and Froont End Bell Microphone & Speaker Cables

❑ Q1 – LOCATION OF REAR PEDESTAL

Note:
Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

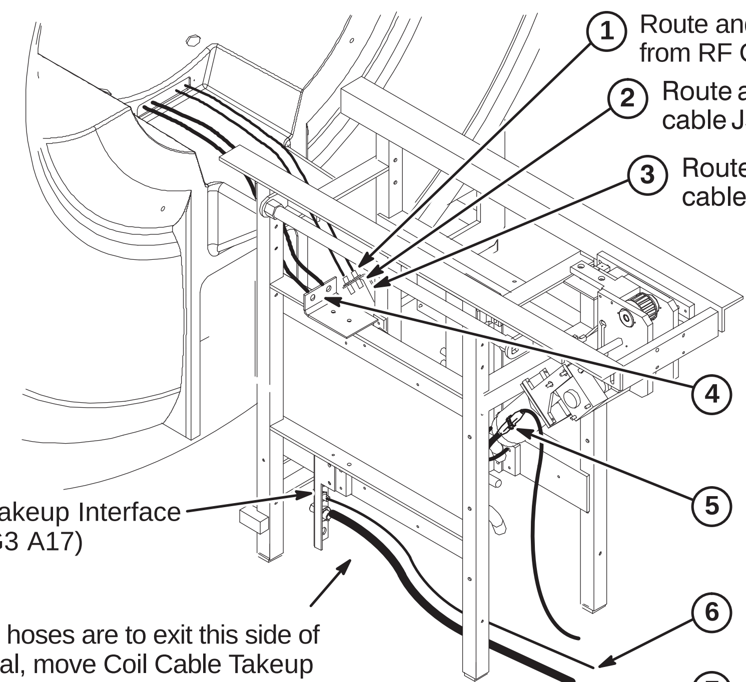
WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AMD MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.



5. For **CV/*i*** install on other side.
4. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
1. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
2. Route gradient cables, water lines, gas lines, and 2" air hose under Rear Pedestal.
3. Route all other cables on top of the above cables and hoses.

❑ Q2 – ROUTE/CONNECT RUNS 258/747, 318, 700, 781, 782, BIAS & DRIVE CABLES



1. Route and connect RF Bias cables from RF Coil to bracket as shown.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
5. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
6. Route/connect Run 700 to J2 on Coil Cable Takeup Interface.
7. Route/connect to J1 on Coil Cable Takeup Interface:
   Run 258 (1.5T, 1/2") or
   Run 747 (1.0T, 3/8")

Coil Cable Takeup Interface
(MG3 A17)

Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

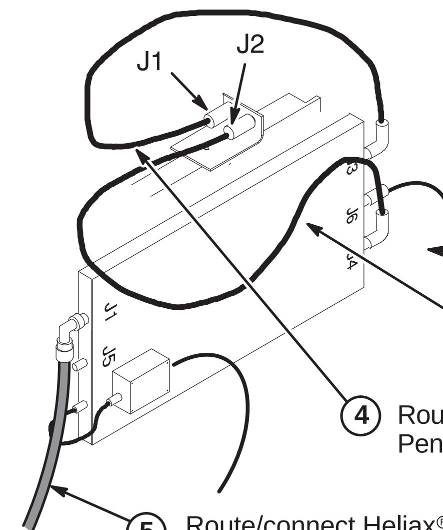
❑ Q3 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

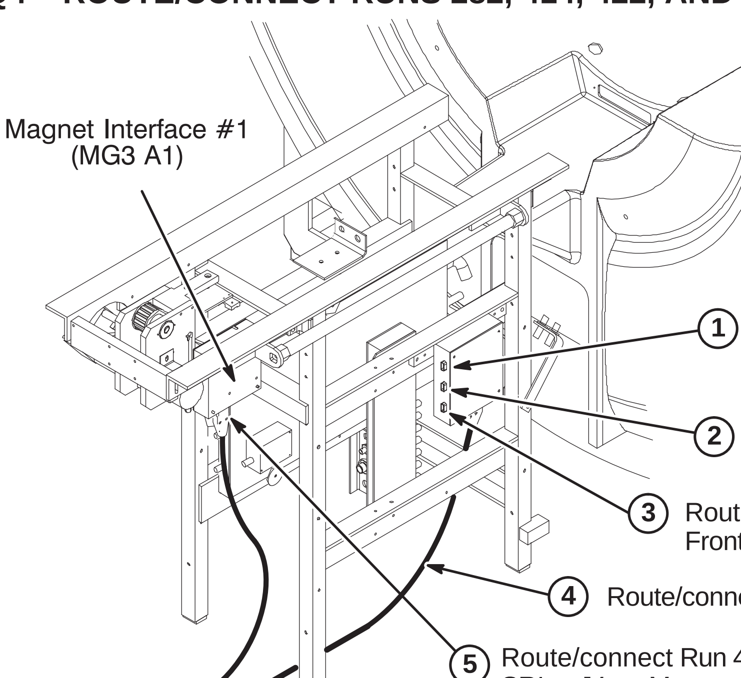
Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.



1. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
2. Route and connect Run 744 from J1 on bracket to J4 on the RF Splitter.
3. Route and connect Run 743 from J2 on the bracket to J3 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier.
5. Route/connect Heliax® to J1 on RF Splitter:
   Run 257 (1.5T, 7/8") or
   Run 746 (1.0T, 1/2")

❑ Q4 – ROUTE/CONNECT RUNS 282, 414, 422, AND FRONT END BELL CABLES

Magnet Interface #1
(MG3 A1)



1. **Do not connect the Microphone cable from Rear End Bell to J7.**
2. Route/connect Front End Bell microphone cable Run 414 to J9.
3. Route/connect speaker cable from Front End Bell to J8.
4. Route/connect Run 282 from J12 on Pen Panel to J6.
5. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

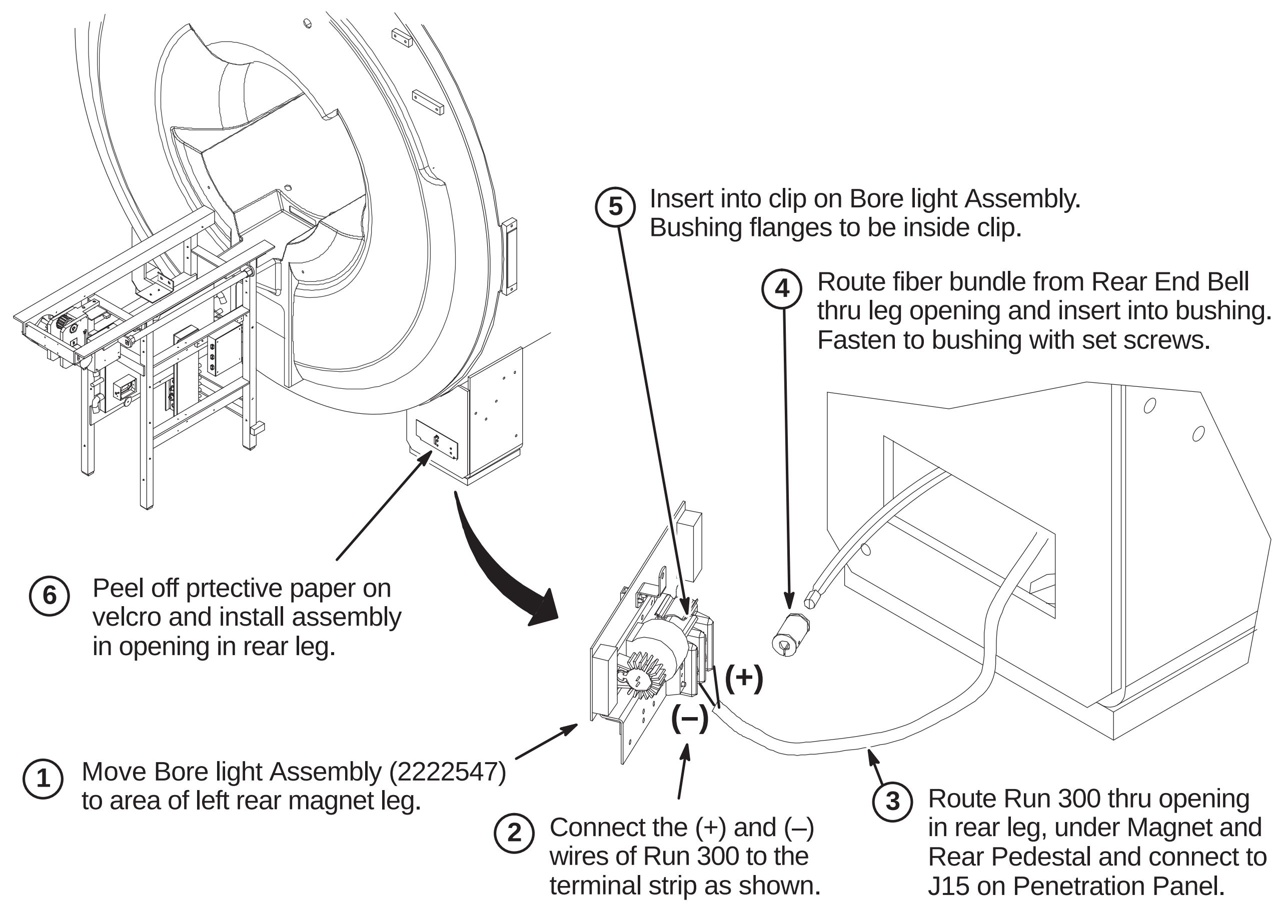
INSTALLATION PROCEDURES SUMMARY

- ❑ **R1 – Install Bore Lamp Assembly**
- ❑ **R2 – Install Arc Cover**
- ❑ **R3 – Fasten Top of Arc Cover**

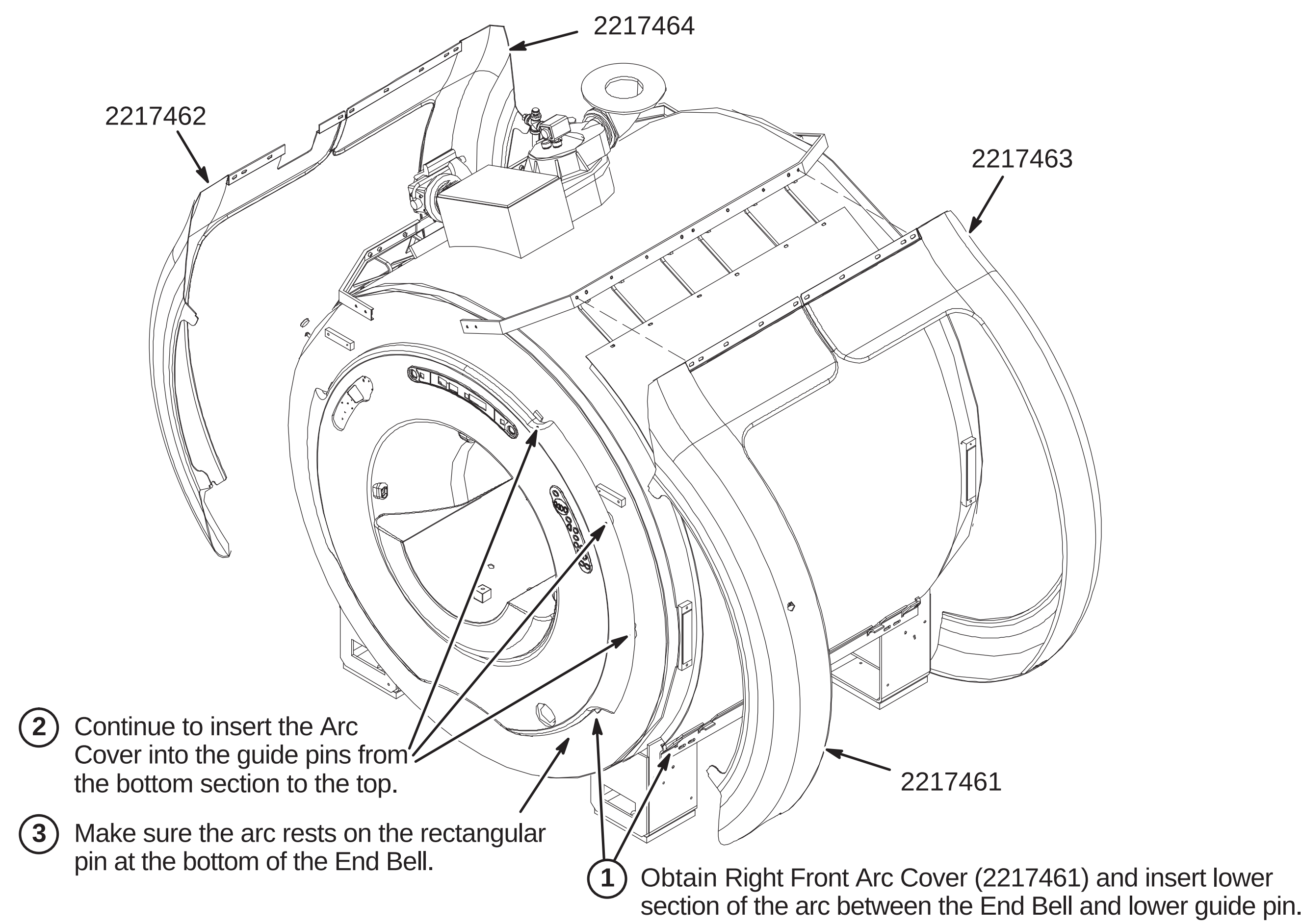
❑ R1 – INSTALL BORE LAMP ASSEMBLY

Note: Cables and hoses routed in Procedure below are not shown for clarity.

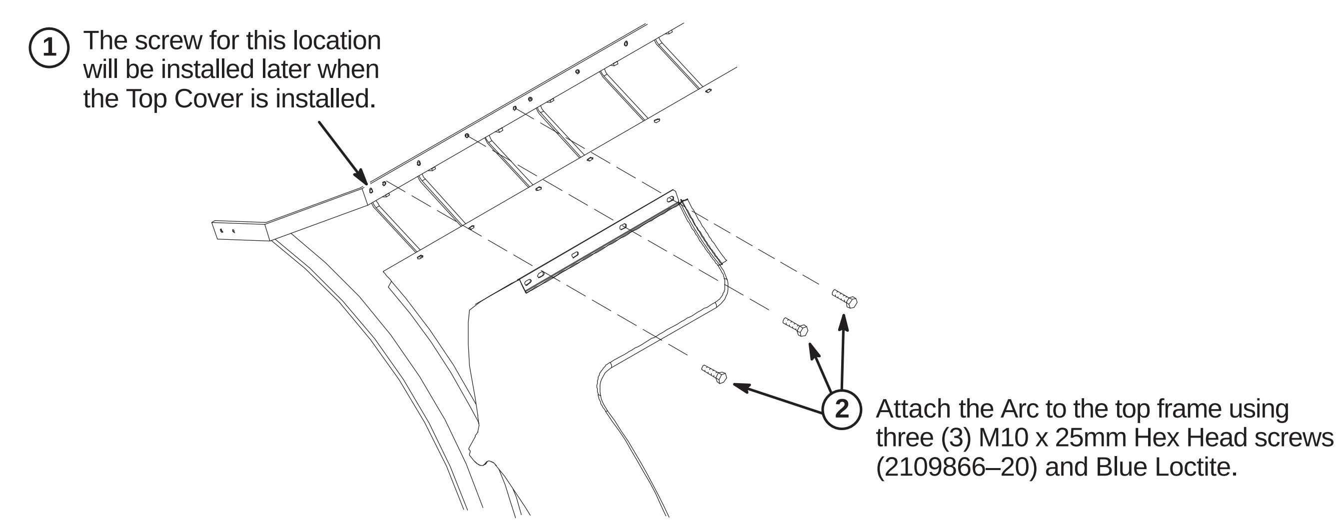
Note: Existing Run 300 can be used in place of Run 300 that is supplied with new Bore Lamp Assembly.



❑ R2 – INSTALL ARC COVER



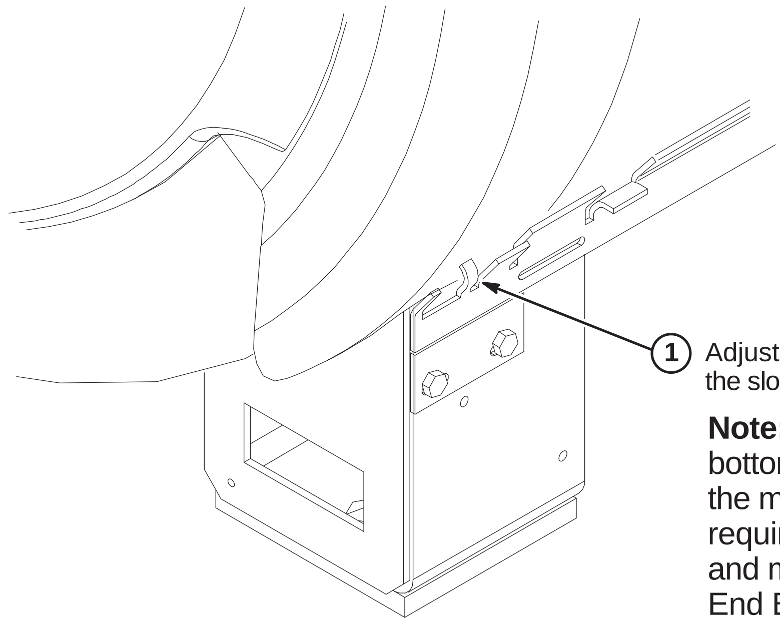
❑ R3 – FASTEN TOP OF ARC COVER



INSTALLATION PROCEDURES SUMMARY

- ❑ S1 – Adjust Bottom of Arc Covers
- ❑ S2 – Complete Arc Cover Installation
- ❑ S3 – Connect Air Hoses to Rear of Magnet

❑ S1 – ADJUST BOTTOM OF ARC COVERS

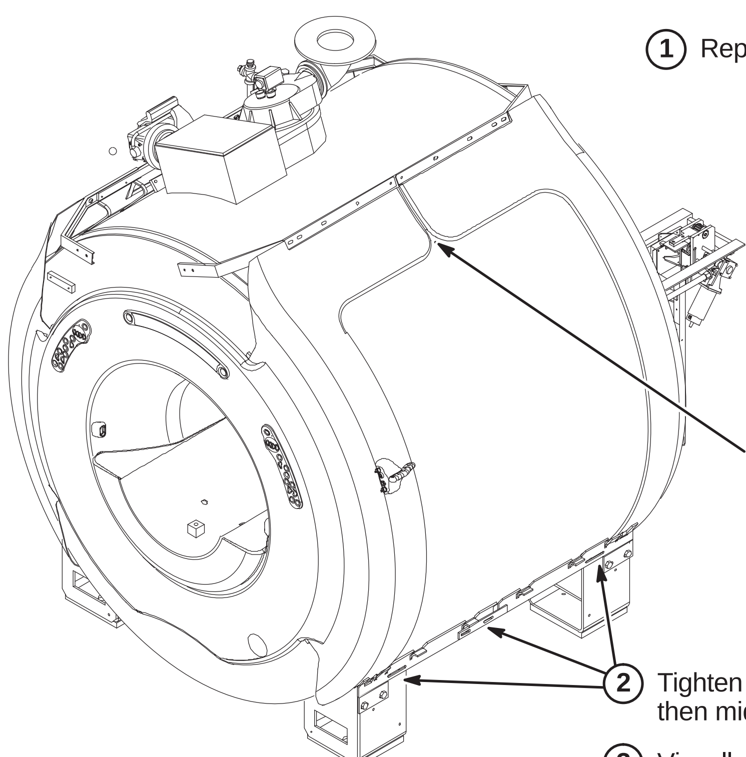


1 Adjust the bottom rails so that the finger fits into the slot in the bottom of the Arc Cover.

Note: The above process will help pull bottom of Rear End Bell flange closer to the magnet. Adjust Rear End Bell as required to keep distance between flange and magnet equal at top and bottom of End Bell.

❑ S2 – COMPLETE ARC COVER INSTALLATION

1 Repeat process for the other three Arc Covers.



Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). Exceptions can be made where esthetic visual enhancement of the component gap is required by industrial design.

Gap uniformity along any continuos seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)

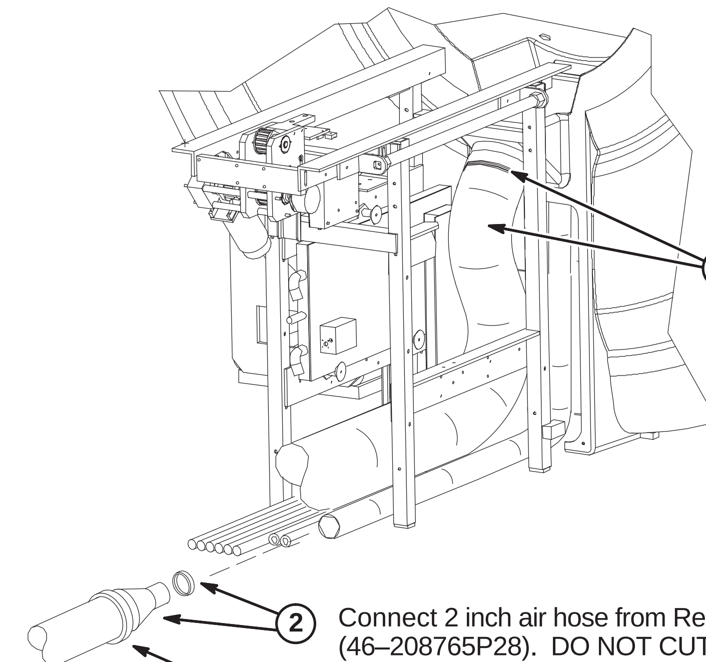
Note: This gap is not included in specification above.

2 Tighten screws in bottom rails. First front rail, then middle connector, then rear of rail.

3 Visually inspect the gap between Arc Cover and End Bell, then the Arc Cover and side Wraps. Adjust as necessary.

❑ S3 – CONNECT AIR HOSES TO REAR OF MAGNET

Note:
The following items are part of Blower Hose Kit, 2129412 – 3.



1 Connect 6 inch air hose (46–282666P11) to Rear End Bell with 6 inch clamp (46–208675P29).

2 Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). DO NOT CUT OR REDUCE THE LENGTH OF THE 2” HOSE.

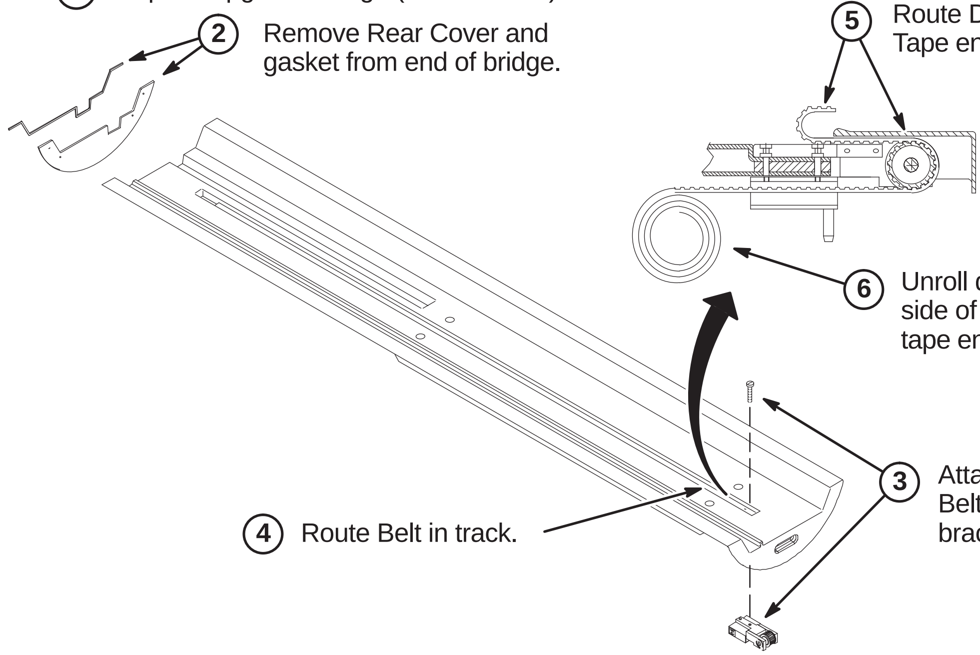
3 Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). IF NECESSARY, 4” HOSE MAY NEED TO BE CUT TO REDUCE LENGTH.

INSTALLATION PROCEDURES SUMMARY

- ❑ **T1 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **T2 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **T3 – Install Bridge**
- ❑ **T4 – Reattach Bridge Mount Assembly**
- ❑ **T5 – Secure Bridge to BRM Saddle Bracket (MR/*i* ONLY)**

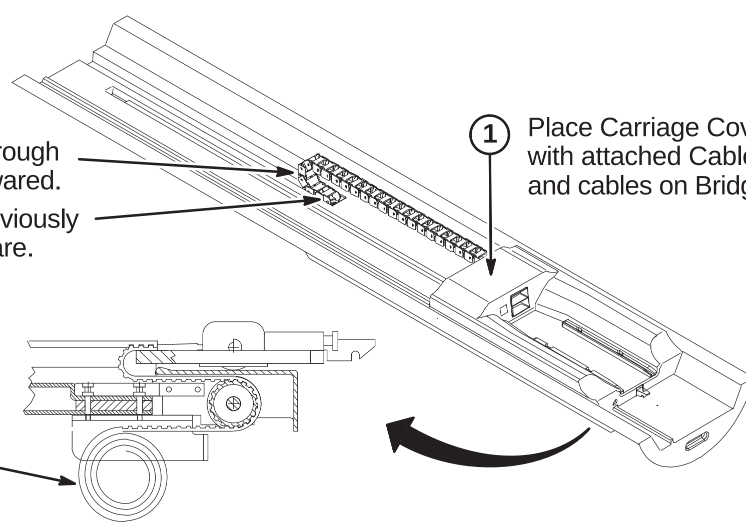
❑ T1 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET

1. Unpack upgrade Bridge (2205100–13).
2. Remove Rear Cover and gasket from end of bridge.
3. Attach previously removed Drive Belt Idler Pulley and Front Support bracket to front of Bridge.
4. Route Belt in track.
5. Route Drive Belt around Idler Pulley. Tape end to top of Bridge.
6. Unroll drive belt along bottom side of Brdige and temporarily tape end of belt to rear of bridge.



❑ T2 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET

1. Place Carriage Cover with attached Cabletrak and cables on Bridge.
2. Feed cables from end of Cabletrak through hole in Bridge and move carriage forwared.
3. Fasten CableTrak to Bridge using previously removed clamp and fastening hardware.
4. If not already done, unroll drive belt along bottom side of Brdige and temporarilyy tape end of belt to rear of bridge.

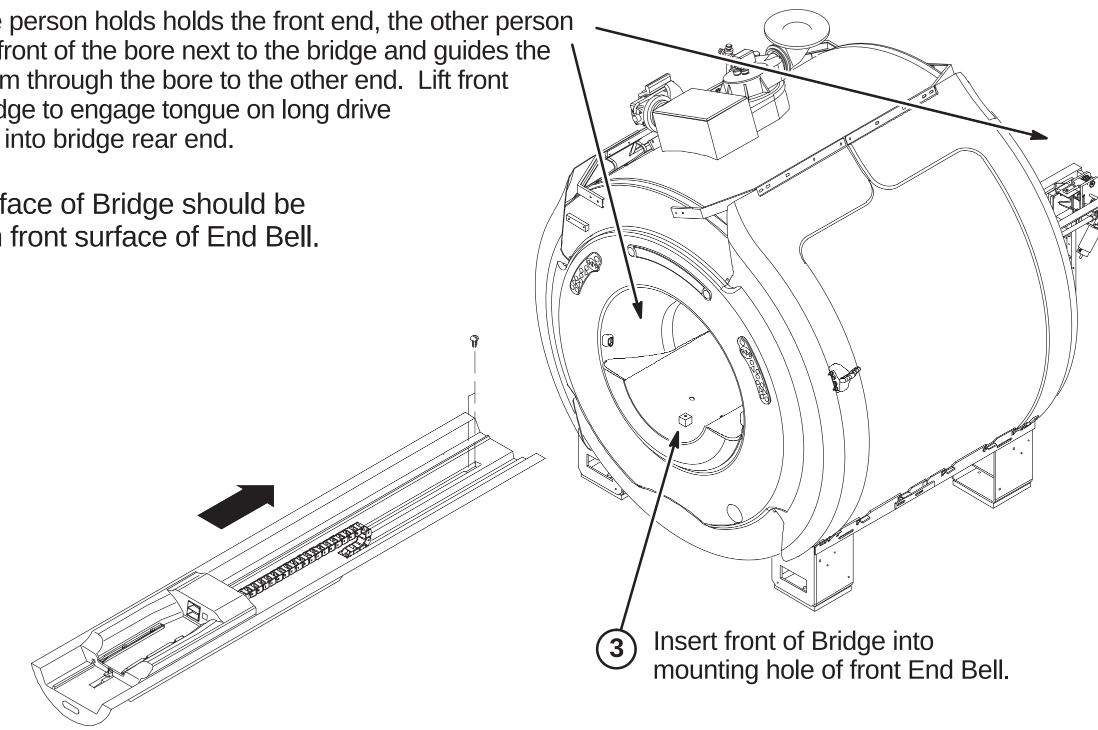


❑ T3 – INSTALL BRIDGE

1. Position rear end of Bridge Asm at front of Body Coil.
2. While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

NOTE:
Front surface of Bridge should be flush with front surface of End Bell.

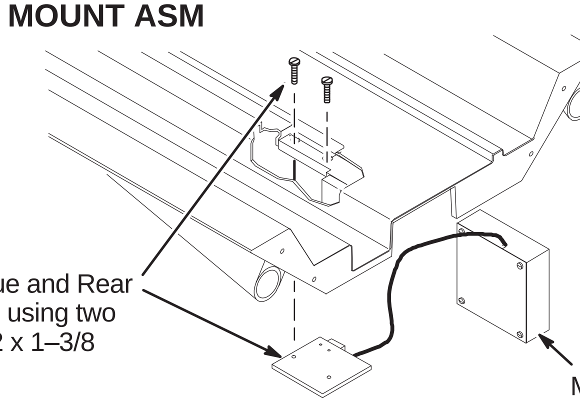
3. Insert front of Bridge into mounting hole of front End Bell.



❑ T4 – REATTACH BRIDGE MOUNT ASM

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

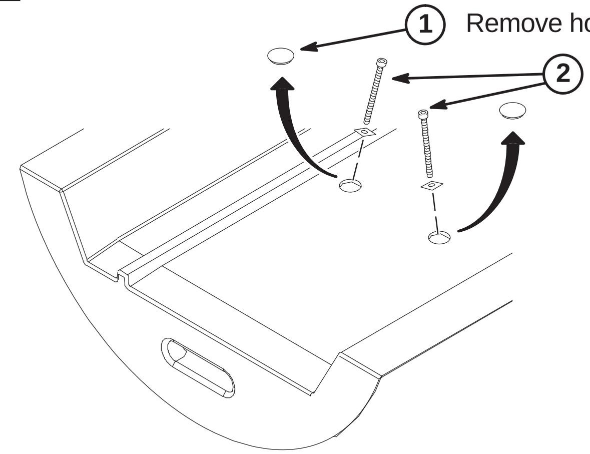
1. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) previously removed 10–32 x 1–3/8 Screws (46–220184P47)



❑ T5 – SECURE BRIDGE TO BRM SADDLE BRACKET (MR/*i* ONLY)

1. Remove hole plugs from Bridge Assembly. Reinstall after bridge is secure.
2. Apply anti–seize compund (2119594) to install four (4) M6 x 90mm Hex Socket screws (2109867–62) and square Bridge Support Washers (2228717). Two thru Front End Bell and two thru Rear End Bell.

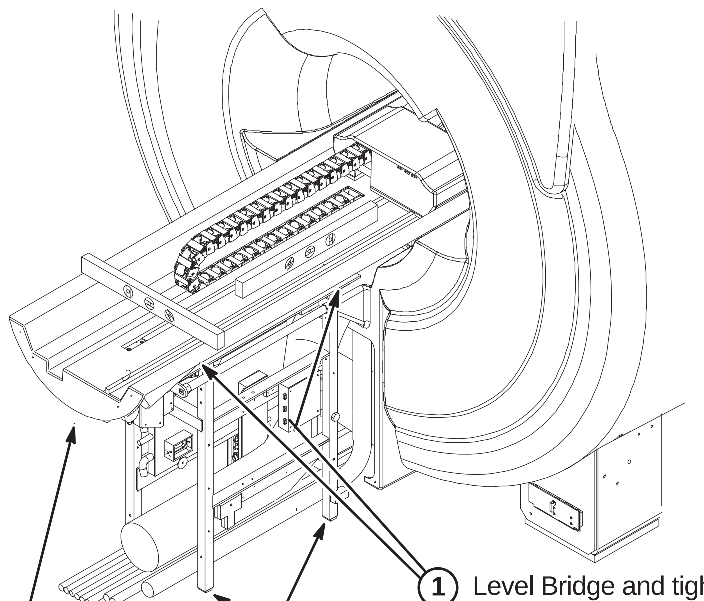
NOTE:
CV/*i* Enclosures does not require the installation of the above screws.



INSTALLATION PROCEDURES SUMMARY

- ❑ U1 – Leveling Bridge
- ❑ U2 – Obtaining Bridge Locators
- ❑ U3 – Aligning Bridge on Rear Pedestal
- ❑ U4 – Installing Bridge Locators
- ❑ U5 – Carriage Cover Removal for Belt Routing

❑ U1 – LEVELING BRIDGE

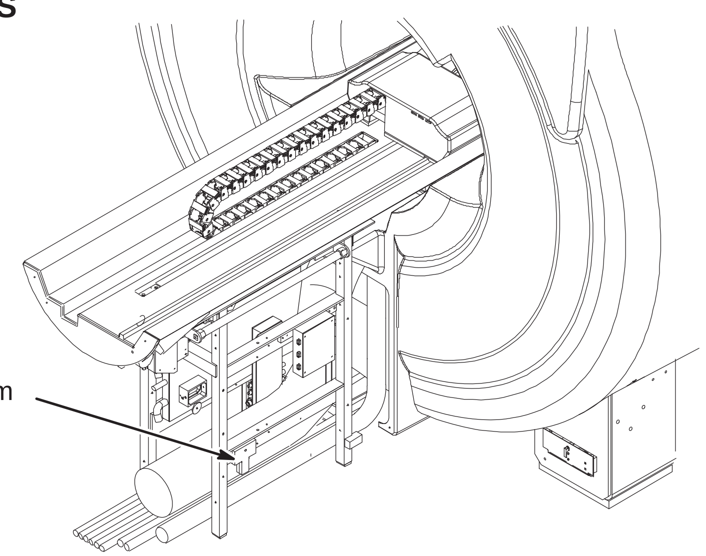


NOTE:
PVC Spacers positioned thru End Bells are to support Bridge. Rear Pedestal to be adjusted to meet bottom of Bridge.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

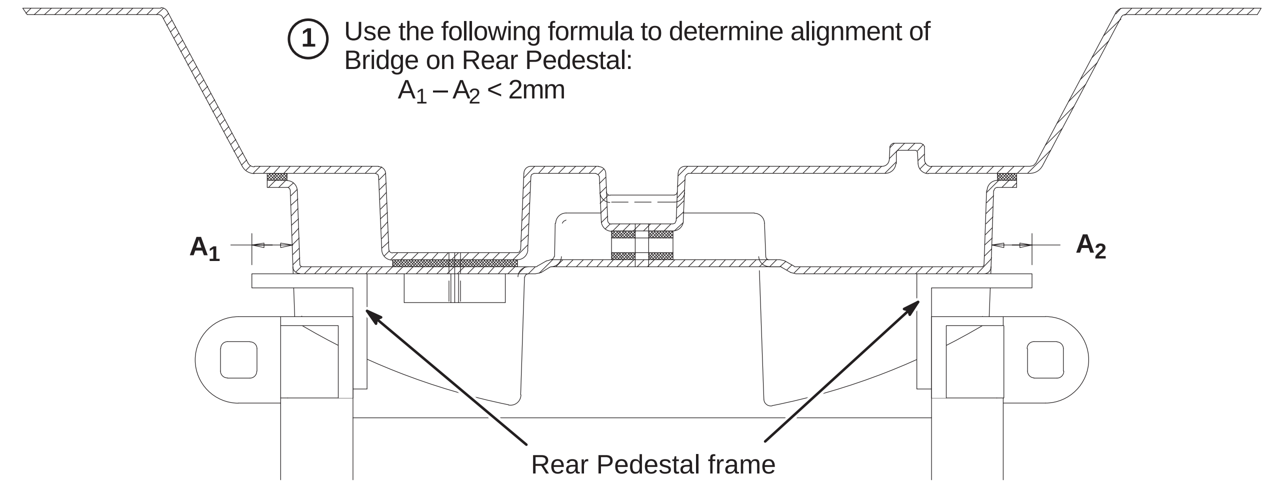
❑ U2 – OBTAINING BRIDGE LOCATORS



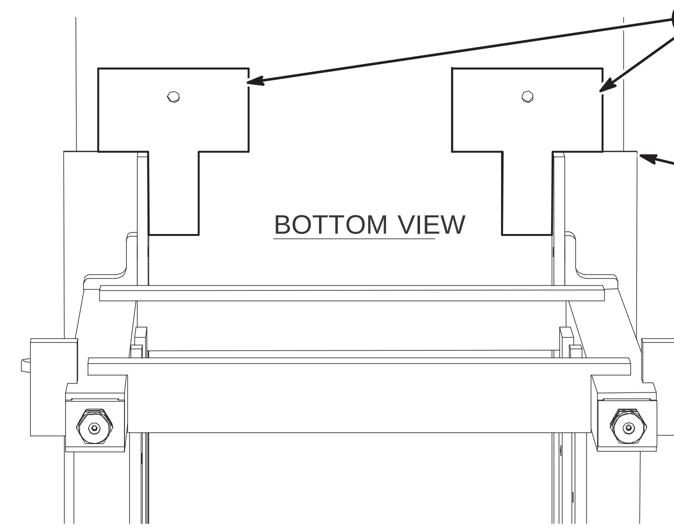
1. Remove two Bridge Locators (2235607) from shipping location on Rear Pedestal.

❑ U3 – ALIGNING BRIDGE ON REAR PEDESTAL

1. Use the following formula to determine alignment of Bridge on Rear Pedestal:
   $A_1 - A_2 < 2mm$



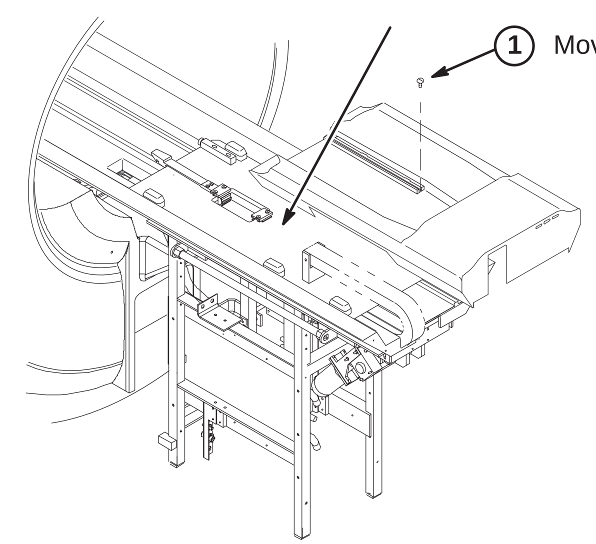
❑ U4 – INSTALLING BRIDGE LOCATORS



1. After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

End of Rear Pedestal frame closest to Magnet.

❑ U5 – CARRIAGE COVER REMOVAL FOR BELT ROUTING

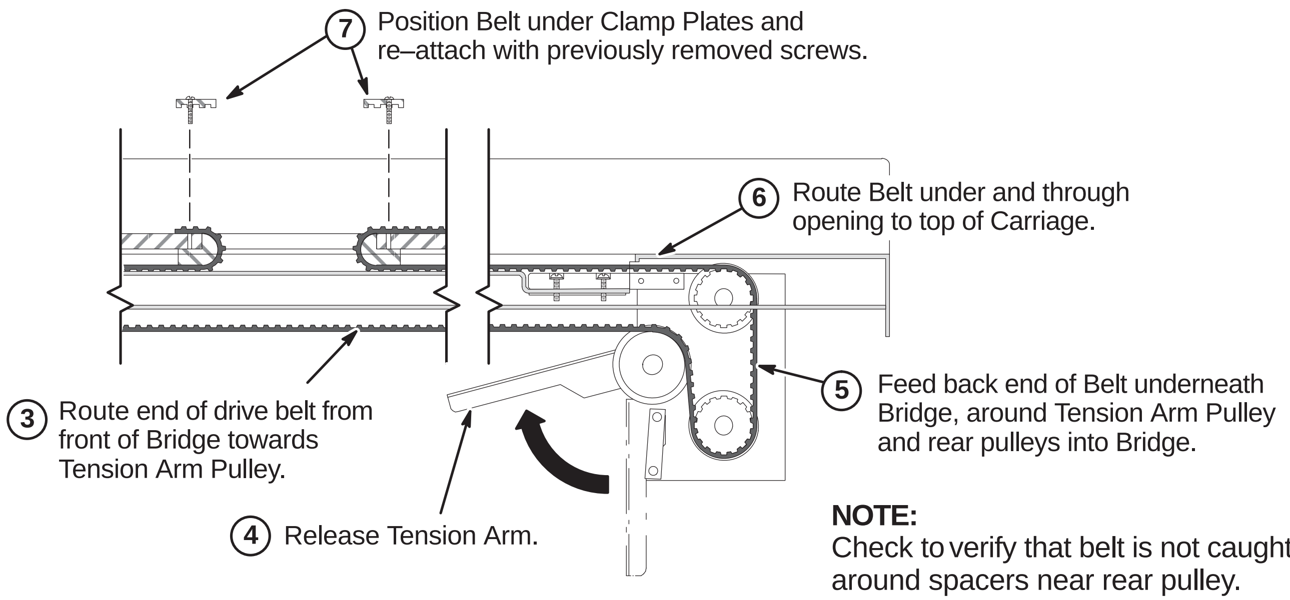


1. Move Carriage to limit of travel at rear of Bridge.
2. To access Belt Clamp Plate, unfasten four screws holding Carriage Cover to Carriage. Leave cables attached.

INSTALLATION PROCEDURES SUMMARY

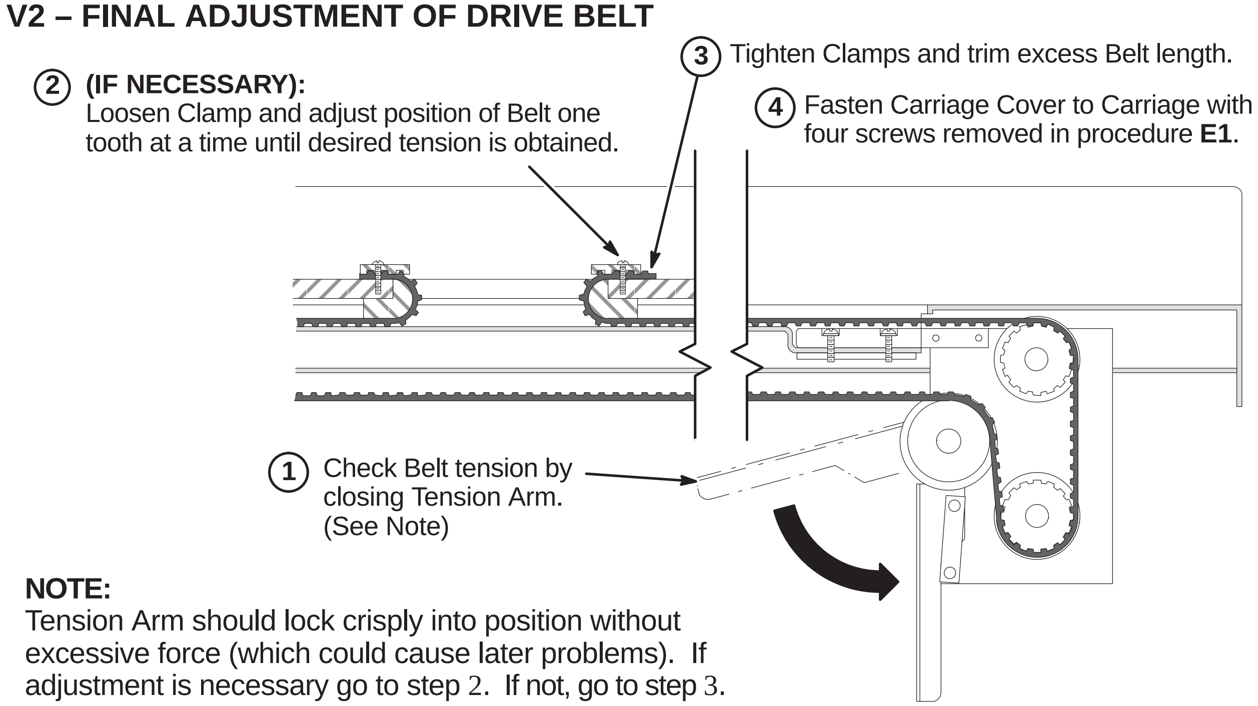
- ❑ **V1 – Routing Drive Belt**
- ❑ **V2 – Final Adjustment of Drive Belt**
- ❑ **V3 – Installation of Sensor Flag**
- ❑ **V4 – Attach Bridge Cover**
- ❑ **V5 – Route/Connect Runs 404 and 485 (and Multi–coil Cables if Required)**

❑ V1 – ROUTING OF DRIVE BELT



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

❑ V2 – FINAL ADJUSTMENT OF DRIVE BELT



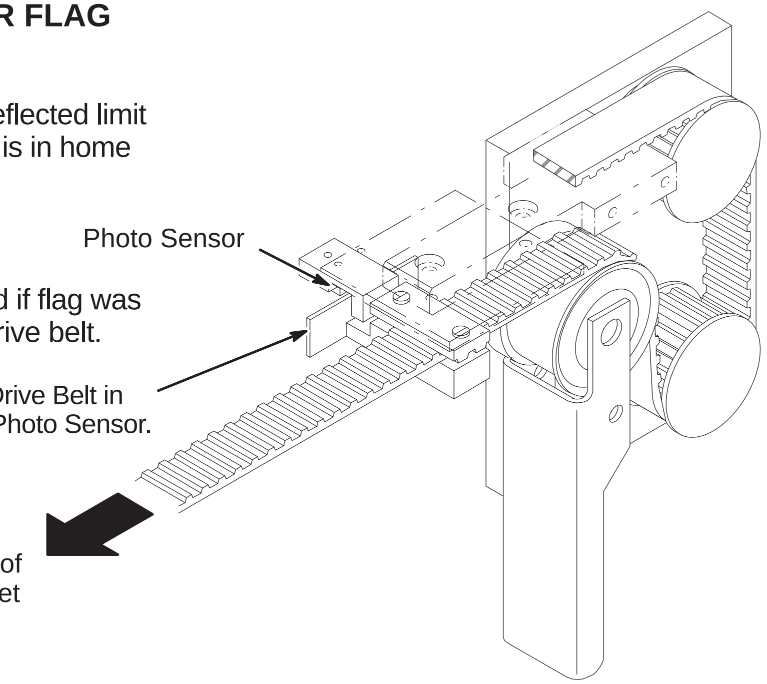
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ V3 – INSTALLATION OF SENSOR FLAG

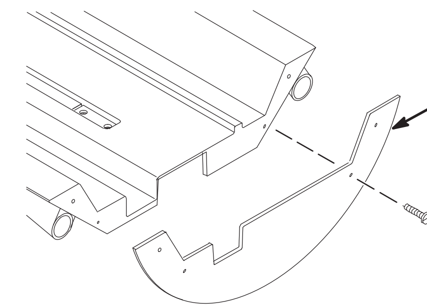
NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1 Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



❑ V4 – ATTACH BRIDGE COVER

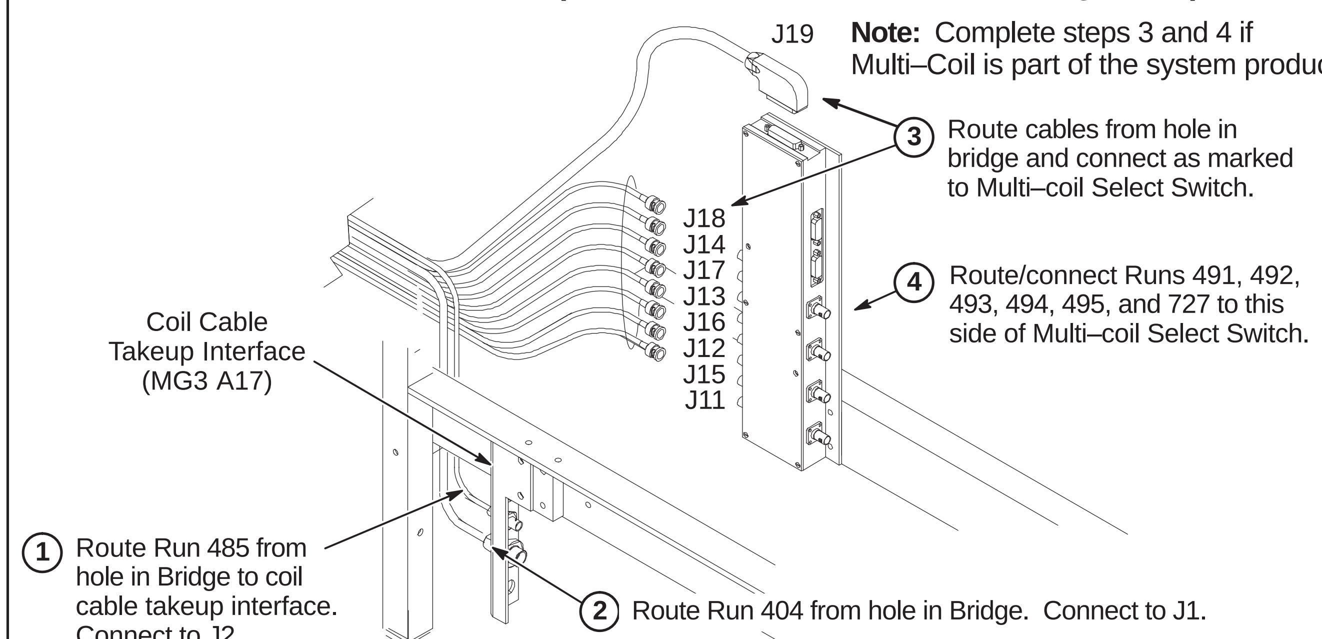


1 Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

NOTE:
Bridge Cover has enclosure rating plates on it.

❑ V5 – ROUTE RUNS 404 AND 485 (AND MULTI–COIL CABLES IF REQUIRED)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



INSTALLATION PROCEDURES SUMMARY

- ❑ **W1 – Prepare Cables and Tubes for Routing**
- ❑ **W2 – Position Remote PAC Interface Mounting Clip**
- ❑ **W3 – Route Remote PAC Interface Cables and Tubes**
- ❑ **W4 – Installing Remote PAC Interface to Arc Cover**
- ❑ **W5 – Install PAC Hanger Bar**

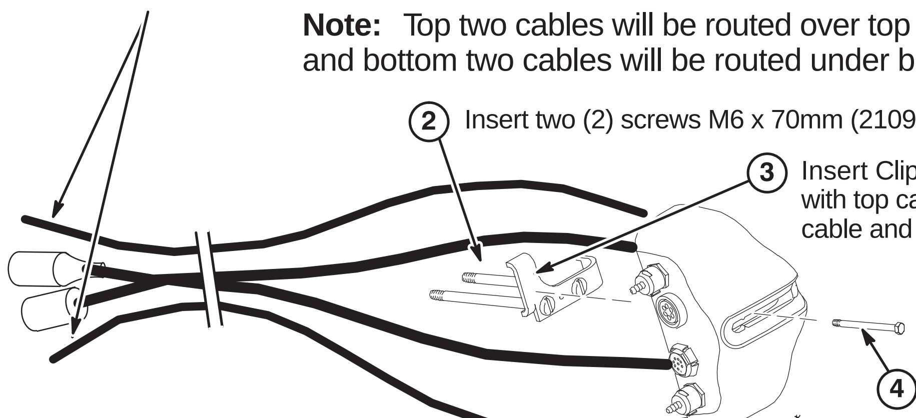
❑ W1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

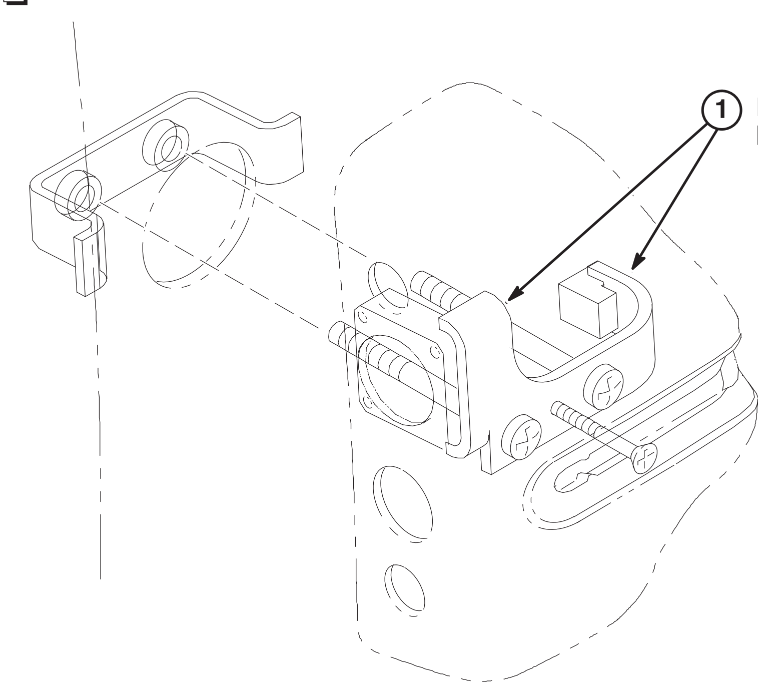
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.



❑ W2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP

1. Make sure each side of clip is mounted behind housing blocks.

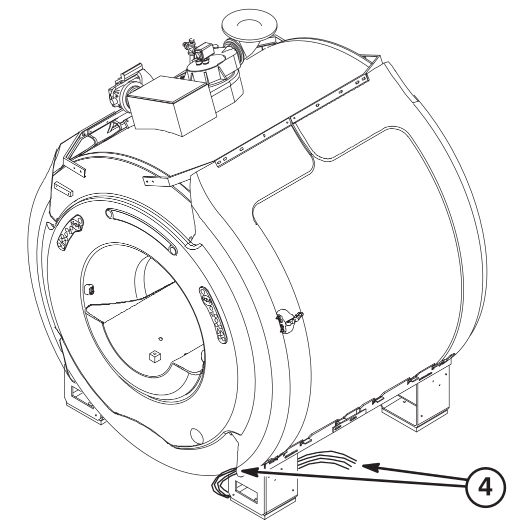


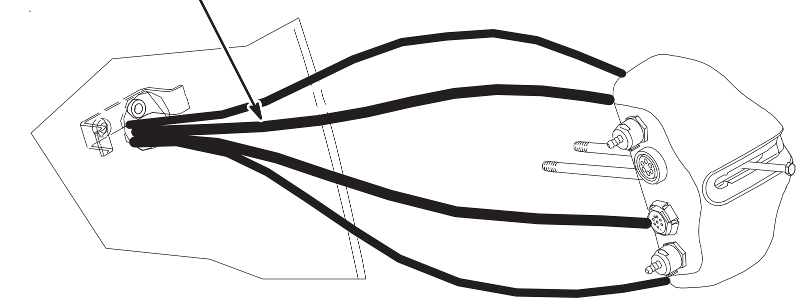
❑ W3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES

1. Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

Note:
Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

2. Insert ECG Cable first. Route this thru pre–mounted foam tube (if available) located on inside of Arc cover.
3. Route Peripheral Pulse cable and tubes next.

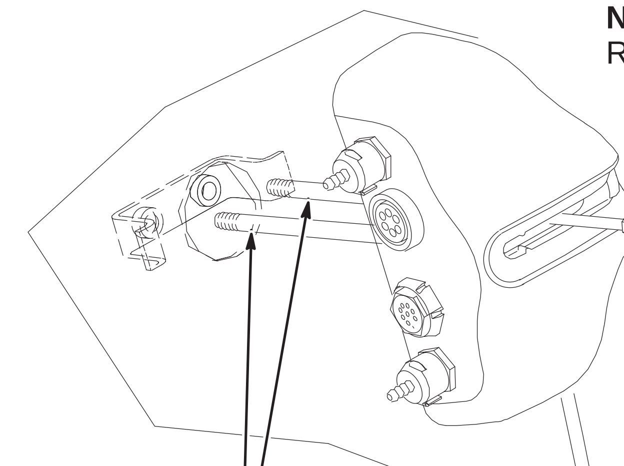




4. Route around foot of magnet to PAC/SRI Platform location.

❑ W4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

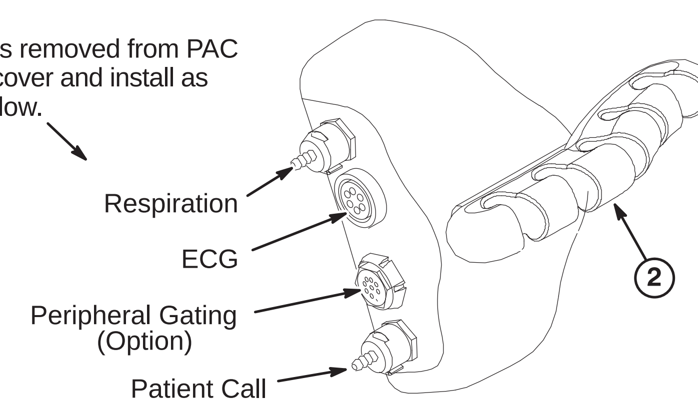
Note:
Routed cables and tubes not shown for clarity.



1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
   OR
   Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.

❑ W5 – INSTALL PAC HANGER BAR

1. Attach cables removed from PAC on old front cover and install as indicated below.

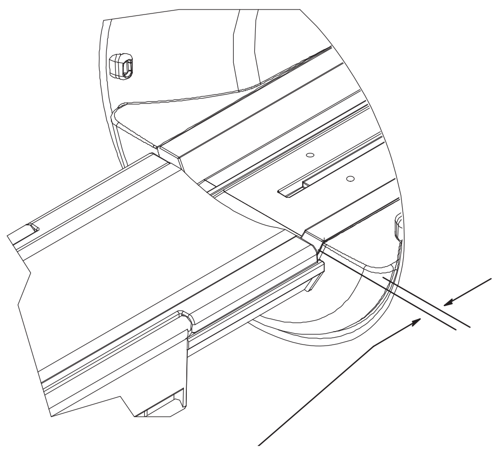


2. Slide PAC Hanger Bar (2221023) into position.

INSTALLATION PROCEDURES SUMMARY

- ❑ X1 – Install Dock Mounting Bracket and Dock
- ❑ X2 – Route and Connect Magnet Enclosure Front Area Cables
- ❑ X3 – Location of Magnet Ground Cable

❑ X1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



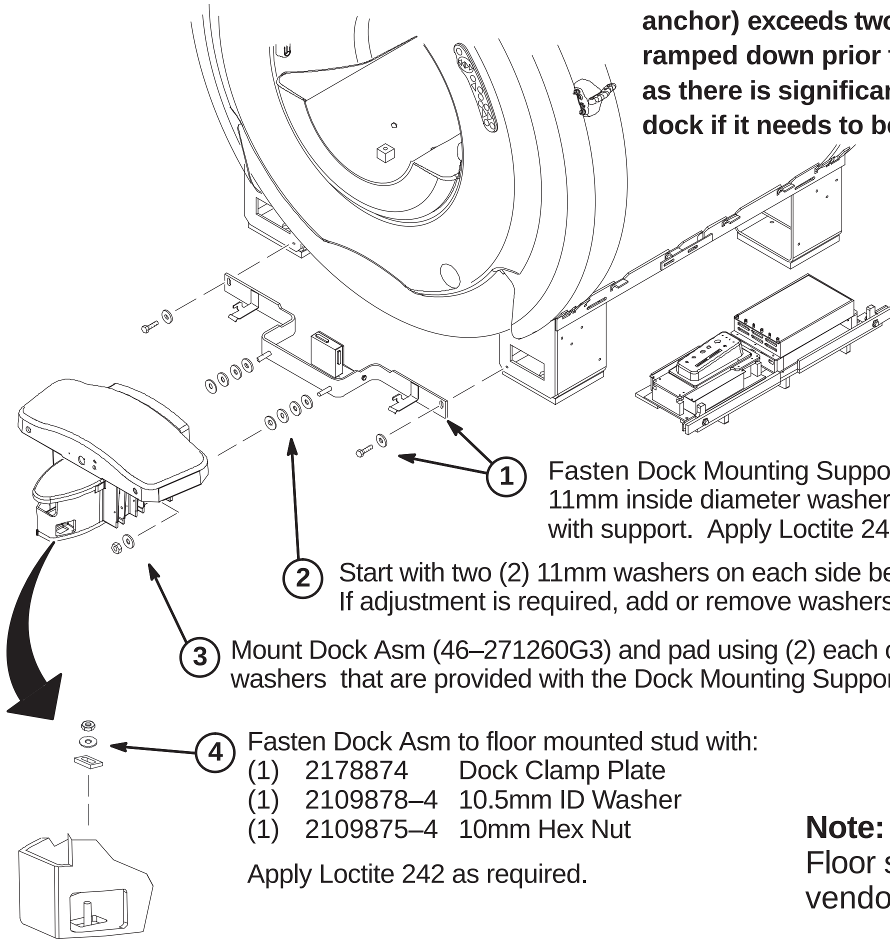
WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY

Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches $\pm$ 0.25 (19.0$\pm$6.4mm). See Step 2 below.

WARNING!

DO NOT LIFT THE DOCK MORE THAN 2 INCHES (5 CM) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installling the dock as there is significant risk of the magnet attracting the dock if it needs to be raised more that 2 inches (5 cm).



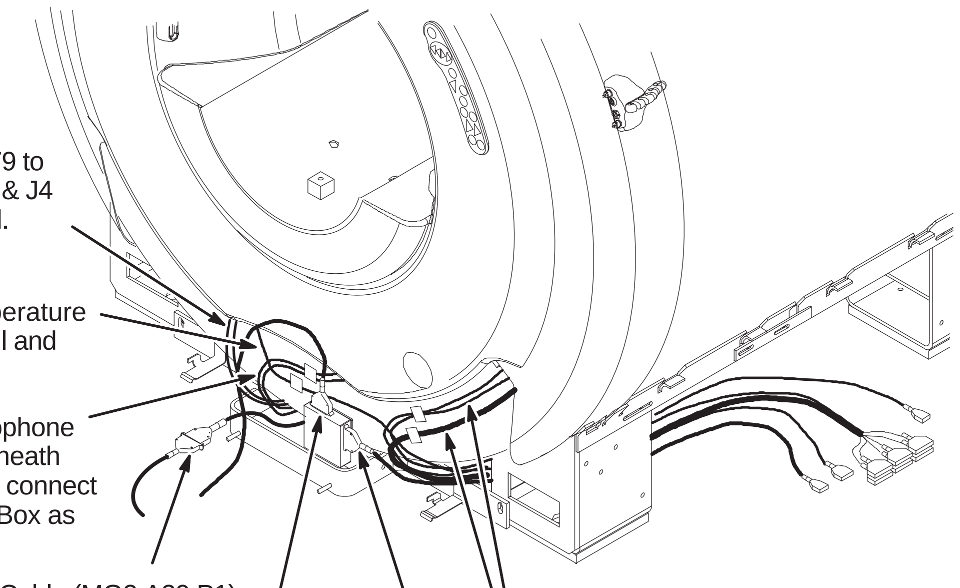
1. Fasten Dock Mounting Support (2226264) to magnet using 11mm inside diameter washers and Hex Screws supplied with support. Apply Loctite 242 as required.
2. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.
3. Mount Dock Asm (46–271260G3) and pad using (2) each of the M10 Hex Nuts and 11mm washers that are provided with the Dock Mounting Support. Apply Loctite 242 as necessary.
4. Fasten Dock Asm to floor mounted stud with:
   (1) 2178874 Dock Clamp Plate
   (1) 2109878–4 10.5mm ID Washer
   (1) 2109875–4 10mm Hex Nut

   Apply Loctite 242 as required.

Note:
Floor stud installed by RF room vendor/mechanical contractor.

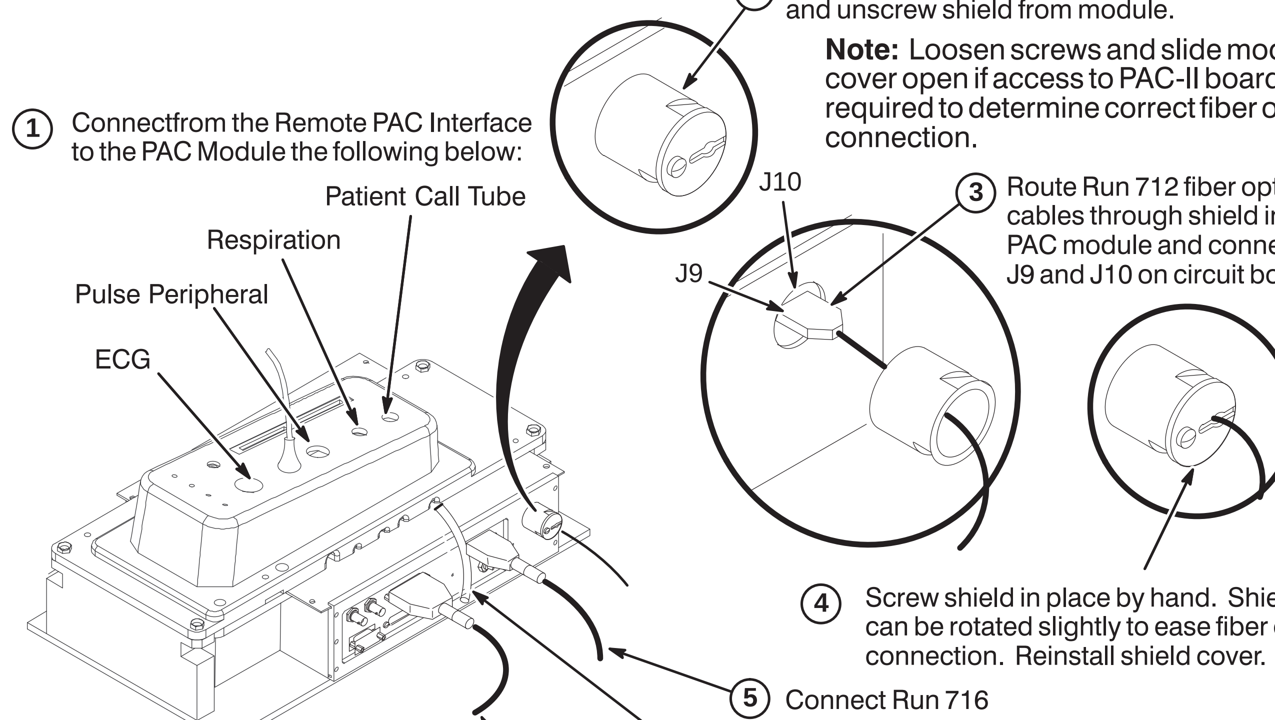
❑ X2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.



1. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
2. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
3. Route Front End Bell Microphone and Speaker cables underneath manet to ReaPedestal and connect to Patient Communication Box as marked.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
5. Connect Dock Cable to J2 on Dock Limit Box
6. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.
7. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

❑ X3 – CABLE CONNECTION TO PAC MODULE



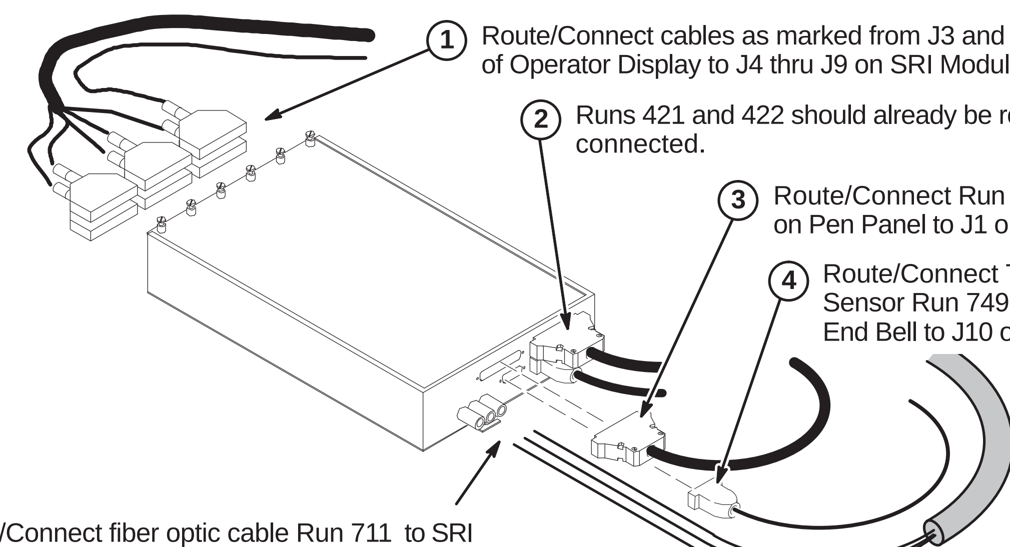
1. Connectfrom the Remote PAC Interface to the PAC Module the following below:
2. Remove shield cover by removing one screw and unscrew shield from module.

   Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.
3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cable from J5 on Operator Display to J4 on PAC Module.

INSTALLATION PROCEDURES SUMMARY

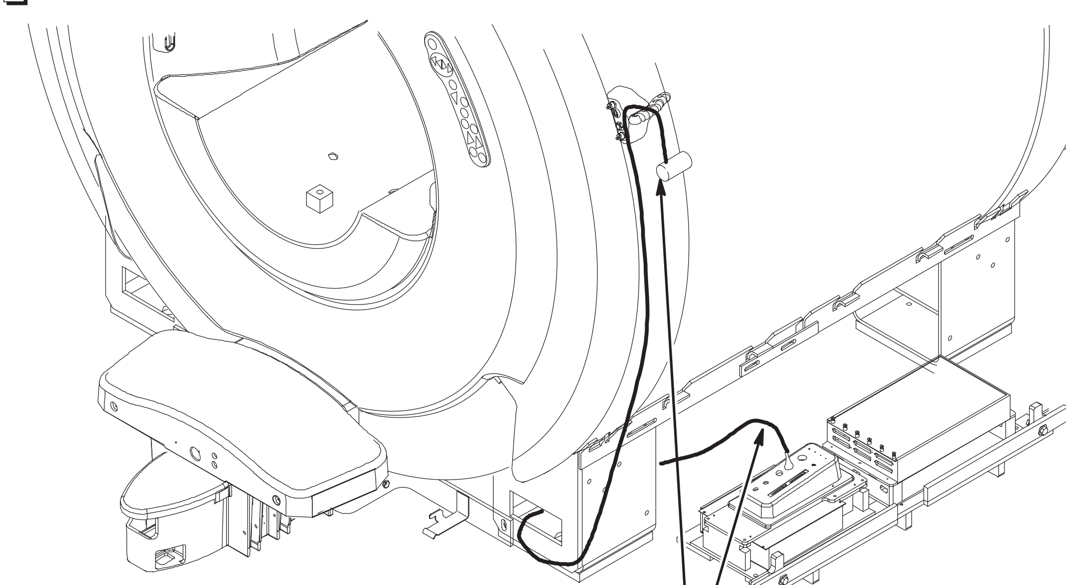
- ❑ **Y1 – Connect Cables to SRI Module**
- ❑ **Y2 – Routing of Photo–Plethysmograph Cable**
- ❑ **Y3 – Route and Tie–wrap Cables to PAC/SRI Platform**
- ❑ **Y4 – Install PAC/SRI Platform under Magnet**

❑ Y1 – CONNECT CABLES TO SRI MODULE



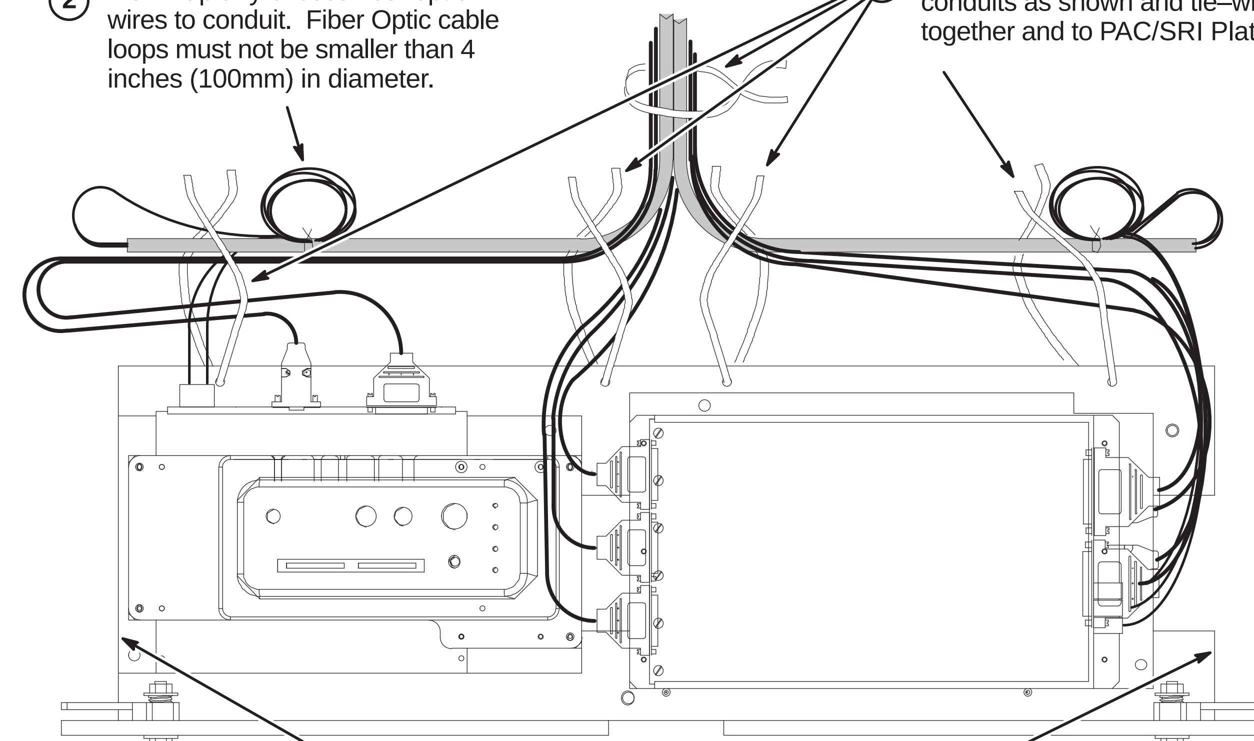
1. Route/Connect cables as marked from J3 and J4 of Operator Display to J4 thru J9 on SRI Module.
2. Runs 421 and 422 should already be routed and connected.
3. Route/Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
4. Route/Connect Temperture Sensor Run 749 from Front End Bell to J10 on SRI Module
5. Route/Connect fiber optic cable Run 711 to SRI circuit board as marked. (If necessary, remove fiber optic cable guides anmd pass fiber optic cables thru) Close SRI cover and tighten captive screws.

❑ Y2 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE



1. Route photo–plethysmograph cable from PAC module thru opening in front leg and up to PAC Hanger Bar.
2. Cable must fit between front Foot Cover and side Trim Cover (which are installed later).

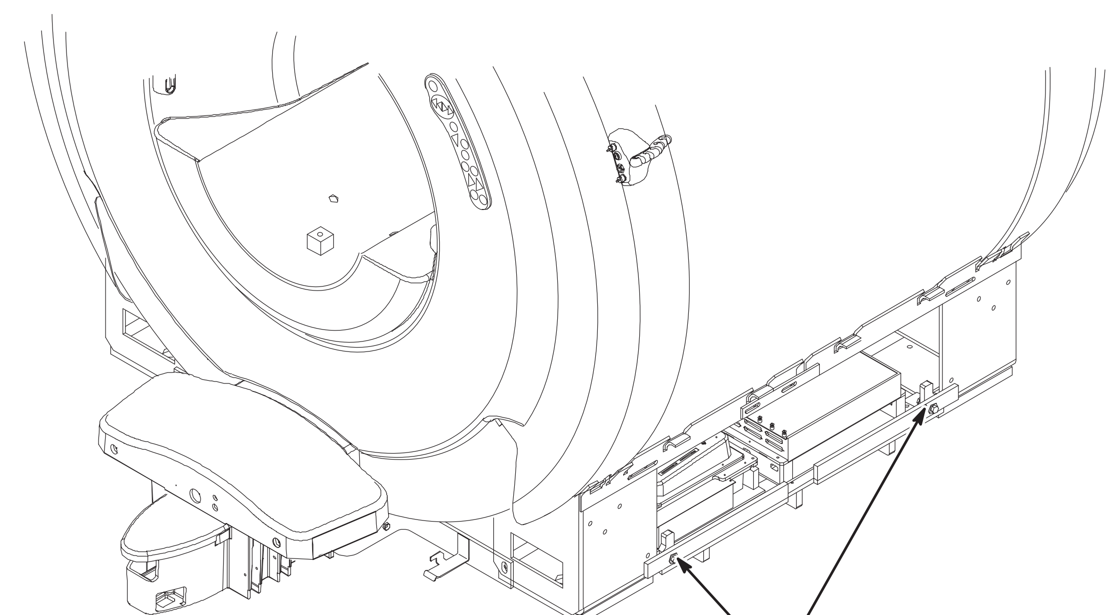
❑ Y3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM



1. Dress all cables and fiber optic conduits as shown and tie–wrap together and to PAC/SRI Platform.
2. Tie–wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.

Note: Keep cables dressed inside edge so unit slides between legs easily for service.

❑ Y4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET

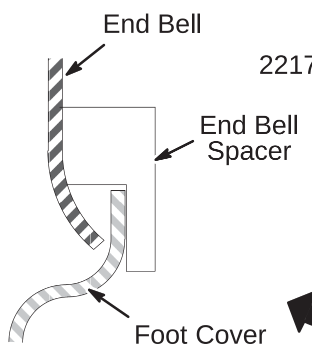


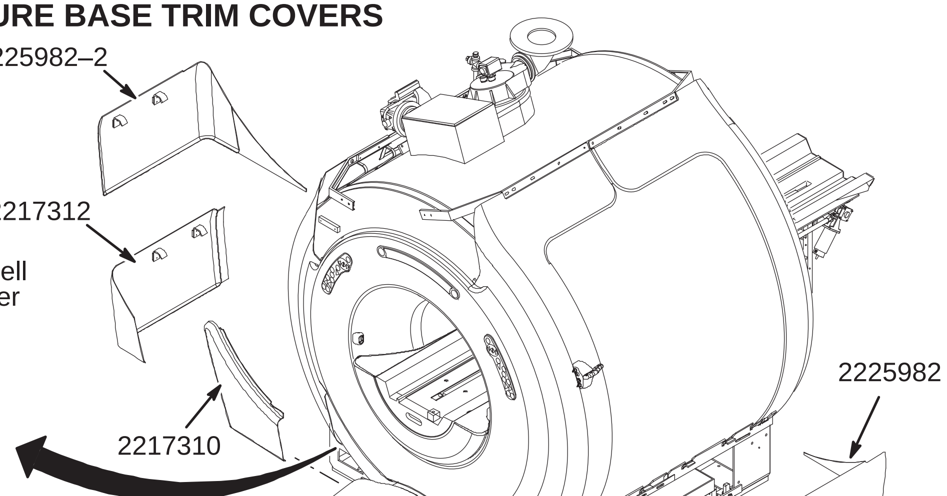
1. Carefully slide PAC/SRI Platform between legs of magnet and lock in place at each end.

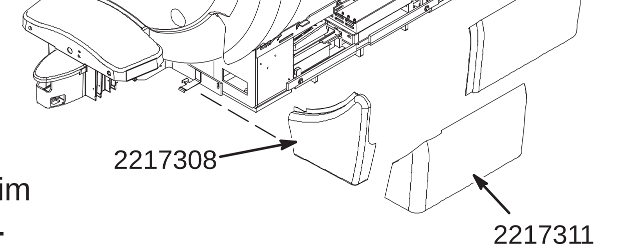
INSTALLATION PROCEDURES SUMMARY

- ❑ **Z1 – Install Enclosure Base Trim Covers**
- ❑ **Z2 – Install Enclosure Top Covers**
- ❑ **Z3 – Install Rear Pedestal Covers**
- ❑ **Z4 – Install Rating Plates**

❑ Z1 – INSTALL ENCLOSURE BASE TRIM COVERS







1. Slide front Foot Covers (2217310 & 2217308) from side, above spring clip, and between End Bell Flange and End Bell Spacers.
2. Install remaining side Base Trim Covers as shown.

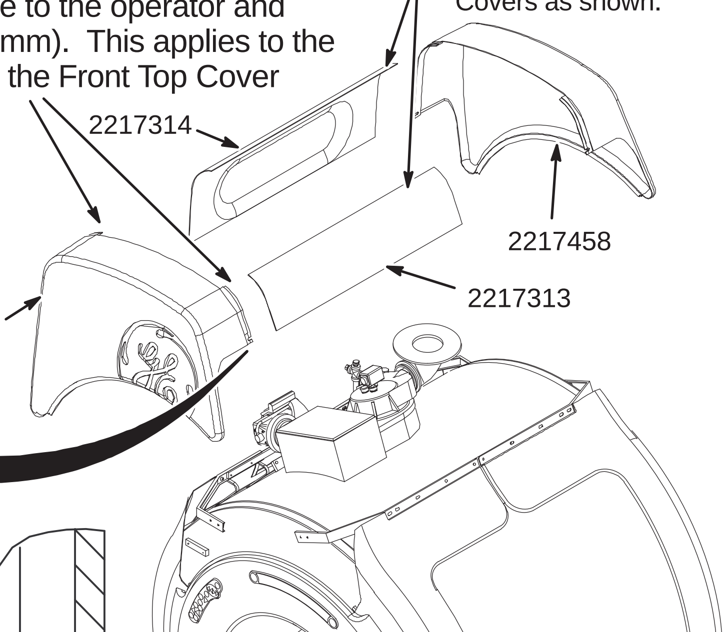
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

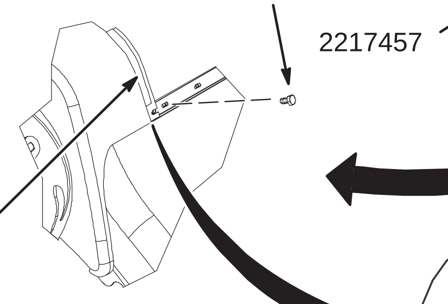
❑ Z2 – INSTALL ENCLOSURE TOP COVERS

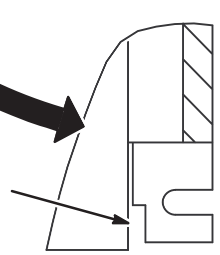
Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)

1. Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.
2. Install side Top Covers as shown.



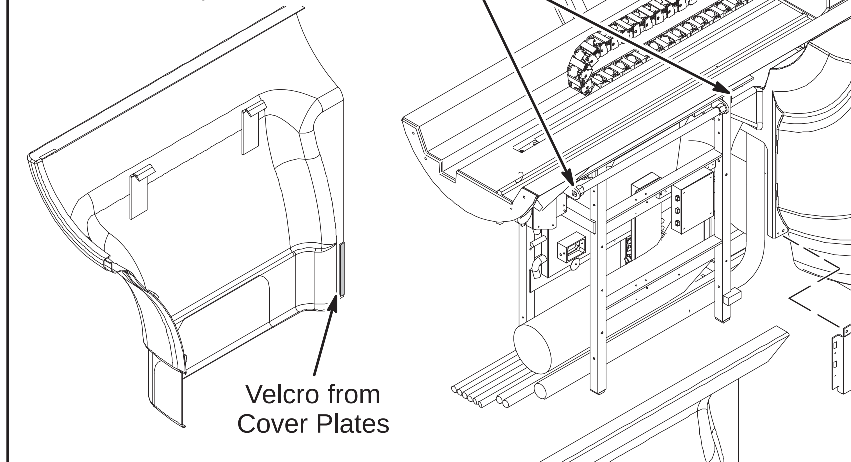


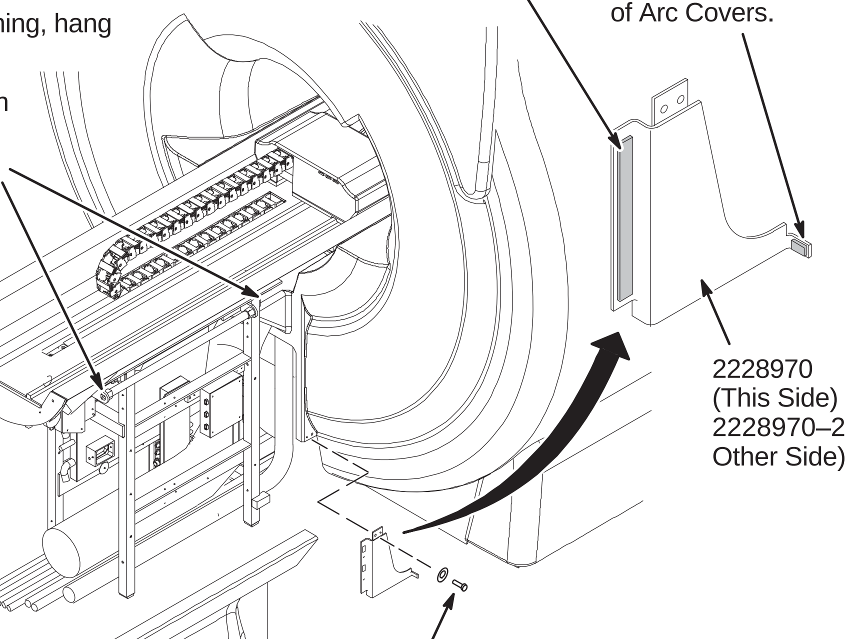


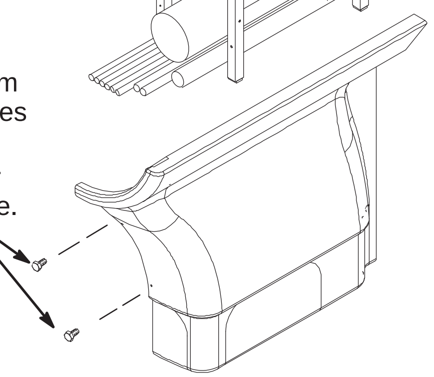
2. Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, cut a notch at this location and reinstall. Repeat as necessary until vertically level.

❑ Z3 – INSTALL REAR PEDESTAL COVERS

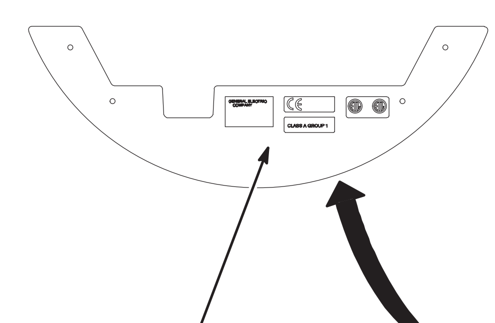
1. Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970–2) and attach to Rear Pedestal Covers
2. Remove mating velcro and attach to back side of Arc Covers.
3. Attach Rear End Bell Cover Plates (2228970 & 2228970–2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873–13) for each plate.
4. Pedestal Covers require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.
5. After cutting out appropriate opening, hang covers on Rear Pedestal.
6. If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.
7. Fasten covers together with furnished hardware.

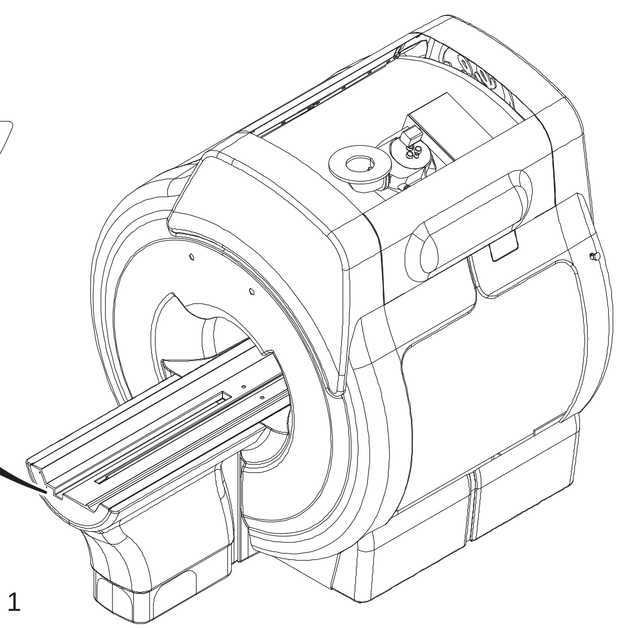






❑ Z4 – INSTALL RATING PLATES





1. If not already applied, place the following Rating Plates on Bridge Rear Plate:

46–302200P7	Serial #, Model #
2142857	ETL List for Global Certification
2117390	CE International Approval
2132620	CE Marking Label Class A Group 1